Lie Algebra

Personal Study Notes
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Preface

These notes are personal study materials based primarily on Howard Georgi's *Lie Algebras in Particle Physics*, together with my own annotations, derivations, and comments.

I welcome feedback, corrections, or discussions regarding the content of these notes. Please feel free to reach out to me via email.

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Disclaimer: These are pedagogical study notes. While I have strived for accuracy, any errors in the calculations, typos, or physical interpretations remain entirely my own. I am actively revising and correcting the notes; the latest version can always be found on the project page at physicsiphile.github.io.

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Chapter 1

Finite Groups

We will begin with an introduction to finite group theory. This is not intended to be a self-contained treatment of this enormous and beautiful subject. We will concentrate on a few simple facts that are useful in understanding the compact Lie algebras. We will introduce a lot of definitions, sometimes proving things, but often relying on the reader to prove them.

1.1 Groups and representations

A Group, G, is a set with a rule for assigning to every (ordered) pair of elements, a third element, satisfying:

(1.A.1) If $f, g \in G$ then $h = fg \in G$

(1.A.2) For $f, g, h \in G$, $f(gh) = (fg)h$

(1.A.3) There is an identity element, e, such that for all $f \in G$, $ef = fe = f$.

(1.A.4) Every element $f \in G$ has an inverse, f^{-1}, such that $ff^{-1} = f^{-1}f = e$

Thus a group is a multiplication table specifying $g_1 g_2$ for all $g_1, g_2 \in G$. If the group elements are discrete, we can write the multiplication table in the form

$$\begin{array}{c|ccc} & e & g_1 & g_2 \\ \hline e & e & g_1 & g_2 \\ g_1 & g_1 & g_1g_1 & g_1g_2 \\ g_2 & g_2 & g_2g_1 & g_2g_2 \\ \vdots & \vdots & \vdots & \vdots \end{array} \tag{1.1}$$

A Representation of G is a mapping, D of the elements of G onto a set of linear operators with the following properties:

1.B.1 $D(e) = 1$, where 1 is the identity operator in the space on which the linear operators act.

1.B.2 $D(g_1)D(g_2) = D(g_1 g_2)$, in other words the group multiplication law is mapped onto the natural multiplication in the linear space on which the linear operators act.

> Each element of the group is mapped to some linear operator. The matrix element of these linear operator will depend on the choice of basis of the vector space on which these operators will act.

1.2 Example - Z_3

A group is finite if it has a finite number of elements. Otherwise it is infinite. The number of elements in a finite group G is called the order of G. Here is a finite group of order 3.

$$\begin{array}{c|ccc} & e & a & b \\ \hline e & e & a & b \\ a & a & b & e \\ b & b & e & a \end{array} \tag{1.2}$$

This is Z_3, the cyclic group of order 3. Notice that every row and column of the multiplication table contains each element of the group exactly once. This must be the case because the inverse exists.

An Abelian group in one in which the multiplication law is commutative

$$g_1 g_2 = g_2 g_1 . \tag{1.3}$$

Evidently, Z_3 is Abelian.

The following is a representation of Z_3

$$D(e) = 1, \quad D(a) = e^{2\pi i/3}, \quad D(b) = e^{4\pi i/3} \tag{1.4}$$

The dimension of a representation is the dimension of the space on which it acts. The representation (1.4) is 1 dimensional.

1.3 The regular representation

Here's another representation of Z_3

$$D(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D(a) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad D(b) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \tag{1.5}$$

This representation was constructed directly from the multiplication table by the following trick. Take the group elements themselves to form an orthonormal basis for a vector space, $|e\rangle$, $|a\rangle$, and $|b\rangle$. Now define

$$D(g_1)|g_2\rangle = |g_1 g_2\rangle \tag{1.6}$$

The reader should show that this is a representation. It is called the regular representation. Evidently, the dimension of the regular representation is the order of the group. The matrices of (1.5) are then constructed as follows.

$$|e_1\rangle \equiv |e\rangle, \quad |e_2\rangle \equiv |a\rangle, \quad |e_3\rangle \equiv |b\rangle \tag{1.7}$$

$$[D(g)]_{ij} = \langle e_i | D(g) | e_j \rangle \tag{1.8}$$

The matrices are the matrix elements of the linear operators. (1.8) is a simple, but very general and very important way of going back and forth from operators to matrices. This works for any representation, not just the regular representation. We will use it constantly. The basic idea here is just the insertion of a complete set of intermediate states. The matrix corresponding to a product of operators is the matrix product of the matrices corresponding to the operators

$$\begin{aligned} [D(g_1 g_2)]_{ij} &= [D(g_1)D(g_2)]_{ij} \\ &= \langle e_i | D(g_1) D(g_2) | e_j \rangle \\ &= \sum_k \langle e_i | D(g_1) | e_k \rangle \langle e_k | D(g_2) | e_j \rangle \\ &= \sum_k [D(g_1)]_{ik} [D(g_2)]_{kj} \end{aligned} \tag{1.9}$$

Note that the construction of the regular representation is completely general for any finite group. For any finite group, we can define a vector space in which the basis vectors are labeled by the group elements. Then (1.6) defines the regular representation. We will see the regular representation of various groups in this chapter.

1.4 Irreducible representations

What makes the idea of group representations so powerful is the fact that they live in linear spaces. And the wonderful thing about linear spaces is we are free to choose to represent the states in a more convenient way by making a linear transformation. As long as the transformation is invertible, the new states are just as good as the old. Such a transformation on the states produces a similarity transformation on the linear operators, so that we can always make a new representation of the form

$$D(g) \to D'(g) = S^{-1}D(g)S \tag{1.10}$$

Because of the form of the similarity transformation, the new set of operators has the same multiplication rules as the old one, so D' is a representation if D is. D' and D are said to be equivalent representations because they differ just by a trivial choice of basis.

Unitary operators (O such that $O^\dagger = O^{-1}$) are particularly important. A representation is unitary if all the $D(g)$s are unitary. Both the representations we have discussed so far are unitary. It will turn out that all representations of finite groups are equivalent to unitary representations (we'll prove this later in theorem 1.1).

A representation is reducible if it has an invariant subspace (called ideals), which means that the action of any $D(g)$ on any vector in the subspace is still in the subspace. In terms of a projection operator P onto the subspace this condition can be written as

$$PD(g)P = D(g)P \quad \forall g \in G \tag{1.11}$$

> This is what defines the ideal or invariant subspace. Therefore, using ladder operators, we can only find states in each invariant subspace and we utilize orthogonality to jump between invariant subspace.

For example, the regular representation of Z_3 (1.5) has an invariant subspace projected on by

$$P = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \tag{1.12}$$

because $D(g)P = P \forall g$. The restriction of the representation to the invariant subspace is itself a representation. In this case, it is the trivial representation for which $D(g) = 1$ (the trivial representation, $D(g) = 1$, is always a representation - every group has one).

A representation is irreducible if it is not reducible.

A representation is completely reducible if it is equivalent to a representation whose matrix elements have the following form:

$$\begin{pmatrix} D_1(g) & 0 & \cdots \\ 0 & D_2(g) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \tag{1.13}$$

where $D_j(g)$ is irreducible $\forall j$. This is called block diagonal form.

A representation in block diagonal form is said to be the direct sum of the subrepresentations, $D_j(g)$,

$$D_1 \oplus D_2 \oplus \cdots \tag{1.14}$$

In transforming a representation to block diagonal form, we are decomposing the original representation into a direct sum of its irreducible components. Thus another way of defining complete reducibility is to say that a completely reducible representation can be decomposed into a direct sum of irreducible representations. This is an important idea. We will use it often.

We will show later that any representation of a finite group is completely reducible. For example, for (1.5), take

$$S = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{pmatrix} \tag{1.15}$$

where $\omega = e^{2\pi i/3}$. Then

$$D'(e) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D'(a) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix} \quad D'(b) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega^2 & 0 \\ 0 & 0 & \omega \end{pmatrix} \tag{1.16}$$

Just transforming to eigenbasis converts the representation to block diagonal form.

1.5 Transformation groups

There is a natural multiplication law for transformations of a physical system. If g_1 and g_2 are two transformations, $g_1 g_2$ means first do g_2 and then do g_1. Note that it is purely convention whether we define our composition law to be right to left, as we have done, or left to right. Either gives a perfectly consistent definition of a transformation group.

If this transformation is a symmetry of a quantum mechanical system, then the transformation takes the Hilbert space into an equivalent one. Then for each group element g, there is a unitary operator $D(g)$ that maps the Hilbert space into an equivalent one. These unitary operators form a representation of the transformation group because the transformed quantum states represent the transformed physical system. Thus for any set of symmetries, there is a representation of the symmetry group on the Hilbert space - we say that the Hilbert space transforms according to some representation of the group. Furthermore, because the transformed states have the same energy as the originals, $D(g)$ commutes with the Hamiltonian, $[D(g), H] = 0$. As we will see in more detail later, this means that we can always choose the energy eigenstates to transform like irreducible representations of the group. It is useful to think about this in a simple example.

1.6 Application: parity in quantum mechanics

Parity is the operation of reflection in a mirror. Reflecting twice gets you back to where you started. If p is a group element representing the parity reflection, this means that $p^2 = e$. Thus this is a transformation that together with the identity transformation (that is, doing nothing) forms a very simple group, with the following multiplication law:

$$\begin{array}{c|cc} & e & p \\ \hline e & e & p \\ p & p & e \end{array} \tag{1.17}$$

This group is called Z_2. For this group there are only two irreducible representations, the trivial one in which $D(p) = 1$ and one in which $D(e) = 1$ and $D(p) = -1$. Any representation is completely reducible. In particular, that means that the Hilbert space of any parity invariant system can be decomposed into states that behave like irreducible representations, that is on which $D(p)$ is either 1 or -1. Furthermore, because $D(p)$ commutes with the Hamiltonian, $D(p)$ and H can be simultaneously diagonalized. That is we can assign each energy eigenstate a definite value of $D(p)$. The energy eigenstates on which $D(p) = 1$ are said to transform according to the trivial representation. Those on which $D(p) = -1$ transform according to the other representation. This should be familiar from nonrelativistic quantum mechanics in one dimension. There you know that a particle in a potential that is symmetric about $x = 0$ has energy eigenfunctions that are either symmetric under $x \to -x$ (corresponding to the trivial representation), or antisymmetric (the representation with $D(p) = -1$).

1.7 Example: S_3

The permutation group (or symmetric group) on 3 objects, called S_3, where

$$\begin{aligned} a_1 &= (1,2,3) \\ a_2 &= (3,2,1) \\ a_3 &= (1,2) \\ a_4 &= (2,3) \\ a_5 &= (3,1) \end{aligned} \tag{1.18}$$

The notation means that a_1 is a cyclic permutation of the things in positions 1, 2 and 3; a_2 is the inverse, anticyclic permutation; a_3 interchanges the objects in positions 1 and 2; and so on. The multiplication law is then determined by the transformation rule that $g_1 g_2$ means first do g_2 and then do g_1. It is

$$\begin{array}{c|cccccc} & e & a_1 & a_2 & a_3 & a_4 & a_5 \\ \hline e & e & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_1 & a_2 & e & a_4 & a_5 & a_3 \\ a_2 & a_2 & e & a_1 & a_5 & a_3 & a_4 \\ a_3 & a_3 & a_5 & a_4 & e & a_2 & a_1 \\ a_4 & a_4 & a_3 & a_5 & a_1 & e & a_2 \\ a_5 & a_5 & a_4 & a_3 & a_2 & a_1 & e \end{array} \tag{1.19}$$

S_3 is non-Abelian because the group multiplication law is not commutative. We will see that it is the lack of commutativity that makes group theory so interesting.

Here is a unitary irreducible representation of S_3

$$\begin{aligned} D(e) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D(a_1) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix} \\ D(a_2) &= \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix}, \quad D(a_3) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \\ D(a_4) &= \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \quad D(a_5) = \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix} \end{aligned} \tag{1.20}$$

The interesting thing is that the irreducible unitary representation is more than 1 dimensional. It is necessary that at least some of the representations of a non-Abelian group must be matrices rather than numbers. Only matrices can reproduce the non-Abelian multiplication law. Not all the operators in the representation can be diagonalized simultaneously. It is this that is responsible for a lot of the power of the theory of group representations.

1.8 Example: addition of integers

The integers form an infinite group under addition. If the group operation is denoted multiplicatively, the law is written as

$$xy = x + y. \tag{1.21}$$

This is the additive group of the integers. Since there are infinitely many integers, there is no finite multiplication table, but the displayed rule fixes the group law completely.

A useful two-dimensional representation is

$$D(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}. \tag{1.22}$$

which respects $D(x).D(y) = D(x + y)$. This representation is reducible, because the subspace projected onto by

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{1.23}$$

is invariant:

$$D(x)P = P. \tag{1.24}$$

But it is not completely reducible, since

$$D(x)(1 - P) \neq (1 - P). \tag{1.25}$$

Thus the complementary subspace is not invariant. This example is important because it shows what can happen for infinite groups. For finite groups this pathology is absent: every representation is equivalent to a unitary one, and every reducible representation is completely reducible.

1.9 Useful theorems

Theorem 1.1. Every representation of a finite group is equivalent to a unitary representation.

Proof. Let $D(g)$ be a representation of a finite group G. Define

$$S = \sum_{g \in G} D(g)^\dagger D(g). \tag{1.26}$$

The operator S is Hermitian and positive. Diagonalize it as

$$S = U^\dagger d U, \tag{1.27}$$

where d is diagonal. The eigenvalues of S are non-negative. They are in fact strictly positive: if $SA = 0$ for some nonzero vector A, then

$$A^\dagger \underbrace{SA}_{=0} = A^\dagger \sum_{g \in G} D(g)^\dagger D(g) A = \sum_{g \in G} \|D(g)A\|^2 = 0, \tag{1.28}$$

so every $D(g)A$ would vanish (sum of square only vanishes when each term vanishes), including $D(e)A = A$, a contradiction. Hence S has an invertible Hermitian square root X,

$$S = X^2. \tag{1.29}$$

Now define an equivalent representation

$$D'(g) = X D(g) X^{-1}. \tag{1.30}$$

Then

$$D'(g)^\dagger D'(g) = X^{-1}D(g)^\dagger X^2 D(g)X^{-1} \tag{1.31}$$
$$= X^{-1}D(g)^\dagger SD(g)X^{-1}. \tag{1.32}$$

But

$$D(g)^\dagger SD(g) = D(g)^\dagger \left(\sum_{h\in G} D(h)^\dagger D(h)\right) D(g) \tag{1.33}$$
$$= \sum_{h\in G} D(hg)^\dagger D(hg) \tag{1.34}$$
$$= \sum_{h\in G} D(h)^\dagger D(h) = S = X^2, \tag{1.35}$$

because hg runs over the whole group as h runs over the whole group. Therefore $D'(g)^\dagger D'(g) = 1$, and the equivalent representation is unitary.

Theorem 1.2. Every representation of a finite group is completely reducible.

Proof. By Theorem 1.1, it is enough to consider unitary representations. If the representation is irreducible, there is nothing to prove. If it is reducible, let P project onto an invariant subspace:

$$PD(g)P = D(g)P. \tag{1.36}$$

Taking the adjoint and using unitarity gives

$$PD(g)P = PD(g) \tag{1.37}$$

for all g, since $D(g)^\dagger = D(g)^{-1} = D(g^{-1})$ and g^{-1} also runs over G. It follows that the complementary projector $1 - P$ also projects onto an invariant subspace:

$$(1 - P)D(g)(1 - P) = D(g)(1 - P). \tag{1.38}$$

Repeating the argument on each invariant subspace gives a direct sum of irreducible representations.

If P projects onto U, then $1 - P$ projects onto U^* such that U and U^* are disjoint.

1.10 Subgroups

A subgroup H of a group G is a group whose elements are all elements of G, with the same multiplication law. The identity subgroup and G itself are the trivial subgroups. Nontrivial subgroups occur frequently; for example S_3 contains the Z_3 subgroup $\{e, a_1, a_2\}$.

A right coset of H in G is a set of elements of the form

$$Hg = \{hg : h \in H\} \tag{1.39}$$

for a fixed $g \in G$. Left cosets are defined analogously. Every element of G belongs to exactly one coset, and every coset has the same number of elements as H. Hence, for a finite group, the order of a subgroup divides the order of the group.

A subgroup H is invariant, or normal, if

$$gH = Hg \qquad \forall g \in G. \tag{1.40}$$

Equivalently,

$$gHg^{-1} = H \qquad \forall g \in G. \tag{1.41}$$

If H is normal, the coset space G/H inherits a group structure and is called the factor group. The product of two cosets is represented by

$$(Hg_1)(Hg_2) = Hg_1g_2. \tag{1.42}$$

For example,

$$S_3/Z_3 \cong Z_2. \tag{1.43}$$

The center of G is the set of all elements that commute with every element of G. The center is always an Abelian normal subgroup.

A related notion is a conjugacy class. For a fixed element $g_1 \in G$, its conjugacy class is the set

$$\{gg_1g^{-1} : g \in G\}. \tag{1.44}$$

For S_3 the conjugacy classes are

$$\{e\}, \qquad \{a_1, a_2\}, \qquad \{a_3, a_4, a_5\}. \tag{1.45}$$

A subgroup that is a union of conjugacy classes is normal.

The map

$$g_1 \mapsto g^{-1}g_1g \tag{1.46}$$

for fixed g is an inner automorphism. It is an automorphism because it preserves multiplication and is one-to-one. An automorphism not expressible in this way is called an outer automorphism.

1.11 Schur's lemma

Theorem 1.3. If

$$D_1(g)A = AD_2(g) \qquad \forall g \in G, \tag{1.47}$$

where D_1 and D_2 are inequivalent irreducible representations, then

$$A = 0. \tag{1.48}$$

Proof. Suppose some vector $|\mu\rangle$ is annihilated by A. Then there is a non zero projector P which maps elements of the space onto $\ker A$ ($AP\,|x\rangle = 0$). The subspace annihilated by A on the right is invariant under D_2, because

$$AD_2(g)P = D_1(g)AP = 0. \tag{1.49}$$

Since $A\,|\mu\rangle = 0$ and $P\,|x\rangle \in \ker A$ which implies $D_2(g)P\,|x\rangle \in \ker A$. Thus making the $\ker A$, an invariant subspace.

Since D_2 is irreducible, this subspace is either zero or the whole space. Hence if A kills one nonzero vector, it kills all vectors. A similar argument applies to vectors annihilating A on the left. If no such vector exists (which we expect since eigenvalues are zero and hence the determinant), A must be square and invertible. Then from (1.49)

$$D_1(g) = AD_2(g)A^{-1}, \tag{1.50}$$

which would make D_1 and D_2 equivalent, contrary to assumption. Therefore $A = 0$.

An irreducible representation has no proper invariant subspace so the only way $\ker A$ is not proper invariant subspace is if $\ker A$ = wholespace and this P projects onto whole space and A annihilates them all.

Theorem 1.4. If

$$D(g)A = AD(g) \qquad \forall g \in G, \tag{1.51}$$

where D is a finite-dimensional irreducible representation, then

$$A = c\,1 \tag{1.52}$$

for some scalar c.

Proof. Since A is finite-dimensional, it has at least one eigenvalue λ. Then $A - \lambda 1$ also commutes with all $D(g)$ and annihilates an eigenvector. The previous invariant-subspace argument implies

$$A - \lambda 1 = 0. \tag{1.53}$$

Thus A is proportional to the identity.

$[A, D(g)] = 0$ so all states are eigenstate of A, thus we can think of it as λI where λ is the eigenvalue of A.

An important interpretation is that, once the matrices of an irreducible representation have been fixed, there is no further nontrivial similarity transformation that leaves all of them unchanged. A change of basis that keeps the representation matrices identical can only multiply all basis states by a common scalar.

A useful consequence of Schur's lemma is that, once an irreducible representation has been placed in a fixed matrix form, its basis is essentially determined. Indeed, Theorem 1.4 may be restated as

$$A^{-1}D(g)A = D(g) \quad \forall\, g \in G \qquad \Longrightarrow \qquad A \propto I. \tag{1.54}$$

Thus, for an irreducible representation D, there is no nontrivial similarity transformation that preserves every matrix $D(g)$. If the change of basis is required to be unitary, the only remaining freedom is an overall phase multiplying all states in the representation.

Schur's lemma also strongly constrains matrix elements of a symmetry-invariant quantum-mechanical operator O. To make this explicit, consider the complete decomposition of the Hilbert space into irreducible representations. The symmetry group acts through a unitary representation according to

$$g \longmapsto D(g), \qquad \forall\, g \in G. \tag{1.55}$$

Here D is generally reducible, since it acts on the full Hilbert space. Because the representation is completely reducible, one may choose a basis in which $D(g)$ is block diagonal, with each block furnishing an irreducible unitary representation of G.

Let the orthonormal basis states be written as

$$|a, j, x\rangle\,, \tag{1.45}$$

States belonging to different irreducible representations are orthogonal.

with inner products

$$\langle a, j, x | b, k, y\rangle = \delta_{ab}\, \delta_{jk}\, \delta_{xy}. \tag{1.56}$$

The label a specifies the irreducible representation (usually the eigenvalue of Casimir Operator), while $j = 1, \ldots, n_a$ labels the states within that representation. The label x denotes any additional quantum numbers needed to distinguish different copies or physical states.

The factor δ_{ab} ensures that states belonging to inequivalent irreducible representations are orthogonal. We now choose the basis so that every occurrence of a given irreducible representation a is described by the same canonical matrices $D_a(g)$. In this basis,

$$\langle a, j, x|D(g)|b, k, y\rangle = \delta_{ab}\,\delta_{xy}\,\big[D_a(g)\big]_{jk}. \tag{1.57}$$

Thus $D(g)$ acts only within states carrying the same representation label a and the same additional label x; the indices j and k are mixed by the irreducible matrix $D_a(g)$.

The identity operator on the complete Hilbert space is

$$I = \sum_{a,j,x} |a, j, x\rangle\,\langle a, j, x|\,. \tag{1.58}$$

Inserting this resolution of the identity on both sides of $D(g)$, one finds

$$\begin{aligned} D(g) &= \sum_{a,j,x} |a, j, x\rangle\,\langle a, j, x|\,D(g)\sum_{b,k,y} |b, k, y\rangle\,\langle b, k, y| \\ &= \sum_{\substack{a,j,x \\ b,k,y}} |a, j, x\rangle\;\delta_{ab}\delta_{xy}\big[D_a(g)\big]_{jk}\,\langle b, k, y| \\ &= \sum_{a,j,k,x} |a, j, x\rangle\,\big[D_a(g)\big]_{jk}\,\langle a, k, x|\,. \end{aligned} \tag{1.59}$$

Equation (1.59) is simply the explicit operator form of a block-diagonal representation. If an irreducible representation occurs only once in D, the extra label x is unnecessary. In a typical quantum-mechanical Hilbert space, however, the same irreducible representation can occur many times. The label x then distinguishes states that transform identically under the symmetry group but correspond to different physical degrees of freedom.

The dependence of $D(g)$ on these additional labels is trivial: the group transformation does not mix different values of x. Consequently, the states remain orthogonal in these labels, while all nontrivial group action is carried by the representation indices.

Under a symmetry transformation, kets and bras transform as

$$|\mu\rangle \longrightarrow D(g)\,|\mu\rangle\,, \qquad \langle\mu| \longrightarrow \langle\mu|\,D(g)^\dagger. \tag{1.60}$$

To preserve matrix elements, an operator must therefore transform according to

$$O \longrightarrow D(g)\,O\,D(g)^\dagger. \tag{1.61}$$

An observable invariant under the symmetry satisfies

$$O \longrightarrow D(g)\,O\,D(g)^\dagger = O. \tag{1.62}$$

Equivalently,

$$[O, D(g)] = 0, \qquad \forall\, g \in G. \tag{1.63}$$

We now use this condition to constrain the matrix element

$$\langle a, j, x|O|b, k, y\rangle\,. \tag{1.64}$$

Since O commutes with every $D(g)$,

$$\begin{aligned} 0 &= \langle a, j, x|[O, D(g)]|b, k, y\rangle \\ &= \sum_{k'} \big\langle a, j, x\big|O\big|b, k', y\big\rangle\,\big\langle b, k', y\big|D(g)\big|b, k, y\big\rangle \end{aligned}$$

$$- \sum_{j'} \left\langle a, j, x \middle| D(g) \middle| a, j', x \right\rangle \left\langle a, j', x \middle| O \middle| b, k, y \right\rangle . \tag{1.65}$$

Using the block-diagonal form of the representation matrices in (1.57), this becomes

$$\begin{aligned} 0 = & \langle a, j, x | [O, D(g)] | b, k, y \rangle \\ = & \sum_{k'} \left\langle a, j, x \middle| O \middle| b, k', y \right\rangle \left[D_b(g) \right]_{k'k} \\ & - \sum_{j'} \left[D_a(g) \right]_{jj'} \left\langle a, j', x \middle| O \middle| b, k, y \right\rangle . \end{aligned} \tag{1.66}$$

For fixed x and y, the matrix

$$\langle a, j, x | O | b, k, y \rangle$$

therefore intertwines the irreducible representations D_b and D_a. Schur's lemma now applies. If $a \neq b$, the matrix element must vanish. If $a = b$, it must be proportional to the identity in the representation indices. Hence

$$\langle a, j, x | O | b, k, y \rangle = f_a(x, y)\, \delta_{ab}\, \delta_{jk}. \tag{1.67}$$

All dependence on the representation labels is therefore fixed by symmetry. The dynamical information is contained entirely in the function $f_a(x, y)$, which depends only on the remaining physical labels. This is the simplest form of the Wigner–Eckart principle.

1.12 * Orthogonality relations

Let D_a and D_b be irreducible representations. Schur's lemma gives powerful orthogonality relations among their matrix elements. For an operator A between the two representation spaces, construct the group-averaged operator

$$\overline{A} = \sum_{g \in G} D_a(g) A D_b(g^{-1}). \tag{1.68}$$

This averaged operator intertwines the two representations. If $a \neq b$, Schur's lemma says $\overline{A} = 0$. If $a = b$, it is proportional to the identity. Choosing A to be a matrix unit gives

$$\sum_{g \in G} [D_a(g)]_{ij} [D_b(g^{-1})]_{k\ell} = \frac{N}{n_a}\, \delta_{ab} \delta_{i\ell} \delta_{jk}, \tag{1.69}$$

where $N = |G|$ and $n_a = \dim D_a$.

For unitary irreducible representations this becomes

$$\sum_{g \in G} [D_a(g)]^*_{ij} [D_b(g)]_{k\ell} = \frac{N}{n_a}\, \delta_{ab} \delta_{ik} \delta_{j\ell}. \tag{1.70}$$

Thus, with normalization, the matrix elements of inequivalent irreducible representations form an orthonormal set of functions on the group.

Completeness follows from the regular representation. Any function $F(g)$ on the group can be written using the regular-representation basis as

$$F(g) = \langle F | g \rangle = \langle F | D_R(g) | e \rangle, \tag{1.71}$$

where

$$\langle F | = \sum_{g' \in G} F(g') \langle g' |. \tag{1.72}$$

Since the regular representation is completely reducible, every function on the group can be expanded in the matrix elements of irreducible representations.

Theorem 1.5. The matrix elements of the unitary irreducible representations of a finite group form a complete orthonormal basis for functions on the group.

A consequence is

$$|G| = N = \sum_a n_a^2, \tag{1.73}$$

where the sum is over inequivalent irreducible representations.

Example: Fourier series. For the cyclic group Z_N with elements a^j, $j = 0, \ldots, N-1$, and multiplication

$$a^j a^k = a^{(j+k) \bmod N}, \tag{1.74}$$

the irreducible representations are one-dimensional:

$$D_n(a^j) = e^{2\pi i n j/N}. \tag{1.75}$$

The orthogonality relation becomes

$$\frac{1}{N}\sum_{j=0}^{N-1} e^{-2\pi i n' j/N} e^{2\pi i n j/N} = \delta_{n'n}, \tag{1.76}$$

which is the basic finite Fourier orthogonality relation.

1.13 Characters

The character of a representation D is the trace of the representing operator:

$$\chi_D(g) = \mathrm{Tr}\, D(g) = \sum_i [D(g)]_{ii}. \tag{1.77}$$

Characters are unchanged by similarity transformations, because traces are cyclic. Hence equivalent representations have the same characters.

From the orthogonality of matrix elements, the characters of inequivalent irreducible representations satisfy

$$\frac{1}{N}\sum_{g\in G} \chi_{D_a}(g)^* \chi_{D_b}(g) = \delta_{ab}. \tag{1.78}$$

Characters are constant on conjugacy classes:

$$\chi_D(g^{-1}hg) = \mathrm{Tr}igl(D(g)^{-1}D(h)D(g)\bigr) = \mathrm{Tr}\, D(h) = \chi_D(h). \tag{1.79}$$

Moreover, the irreducible characters form a complete orthonormal basis for functions constant on conjugacy classes. Therefore,

$$\boxed{\text{number of irreducible representations} = \text{number of conjugacy classes.}} \tag{1.80}$$

If the conjugacy classes are labeled by α and k_α is the number of elements in class α, one may write the character orthogonality as

$$\sum_\alpha \frac{k_\alpha}{N} \chi_a(C_\alpha)^* \chi_b(C_\alpha) = \delta_{ab}. \tag{1.81}$$

There is also the dual orthogonality relation over representations,

$$\sum_a \chi_a(C_\alpha)^* \chi_a(C_\beta) = \frac{N}{k_\alpha}\delta_{\alpha\beta}. \tag{1.82}$$

If D is a reducible representation containing the irreducible representation D_a with multiplicity m_a, then

$$m_a = \frac{1}{N} \sum_{g \in G} \chi_{D_a}(g)^* \chi_D(g). \tag{1.83}$$

For the regular representation,

$$\chi_R(e) = N, \qquad \chi_R(g) = 0 \quad (g \neq e), \tag{1.84}$$

so each irreducible representation appears in the regular representation a number of times equal to its dimension.

For S_3, with conjugacy classes $\{e\}$, $\{a_1, a_2\}$, and $\{a_3, a_4, a_5\}$, the character table is

$$\begin{array}{c|ccc} & e & \{a_1,a_2\} & \{a_3,a_4,a_5\} \\ \hline D_0 & 1 & 1 & 1 \\ D_1 & 1 & 1 & -1 \\ D_2 & 2 & -1 & 0 \end{array} \tag{1.85}$$

The character projection operator onto the part of a representation transforming as D_a is

$$P_a = \frac{n_a}{N} \sum_{g \in G} \chi_{D_a}(g)^* D(g). \tag{1.86}$$

This gives the projector in the original basis, without first explicitly block-diagonalizing the representation.

For the defining three-dimensional representation of S_3, the projectors are

$$P_0 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \tag{1.87}$$

which projects onto the invariant combination

$$\frac{1}{\sqrt{3}} \big(|1\rangle + |2\rangle + |3\rangle \big), \tag{1.88}$$

and

$$P_2 = 1 - P_0, \tag{1.89}$$

which projects onto the two-dimensional subspace spanned by differences such as $|1\rangle - |2\rangle$. Thus

$$D_3 = D_0 \oplus D_2. \tag{1.90}$$

1.14 Eigenstates

In quantum mechanics, one often wants the eigenstates of an invariant Hermitian operator, especially the Hamiltonian. If

$$[D(g), H] = 0 \qquad \forall g \in G, \tag{1.91}$$

then each eigenspace of H is invariant under the symmetry group. Therefore each eigenspace carries a representation of the group, and that representation can be completely reduced into irreducible representations.

If an irreducible representation appears only once in the Hilbert space, then every state in that irreducible representation is automatically an eigenstate of any invariant operator. Schematically,

$$H|a, j, x\rangle = \sum_y c_y |a, j, y\rangle, \tag{1.92}$$

and if the multiplicity label x has only one value, the action of H is just multiplication by a number.

Theorem 1.6. If a Hermitian operator H commutes with all $D(g)$, then the eigenstates of H can be chosen to transform according to irreducible representations of G. If an irreducible representation appears only once, all states in that irreducible representation have the same eigenvalue of H.

For Abelian groups this resembles simultaneous diagonalization. In fact:

Theorem 1.7. All irreducible representations of a finite Abelian group are one-dimensional.

For an Abelian group, each element is its own conjugacy class. Hence the number of irreducible representations equals the order of the group. Since $|G| = \sum_a n_a^2$, all n_a must be 1.

For non-Abelian groups, the representation matrices cannot generally all be diagonalized at once. The replacement for simultaneous diagonalization is complete reduction into irreducible representations. The classical analogue is the normal-mode problem for small oscillations, where the normal modes are eigenvectors of the relevant dynamical matrix.

1.15 Tensor products

Given an m-dimensional representation D_1 with basis $|j\rangle$, and an n-dimensional representation D_2 with basis $|x\rangle$, the tensor product space has basis

$$|j, x\rangle, \tag{1.93}$$

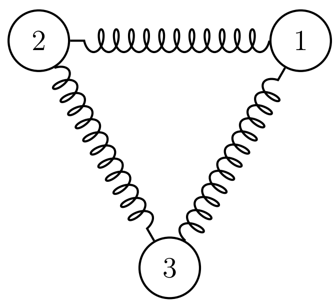
with mn basis vectors. The tensor product representation $D_1 \otimes D_2$ is defined by

$$[D_1 \otimes D_2(g)]_{jx,ky} = [D_1(g)]_{jk}[D_2(g)]_{xy}. \tag{1.94}$$

This is a representation of G, but it is generally reducible. Decomposing tensor products into irreducible representations is one of the standard uses of character theory.

1.16 Example of tensor products

Consider three identical blocks connected by springs in a triangular configuration on a frictionless surface.



The system has six degrees of freedom corresponding to the x and y coordinates of the three masses:

$$(x_1,\, y_1,\, x_2,\, y_2,\, x_3,\, y_3) \tag{1.95}$$

This configuration space decomposes as a tensor product $\mathbb{R}^3 \otimes \mathbb{R}^2$, where the 3D space tracks the block positions and the 2D space tracks spatial coordinates. We define components as $r_{j\mu}$, where $j \in \{1, 2, 3\}$ labels the mass and $\mu \in \{1, 2\}$ labels the coordinate direction ($1 \to x$, $2 \to y$):

$$(x_1,\, y_1,\, x_2,\, y_2,\, x_3,\, y_3) = (r_{11},\, r_{12},\, r_{21},\, r_{22},\, r_{31},\, r_{32}) \tag{1.96}$$

The system possesses S_3 permutation symmetry. The 3D positional space transforms under the standard 3D permutation representation D_3, while the 2D spatial coordinate space transforms under D_2 as given by:

$$\begin{aligned} D_2(e) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & D_2(a_1) &= \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \\ D_2(a_2) &= \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, & D_2(a_3) &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \\ D_2(a_4) &= \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, & D_2(a_5) &= \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \end{aligned} \tag{1.97}$$

Using (1.94), the 6-dimensional representation of coordinates is the tensor product of the 3-dimensional positional representation D_3 and the 2-dimensional coordinate representation D_2:

$$[D_6(g)]_{j\mu,k\nu} = [D_3(g)]_{jk}[D_2(g)]_{\mu\nu} \tag{1.98}$$

For element a_1, substituting $D_2(a_1)$ into $D_3(a_1)$ gives:

$$D_6(a_1) = \begin{pmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 0 \end{pmatrix} \tag{1.99}$$

This replaces each non-zero entry (1) in $D_3(a_1)$ with a copy of $D_2(a_1)$. The remaining group generators follow the same block matrix structure.

Due to S_3 symmetry, the normal modes transform under irreducible representations (irreps) of S_3. Projecting onto these irreps isolates the normal modes: an irrep occurring once in D_6 directly corresponds to a single normal mode (by Theorem 1.6), whereas repeated irreps require further diagonalization.

To count irrep multiplicities in D_6, we evaluate the character of the tensor product. The character of a product representation equals the product of individual characters ($\chi_{D_1\times D_2} = \chi_{D_1}\chi_{D_2}$):

$$\chi_{D_1\times D_2} = \chi_{D_1}\chi_{D_2} \tag{1.100}$$

Taking the trace over the combined indices yields:

$$\begin{aligned} \chi_6(g) &= \sum_{j,\mu}[D_6(g)]_{j\mu,j\mu} \\ &= [D_3(g)]_{jj}[D_2(g)]_{\mu\mu} = \chi_3(g)\chi_2(g) \end{aligned} \tag{1.101}$$

Multiplying the characters of D_3 and D_2 component-wise yields the character table for D_6:

(1.102)

	e	a_1 a_2	a_3 a_4 a_5
D_3	3	0	1
D_2	2	-1	0
D_6	6	0	0

These characters match those of the regular representation of S_3. Consequently, D_6 decomposes into irreducible representations as:

$$D_6 \cong D_0 \oplus D_1 \oplus 2D_2$$

where D_0 is the trivial representation, D_1 is the sign representation, and D_2 is the two-dimensional irreducible representation.

Theorem 1.8. *The characters of a tensor product representation are the products of the characters of its constituent factors.*

1.17 Finding the normal modes

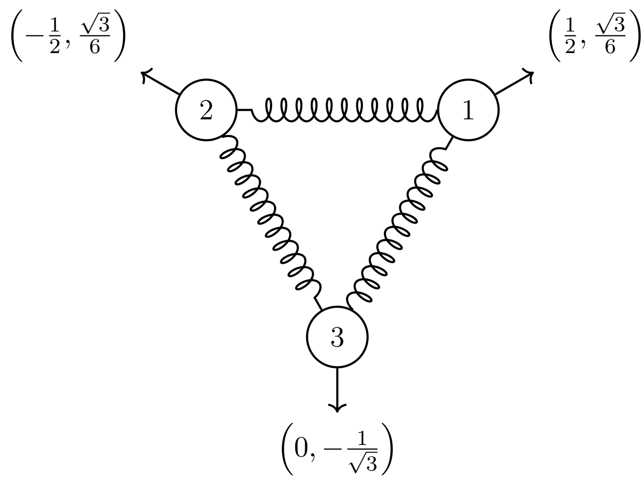
The projection operators onto D_0 and D_1 project onto 1-dimensional subspaces. The projector P_0 onto the trivial representation D_0 is defined as:

$$\begin{aligned} P_0 &= \frac{1}{6}\sum_{g\in G}\chi_0(g)^* D_6(g) \\ &= \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{12} & -\frac{1}{4} & \frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{12} & \frac{1}{12} & -\frac{\sqrt{3}}{12} & \frac{1}{12} & 0 & -\frac{\sqrt{3}}{6} \\ -\frac{1}{4} & -\frac{\sqrt{3}}{12} & \frac{1}{4} & -\frac{\sqrt{3}}{12} & 0 & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{12} & \frac{1}{12} & -\frac{\sqrt{3}}{12} & \frac{1}{12} & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & -\frac{1}{6} & \frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} \end{pmatrix} \end{aligned} \tag{1.103}$$

Factoring P_0 as an outer product of its normalized eigenvector gives:

$$P_0 = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{6} \\ -\frac{1}{2} \\ \frac{\sqrt{3}}{6} \\ 0 \\ -\frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{6} & -\frac{1}{2} & \frac{\sqrt{3}}{6} & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix} \tag{1.104}$$

corresponding to the motion



This vector describes the "breathing mode," where the triangle expands and contracts while maintaining its shape.

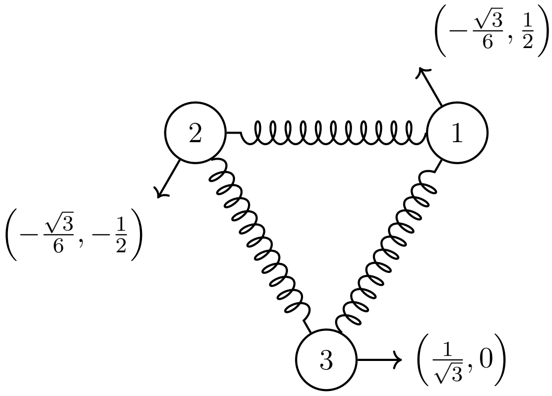
The projection operator P_1 onto the D_1 representation is:

$$
\begin{aligned}
P_1 &= \frac{1}{6}\sum_{g\in G}\chi_1(g)^* D_6(g) \\
&= \begin{pmatrix}
\frac{1}{12} & -\frac{\sqrt{3}}{12} & \frac{1}{12} & \frac{\sqrt{3}}{12} & -\frac{1}{6} & 0 \\
-\frac{\sqrt{3}}{12} & \frac{1}{4} & -\frac{\sqrt{3}}{12} & -\frac{1}{4} & \frac{\sqrt{3}}{6} & 0 \\
\frac{1}{12} & -\frac{\sqrt{3}}{12} & \frac{1}{12} & \frac{\sqrt{3}}{12} & -\frac{1}{6} & 0 \\
\frac{\sqrt{3}}{12} & -\frac{1}{4} & \frac{\sqrt{3}}{12} & \frac{1}{4} & -\frac{\sqrt{3}}{6} & 0 \\
-\frac{1}{6} & \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & \frac{1}{3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}
\end{aligned}
\tag{1.105}
$$

Factoring P_1 as an outer product gives:

$$
P_1 = \begin{pmatrix} -\frac{\sqrt{3}}{6} \\ \frac{1}{2} \\ -\frac{\sqrt{3}}{6} \\ -\frac{1}{2} \\ \frac{1}{\sqrt{3}} \\ 0 \end{pmatrix}
\begin{pmatrix} -\frac{\sqrt{3}}{6} & \frac{1}{2} & -\frac{\sqrt{3}}{6} & -\frac{1}{2} & \frac{1}{\sqrt{3}} & 0 \end{pmatrix}
\tag{1.106}
$$

corresponding to the motion



This corresponds to rigid rotation of the triangle—a zero-frequency normal mode since no restoring forces act on it. These two normal modes were determined purely using group-theoretic symmetry without solving the equations of motion.

The projector P_2 onto the 2-dimensional representation D_2 is:

$$
\begin{aligned}
P_2 &= \frac{2}{6}\sum_{g\in G}\chi_2(g)^* D_6(g) \\
&= \begin{pmatrix}
\frac{2}{3} & 0 & \frac{1}{6} & -\frac{\sqrt{3}}{6} & \frac{1}{6} & \frac{\sqrt{3}}{6} \\
0 & \frac{2}{3} & \frac{\sqrt{3}}{6} & \frac{1}{6} & -\frac{\sqrt{3}}{6} & \frac{1}{6} \\
\frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{2}{3} & 0 & \frac{1}{6} & -\frac{\sqrt{3}}{6} \\
-\frac{\sqrt{3}}{6} & \frac{1}{6} & 0 & \frac{2}{3} & \frac{\sqrt{3}}{6} & \frac{1}{6} \\
\frac{1}{6} & -\frac{\sqrt{3}}{6} & \frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{2}{3} & 0 \\
\frac{\sqrt{3}}{6} & \frac{1}{6} & -\frac{\sqrt{3}}{6} & \frac{1}{6} & 0 & \frac{2}{3}
\end{pmatrix}
\end{aligned}
\tag{1.107}
$$

Since $\operatorname{Tr} P_2 = 4$, P_2 is a rank-4 projector spanning four modes. Isolating these modes requires basic dynamical considerations. Two correspond to rigid translations of the whole system. The projector for uniform x-translations is:

$$T_x = \begin{pmatrix} \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.108}$$

The projector for uniform translations in the y direction is:

$$T_y = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \end{pmatrix} \tag{1.109}$$

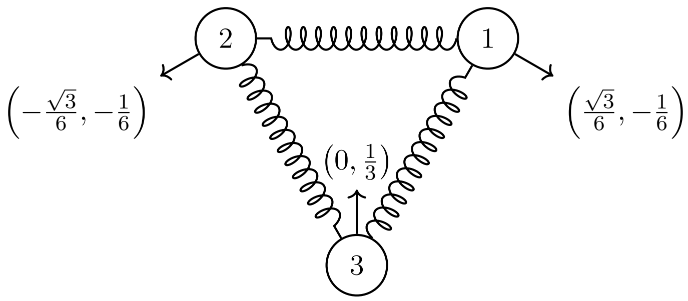
Subtracting the translation operators T_x (1.108) and T_y (1.109) from P_2 (1.107) yields the projector onto the non-trivial vibrational modes:

$$P_2 - T_x - T_y = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & \frac{\sqrt{3}}{6} \\ 0 & \frac{1}{3} & \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{1}{3} & 0 & -\frac{1}{6} & -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} & \frac{\sqrt{3}}{6} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & \frac{\sqrt{3}}{6} & \frac{1}{3} & 0 \\ \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} \end{pmatrix} \tag{1.110}$$

Applying this projected operator to the basis vector $(0, 0, 0, 0, 0, 1)^\top$ isolates a representative vibrational mode vector:

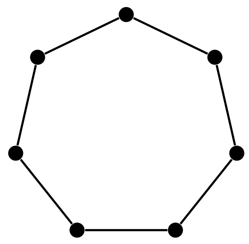
$$\begin{pmatrix} \frac{\sqrt{3}}{6} & -\frac{1}{6} & -\frac{\sqrt{3}}{6} & -\frac{1}{6} & 0 & \frac{1}{3} \end{pmatrix} \tag{1.111}$$

which describes the motion.



1.17 * Symmetries of $2n + 1$-gons

This is a nice simple example of a transformation group for which we can work out the characters (and actually the whole set of irreducible representations) easily. Consider a regular polygon with $2n + 1$ vertices, like the 7-gon shown below.

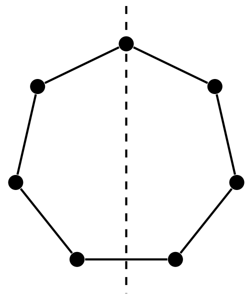


(1.112)

The group of symmetries of the $2n+1$-gon consists of the identity, the $2n$ rotations by $\frac{\pm 2\pi j}{2n+1}$ for $j = 1$ to n,

$$\text{rotations by } \frac{\pm 2\pi j}{2n+1} \text{ for } j = 1 \text{ to } n \tag{1.113}$$

and the $2n+1$ reflections about lines through the center and a vertex, as shown below:



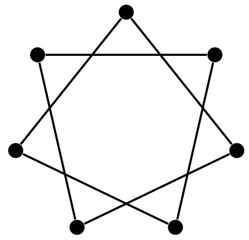
reflections about lines through center and vertex (1.114)

Thus the order of the group of symmetries is $N = 2 \times (2n+1)$. There are $n+2$ conjugacy classes:

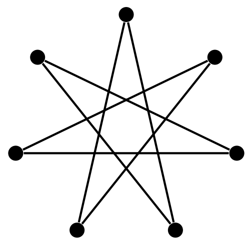
1. the identity, e;
2. the $2n+1$ reflections;
3. to $n+2-$ the rotations by $\frac{\pm 2\pi j}{2n+1}$ for $j = 1$ to n each value of j is a separate conjugacy class.

The way this works is that the reflections are all in the same conjugacy class because by conjugating with rotations, you can get from any one reflection to any other. The rotations are unchanged by conjugation by rotations, but a conjugation by a reflection changes the sign of the rotation, so there is a $\pm$ pair in each conjugacy class.

Furthermore, the n conjugacy classes of rotations are equivalent under cyclic permutations and relabeling of the vertices, as shown below:



(1.115)



(1.116)

The characters look like

$$\begin{array}{c|c|cccccc} & e & r & j=1 & j=2 & \cdots & j=n \\ \hline 1 & 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & -1 & 1 & 1 & \cdots & 1 \\ 2 & 2 & 0 & 2\cos\frac{2\pi m}{2n+1} & 2\cos\frac{4\pi m}{2n+1} & \cdots & 2\cos\frac{2n\pi m}{2n+1} \end{array} \tag{1.117}$$

In the last line, the different values of m give the characters of the n different 2-dimensional representations.

1.18 Permutation group on n objects

Any element of the permutation group on n objects, called S_n, can be written in term of cycles, where a cycle is a cyclic permutation of a subset. We will use a notation that makes use of this, where each cycle is written as a set of numbers in parentheses, indicating the set of things that are cyclicly permuted. For example:

$$(1) \text{ means } x_1 \to x_1 \tag{1.118}$$

$$(1372) \text{ means } x_1 \to x_3 \to x_7 \to x_2 \to x_1 \tag{1.119}$$

Each element of S_n involves each integer from 1 to n in exactly one cycle.

Examples: The identity element looks like $e = (1)(2)\cdots(n)$ - n 1-cycles - there is only one of these. An interchange of two elements looks like $(12)(3)\cdots(n)$ - a 2-cycle and $n-2$ 1-cycles and there are $n(n-1)/2$ of these $(j_1j_2)(j_3)\cdots(j_n)$.

An arbitrary element has k_j j-cycles, where

$$\sum_{j=1}^{n} jk_j = n \tag{1.120}$$

For example, the permutation (123)(456)(78)(9) has two 3-cycles, 1 2-cycle and a 1-cycle, so $k_1 = 1$, $k_2 = 1$, and $k_3 = 2$.

There is an simple (but reducible) n dimensional representation of S_n called the defining representation where the "objects" being permuted are just the basis vectors of an n dimensional vector space,

$$|1\rangle, |2\rangle, \ldots, |n\rangle \tag{1.121}$$

If the permutation takes x_j to x_k, the corresponding representation operator D takes $|j\rangle$ to $|k\rangle$, so that

$$D|j\rangle = |k\rangle \tag{1.122}$$

and thus

$$\langle l|D|j\rangle = \delta_{kl} \tag{1.123}$$

Each matrix in the representation has a single 1 in each row and column.

1.19 Conjugacy classes

The conjugacy classes are just the cycle structure, that is they can be labeled by the integers k_j. For example, all interchanges are in the same conjugacy class. It is enough to check that the inner automorphism gg_1g^{-1} doesn't change the cycle structure of g_1 when g is an interchange, because we can build up any permutation from interchanges. Let us see how this works in some examples. In particular, we will see that conjugating an arbitrary permutation by the interchange (12)(3) just interchanges 1 and 2 without changing the cycle structure.

Examples:

$$(12)(3)(4) \cdot (1)(23)(4) \cdot (12)(3)(4) \tag{1.124}$$

(note that an interchange is its own inverse)

$$\begin{array}{cl} 1234 & \\ \downarrow & (12)(3)(4) \\ 2134 & \\ \downarrow & (\mathbf{1})(\mathbf{23})(\mathbf{4}) \\ 2314 & \\ \downarrow & (12)(3)(4) \\ 3214 & \\ & \\ = & \\ & \\ 1234 & \\ \downarrow & (\mathbf{2})(\mathbf{13})(\mathbf{4}) \\ 3214 & \end{array} \tag{1.125}$$

$(12)(3)(4) \cdot (1)(234) \cdot (12)(3)(4)$

$$\begin{array}{cl} 1234 & \\ \downarrow & (12)(3)(4) \\ 2134 & \\ \downarrow & (\mathbf{1})(\mathbf{234}) \\ 2341 & \\ \downarrow & (12)(3)(4) \\ 3241 & \\ & \\ = & \\ & \\ 1234 & \\ \downarrow & (\mathbf{2})(\mathbf{134}) \\ 3241 & \end{array} \tag{1.126}$$

If 1 and 2 are in different cycles, they just get interchanged by conjugation by (12), as promised.

The same thing happens when 1 and 2 are in the same cycle. For example

$$\begin{array}{cl} 1234 & \\ \downarrow & (12)(3)(4) \\ 2134 & \\ \downarrow & (\mathbf{123})(\mathbf{4}) \\ 1324 & \\ \downarrow & (12)(3)(4) \\ 3124 & \\ & \\ = & \\ & \\ 1234 & \\ \downarrow & (\mathbf{213})(\mathbf{4}) \\ 3124 & \end{array} \tag{1.127}$$

Again, in the same cycle this time, 1 and 2 just get interchanged.

Another way of seeing this is to notice that the conjugation is analogous to a similarity transformation. In fact, in the defining, n dimensional representation of (1.135) the conjugation by the interchange (12) is just a change of basis that switches $|1\rangle \leftrightarrow |2\rangle$. Then it is clear that conjugation does not change the cycle structure, but simply interchanges what the permutation does to 1 and 2. Since we can put interchanges together to form an arbitrary permutation, and since by repeated conjugations by interchanges, we can get from any ordering of the integers in the given cycle structure to any other, the conjugacy classes must consist of all possible permutations with a particular cycle structure.

Now let us count the number of group elements in each conjugacy class. Suppose a conjugacy class consists of permutations of the form of k_1 1-cycles, k_2 2-cycles, etc, satisfying (1.134). The number of different permutations in the conjugacy class is

$$\frac{n!}{\prod_j j^{k_j} k_j!} \tag{1.128}$$

because each permutation of number 1 to n gives a permutation in the class, but cyclic order doesn't matter within a cycle

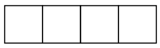
$$(123) \text{ is the same as } (231) \tag{1.129}$$

and order doesn't matter at all between cycles of the same length

$$(12)(34) \text{ is the same as } (34)(12) \tag{1.130}$$

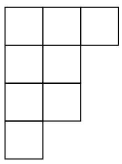
1.20 Young tableaux

It is useful to represent each j-cycle by a column of boxes of length j, top-justified and arranged in order of decreasing j as you go to the right. The total number of boxes is n. Here is an example:



(1.131)

is four 1-cycles in S_4 - that is the identity element - always a conjugacy class all by itself. Here's another:

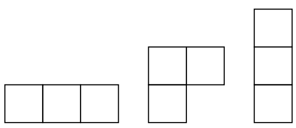


(1.132)

is a 4-cycle, a 3-cycle and a 1-cycle in S_8. These collections of boxes are called Young tableaux. Each different tableaux represents a different conjugacy class, and therefore the tableaux are in one-to-one correspondence with the irreducible representations.

1.21 Example - our old friend S_3

The conjugacy classes are



(1.133)

with numbers of elements

$$\frac{3!}{3!} = 1 \quad \frac{3!}{2} = 3 \quad \frac{3!}{3} = 2 \tag{1.134}$$

1.22 Another example - S_4

$$\ydiagram{4} \quad \ydiagram{3,1} \quad \ydiagram{2,2} \quad \ydiagram{2,1,1} \quad \ydiagram{1,1,1,1} \tag{1.135}$$

with numbers of elements

$$\frac{4!}{4!} = 1 \quad \frac{4!}{4} = 6 \quad \frac{4!}{8} = 3 \quad \frac{4!}{3} = 8 \quad \frac{4!}{4} = 6 \tag{1.136}$$

The characters of S_4 look like this:

$$\begin{array}{ccccc} 1 & 1 & 1 & 1 & 1 \\ 3 & 1 & -1 & 0 & -1 \\ 2 & 0 & 2 & -1 & 0 \\ 3 & -1 & -1 & 0 & 1 \\ 1 & -1 & 1 & 1 & -1 \end{array} \tag{1.137}$$

The first row represents the trivial representation.

1.23 * Young tableaux and representations of S_n

We have seen that a Young tableau with n boxes is associated with an irreducible representation of S_n. We can actually use the tableau to explicitly construct the irreducible representation by identifying an appropriate subspace of the regular representation of S_n. To see what the irreducible representation is, we begin by putting the integers from 1 to n in the boxes of the tableau in all possible ways. There are $n!$ ways to do this. We then identify each assignment of integers 1 to n to the boxes with a state in the regular representation of S_n by defining a standard ordering, say from left to right and then top down (like reading words on a page) to translate from integers in the boxes to a state associated with a particular permutation. So for example

$$\begin{ytableau} 6 & 5 & 3 & 2 \\ 1 & 7 \\ 4 \end{ytableau} \quad \to |6532174\rangle \tag{1.138}$$

where $|6532174\rangle$ is the state corresponding to the permutation

$$1234567 \to 6532174 \tag{1.139}$$

Now each of the $n!$ assignment of boxes to the tableau describes one of the $n!$ states of the regular representation.

Next, for a particular tableau, symmetrize the corresponding state in the numbers in each row, and antisymmetrize in the numbers in each column. For example

$$\begin{ytableau} 1 & 2 \\ 3 \end{ytableau} \quad \to |123\rangle + |213\rangle - |321\rangle - |231\rangle \tag{1.140}$$

Now the set of states constructed in this way spans some subspace of the regular representation. We can construct the states explicitly, and we know how permutations act on these states. That the subspace constructed in this way is a representation of S_n, because a permutation just corresponds to starting with a different assignment of numbers to the tableau, so acting with the permutation on any state in the subspace gives another state in the subspace. In fact, this representation is irreducible, and is the irreducible representation we say is associated with the Young tableau.

Consider the example of S_3. The tableau

$$\begin{array}{|c|c|c|}\hline & & \\ \hline \end{array} \tag{1.141}$$

gives completely symmetrized states, and so is associated with a one dimensional subspace that transforms under the trivial representation. The tableau

$$\begin{array}{|c|}\hline \\ \hline \\ \hline \\ \hline \end{array} \tag{1.142}$$

gives completely antisymmetrized states, and so, again is associated with a one dimensional subspace, this time transforming under the representation in which interchanges are represented by -1. Finally

$$\begin{array}{|c|c|}\hline & \\ \hline \multicolumn{1}{|c|}{} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \tag{1.143}$$

gives the following states:

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline \multicolumn{1}{|c|}{3} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |123\rangle + |213\rangle - |321\rangle - |231\rangle \tag{1.144}$$

$$\begin{array}{|c|c|}\hline 1 & 3 \\ \hline \multicolumn{1}{|c|}{2} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |132\rangle + |312\rangle - |231\rangle - |321\rangle \tag{1.145}$$

$$\begin{array}{|c|c|}\hline 2 & 1 \\ \hline \multicolumn{1}{|c|}{3} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |213\rangle + |123\rangle - |312\rangle - |132\rangle \tag{1.146}$$

$$\begin{array}{|c|c|}\hline 2 & 3 \\ \hline \multicolumn{1}{|c|}{1} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |231\rangle + |321\rangle - |132\rangle - |312\rangle \tag{1.147}$$

$$\begin{array}{|c|c|}\hline 3 & 1 \\ \hline \multicolumn{1}{|c|}{2} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |312\rangle + |132\rangle - |213\rangle - |123\rangle \tag{1.148}$$

$$\begin{array}{|c|c|}\hline 3 & 2 \\ \hline \multicolumn{1}{|c|}{1} & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \to |321\rangle + |231\rangle - |123\rangle - |213\rangle \tag{1.149}$$

Note that interchanging two numbers in the same column of a tableau just changes the sign of the state. This is generally true. Furthermore, you can see explicitly that the sum of three states related by cyclic permutations vanishes. Thus the subspace is two dimensional and transforms under the two dimensional irreducible representation of S_3.

It turns out that the dimension of the representation constructed in this way is

$$\frac{n!}{H} \tag{1.150}$$

where the quantity H is the "hooks" factor for the Young tableau, computed as follows. A hook is a line passing vertically up through the bottom of some column of boxes, making a right hand turn in some box and passing out through the row of boxes. There is one hook for each box. Call the number of boxes the hook passes through h. Then H is the product of the hs for all hooks. We will come back to hooks when we discuss the application of Young tableaux to the representations of $SU(N)$ in chapter XIII.

This procedure for constructing the irreducible representations of S_n is entirely mechanical (if somewhat tedious) and can be used to construct all the representations of S_n from the Young tableaux with n boxes.

We could say much more about finite groups and their representations, but our primary subject is continuous groups, so we will leave finite groups for now. We will see, however, that the representations of the permutation groups play an important role in the representations of continuous groups. So we will come back to S_n now and again.

Problems

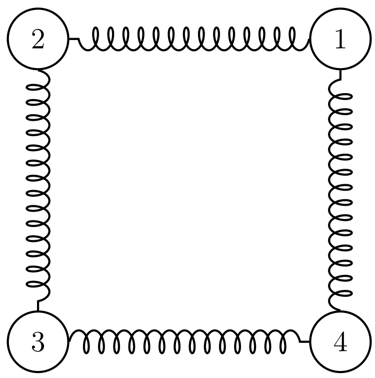
1.A. Find the multiplication table for a group with three elements and prove that it is unique.
1.B. Find all essentially different possible multiplication tables for groups with four elements (which cannot be related by renaming elements).
1.C. Show that the representation (1.135) of the permutation group is reducible.
1.D. Suppose that D_1 and D_2 are equivalent, irreducible representations of a finite group G, such that

$$D_2(g) = SD_1(g)S^{-1}\forall g \in G \tag{1.151}$$

What can you say about an operator A that satisfies

$$AD_1(g) = D_2(g)A\forall g \in G? \tag{1.152}$$

1.E. Find the group of all the discrete rotations that leave a regular tetrahedron invariant by labeling the four vertices and considering the rotations as permutations on the four vertices. This defines a four dimensional representation of a group. Find the conjugacy classes and the characters of the irreducible representations of this group.
***1.F.** Analyze the normal modes of the system of four blocks sliding on a frictionless plane, connected by springs as shown below:



just as we did for the triangle, but using the 8-element symmetry group of the square. Assume that the springs are rigidly attached to the masses (rather than pivoted, for example), so that the square has some rigidity.

Chapter 2

Lie Groups

A Lie group is a group whose elements can be described, at least locally, by continuous parameters. We write this schematically as

$$g(\alpha). \tag{2.1}$$

Smoothness means that nearby group elements correspond to nearby parameter values. The point of this chapter is that near the identity the group can be studied through linear objects: its generators and their commutators.

2.1 Generators

Choose parameters so that the origin in parameter space gives the identity. With N real parameters α_a, $a = 1, \ldots, N$, impose

$$g(\alpha)\big|_{\alpha=0} = e. \tag{2.2}$$

In a representation, the corresponding linear operators satisfy

$$D(\alpha)\big|_{\alpha=0} = 1. \tag{2.3}$$

For an infinitesimal displacement $d\alpha$, Taylor expansion gives

$$D(d\alpha) = 1 + i\, d\alpha_a X_a + \cdots . \tag{2.4}$$

Repeated indices are summed. The generators are therefore

$$X_a \equiv -i\, \frac{\partial}{\partial \alpha_a} D(\alpha)\bigg|_{\alpha=0} . \tag{2.5}$$

The factor of i is chosen so that X_a is Hermitian in a unitary representation.

For an infinitesimal element in a fixed direction,

$$D(d\alpha) = 1 + i\, d\alpha_a X_a. \tag{2.6}$$

Raising such an element to a large power motivates the exponential parametrization

$$D(\alpha) = \lim_{k\to\infty} (1 + i\alpha_a X_a/k)^k = e^{i\alpha_a X_a}. \tag{2.7}$$

Thus near the identity the group is encoded in the real vector space spanned by the generators.

2.2 Lie algebras

Along one fixed direction the multiplication law is simple. Define

$$U(\lambda) = e^{i\lambda\alpha_a X_a}. \tag{2.8}$$

Then

$$U(\lambda_1)U(\lambda_2) = U(\lambda_1 + \lambda_2). \tag{2.9}$$

For two different directions the naive addition rule fails in general:

$$e^{i\alpha_a X_a} e^{i\beta_b X_b} \neq e^{i(\alpha_a+\beta_a)X_a}. \tag{2.10}$$

Nevertheless, near the identity the product is again a group element, so it can be written as

$$e^{i\alpha_a X_a} e^{i\beta_b X_b} = e^{i\delta_a X_a}. \tag{2.11}$$

To find the first nontrivial correction, write

$$i\delta_a X_a = \ln\left(1 + e^{i\alpha_a X_a} e^{i\beta_b X_b} - 1\right). \tag{2.12}$$

Let

$$\begin{aligned} K &= e^{i\alpha_a X_a} e^{i\beta_b X_b} - 1 \\ &= \left(1 + i\alpha_a X_a - \frac{1}{2}(\alpha_a X_a)^2 + \cdots\right)\left(1 + i\beta_b X_b - \frac{1}{2}(\beta_b X_b)^2 + \cdots\right) - 1 \\ &= i\alpha_a X_a + i\beta_a X_a - \alpha_a X_a \beta_b X_b - \frac{1}{2}(\alpha_a X_a)^2 - \frac{1}{2}(\beta_a X_a)^2 + \cdots. \end{aligned} \tag{2.13}$$

Using $\ln(1+K) = K - \frac{1}{2}K^2 + \cdots$ gives

$$\begin{aligned} i\delta_a X_a &= K - \frac{1}{2}K^2 + \cdots \\ &= i\alpha_a X_a + i\beta_a X_a - \alpha_a X_a \beta_b X_b - \frac{1}{2}(\alpha_a X_a)^2 - \frac{1}{2}(\beta_a X_a)^2 + \frac{1}{2}(\alpha_a X_a + \beta_a X_a)^2 + \cdots. \end{aligned} \tag{2.14}$$

The second-order pieces cancel except for the noncommuting part, so

$$\begin{aligned} i\delta_a X_a &= K - \frac{1}{2}K^2 + \cdots \\ &= i\alpha_a X_a + i\beta_a X_a - \frac{1}{2}[\alpha_a X_a, \beta_b X_b] + \cdots. \end{aligned} \tag{2.15}$$

Therefore

$$[\alpha_a X_a, \beta_b X_b] = -2i(\delta_c - \alpha_c - \beta_c)X_c + \cdots \equiv i\gamma_c X_c. \tag{2.16}$$

Since this must hold for arbitrary α and β,

$$\gamma_c = \alpha_a \beta_b f_{abc} \tag{2.17}$$

for constants f_{abc}. Hence the generators close under commutation:

$$[X_a, X_b] = i f_{abc} X_c. \tag{2.18}$$

The antisymmetry of the commutator gives

$$f_{abc} = -f_{bac}. \tag{2.19}$$

The product parameter begins as

$$\delta_a = \alpha_a + \beta_a - \frac{1}{2}\gamma_a + \cdots . \tag{2.20}$$

The constants f_{abc} are the structure constants. They determine the Lie algebra and encode the local multiplication law of the group near the identity.

For Hermitian generators in a unitary representation, taking the adjoint of the commutator gives

$$\begin{aligned} [X_a, X_b]^\dagger &= -if^*_{abc}X_c \\ &= [X_b, X_a] = if_{bac}X_c = -if_{abc}X_c. \end{aligned} \tag{2.21}$$

Thus, for the compact/unitary setting considered here, the structure constants may be taken real.

2.3 The Jacobi identity

Matrix commutators satisfy the Jacobi identity

$$[X_a, [X_b, X_c]] + \text{cyclic permutations} = 0. \tag{2.22}$$

Equivalently,

$$[X_a, [X_b, X_c]] = [[X_a, X_b], X_c] + [X_b, [X_a, X_c]]. \tag{2.23}$$

This is the commutator analogue of the product rule

$$[X_a, X_bX_c] = [X_a, X_b]X_c + X_b[X_a, X_c]. \tag{2.24}$$

2.4 The adjoint representation

Use the algebra to compute a double commutator:

$$\begin{aligned} [X_a, [X_b, X_c]] &= if_{bcd}[X_a, X_d] \\ &= -f_{bcd}f_{ade}X_e. \end{aligned} \tag{2.25}$$

Since the X_a are independent, the Jacobi identity implies

$$f_{bcd}f_{ade} + f_{abd}f_{cde} + f_{cad}f_{bde} = 0. \tag{2.26}$$

Define matrices T_a by

$$[T_a]_{bc} = -if_{abc}. \tag{2.27}$$

Then the preceding identity is equivalent to

$$[T_a, T_b] = if_{abc}T_c. \tag{2.28}$$

Thus the structure constants themselves furnish a representation: the adjoint representation. Its dimension is the number of independent generators.

A useful bilinear form on the space of generators is the trace in the adjoint representation,

$$\mathrm{Tr}(T_aT_b). \tag{2.29}$$

Now change basis in the generator space:

$$X_a \to X'_a = L_{ab}X_b. \tag{2.30}$$

Then

$$\begin{aligned}[X'_a, X'_b] &= iL_{ad}L_{be}f_{dec}X_c \\ &= iL_{ad}L_{be}f_{deg}L^{-1}_{gh}L_{hc}X_c \\ &= iL_{ad}L_{be}f_{deg}L^{-1}_{gc}X'_c. \end{aligned} \tag{2.31}$$

So

$$f_{abc} \to f'_{abc} = L_{ad}L_{be}f_{deg}L^{-1}_{gc}. \tag{2.32}$$

With the transformed structure constants,

$$[T_a]_{bc} \to [T'_a]_{bc} = L_{ad}L_{be}[T_d]_{eg}L^{-1}_{gc}. \tag{2.33}$$

Equivalently,

$$[T_a] \to [T'_a] = L_{ad}L[T_d]L^{-1}. \tag{2.34}$$

Therefore, under the same basis change,

$$\mathrm{Tr}(T_aT_b) \to \mathrm{Tr}(T'_aT'_b) = L_{ac}L_{bd}\,\mathrm{Tr}(T_cT_d). \tag{2.35}$$

One may diagonalize the trace form. In such a basis,

$$\mathrm{Tr}(T_aT_b) = k^a\delta_{ab}, \qquad \text{no sum on } a. \tag{2.36}$$

For compact Lie algebras, take the nonzero k^a positive and choose the normalization

$$\mathrm{Tr}(T_aT_b) = \lambda\delta_{ab} \tag{2.37}$$

with $\lambda > 0$. In this basis,

$$f_{abc} = -i\lambda^{-1}\,\mathrm{Tr}([T_a, T_b]T_c). \tag{2.38}$$

By cyclicity of the trace,

$$\begin{aligned}\mathrm{Tr}([T_a, T_b]T_c) &= \mathrm{Tr}(T_aT_bT_c - T_bT_aT_c) \\ &= \mathrm{Tr}(T_bT_cT_a - T_cT_bT_a) = \mathrm{Tr}([T_b, T_c]T_a). \end{aligned} \tag{2.39}$$

Thus

$$f_{abc} = f_{bca}. \tag{2.40}$$

Together with $f_{abc} = -f_{bac}$ this gives complete antisymmetry:

$$f_{abc} = f_{bca} = f_{cab} = -f_{bac} = -f_{acb} = -f_{cba}. \tag{2.41}$$

With this normalization the adjoint generators are imaginary and antisymmetric, hence Hermitian.

2.5 Simple algebras and groups

An invariant subalgebra is a subspace of generators closed under commutation with every generator of the full algebra. If X lies in the invariant subalgebra and Y lies in the full algebra, then $[Y, X]$ must remain in the invariant subalgebra. Exponentiation turns such a subalgebra into an invariant subgroup. Write

$$h = e^{iX}, \qquad g = e^{iY}. \tag{2.42}$$

Then

$$g^{-1}hg = e^{iX'} \tag{2.43}$$

where

$$X' = e^{-iY} X e^{iY} = X - i[Y, X] - \frac{1}{2}[Y, [Y, X]] + \cdots . \tag{2.44}$$

A quick derivation is to introduce

$$X'(\epsilon) = e^{-i\epsilon Y} X e^{i\epsilon Y} \tag{2.45}$$

then expand in ϵ and set $\epsilon = 1$.

An algebra with no nontrivial invariant subalgebra is simple. A simple algebra generates a simple group. For a simple compact Lie algebra satisfying the trace normalization above, the adjoint representation is irreducible. If the adjoint had an invariant subspace spanned by T_r, $r = 1, \ldots, K$, with the remaining generators T_x, $x = K+1, \ldots, N$, invariance would require

$$[T_a]_{xr} = -if_{axr} = 0. \tag{2.46}$$

Complete antisymmetry then forces all mixed structure constants to vanish, splitting the algebra into two invariant pieces, contrary to simplicity.

A one-generator Abelian invariant subalgebra, commuting with all generators, is called a $U(1)$ factor. Such factors do not appear in the structure constants. Algebras with no Abelian invariant subalgebras are semisimple. In the rest of these notes, unless stated otherwise, representations are taken unitary and the algebras semisimple.

2.6 States and operators

Generators in a representation can be viewed either as linear operators or as matrices. With a basis $\{|i\rangle\}$,

$$X_a|i\rangle = |j\rangle\langle j|X_a|i\rangle = |j\rangle[X_a]_{ji}. \tag{2.47}$$

The finite group element maps kets by

$$|i\rangle \to |i'\rangle = e^{i\alpha_a X_a}|i\rangle. \tag{2.48}$$

The corresponding bras transform as

$$\langle i| \to \langle i'| = \langle i|e^{-i\alpha_a X_a}. \tag{2.49}$$

If O is an operator, then $O|i\rangle$ must transform as a ket:

$$\begin{aligned} O|i\rangle &\to e^{i\alpha_a X_a} O|i\rangle \\ &= e^{i\alpha_a X_a} O e^{-i\alpha_a X_a} e^{i\alpha_a X_a}|i\rangle = O'|i'\rangle. \end{aligned} \tag{2.50}$$

Therefore

$$O \to O' = e^{i\alpha_a X_a} O e^{-i\alpha_a X_a}. \tag{2.51}$$

Infinitesimally,

$$-i\delta|i\rangle = -i\left((1 + i\alpha_a X_a)|i\rangle - |i\rangle\right) = \alpha_a X_a|i\rangle, \tag{2.52}$$

$$-i\delta\langle i| = -\langle i|\alpha_a X_a, \tag{2.53}$$

$$-i\delta O = [\alpha_a X_a, O]. \tag{2.54}$$

Thus the generator acts on a ket as

$$X_a|i\rangle, \tag{2.55}$$

on a bra as

$$-\langle i|X_a, \tag{2.56}$$

and on an operator by the commutator

$$[X_a, O]. \tag{2.57}$$

The invariance of the matrix element $\langle i|O|i\rangle$ is expressed as

$$\langle i|O(X_a|i\rangle) + \langle i|[X_a, O]|i\rangle - (\langle i|X_a)O|i\rangle = 0. \tag{2.58}$$

2.7 Fun with exponentials

Consider

$$e^{i\alpha_a X_a}. \tag{2.59}$$

As a power series,

$$e^{i\alpha_a X_a} = \sum_{n=0}^{\infty} \frac{(i\alpha_a X_a)^n}{n!}. \tag{2.60}$$

Differentiation is not generally as simple as in the commuting case:

$$\frac{\partial}{\partial \alpha_b} e^{i\alpha_a X_a} \neq iX_b e^{i\alpha_a X_a}. \tag{2.61}$$

But along a fixed direction,

$$\frac{d}{ds} e^{is\alpha_a X_a} = i\alpha_b X_b e^{is\alpha_a X_a} = ie^{is\alpha_a X_a} \alpha_b X_b. \tag{2.62}$$

At the origin,

$$\left. \frac{\partial}{\partial \alpha_b} e^{i\alpha_a X_a} \right|_{\alpha=0} = iX_b. \tag{2.63}$$

The general derivative formula is

$$\frac{\partial}{\partial \alpha_b} e^{i\alpha_a X_a} = \int_0^1 ds\ e^{is\alpha_a X_a} (iX_b) e^{i(1-s)\alpha_c X_c}. \tag{2.64}$$

A useful limiting form of the exponential is

$$e^{i\alpha_a X_a} = \lim_{k\to\infty} \left(1 + i\alpha_a X_a / k\right)^k. \tag{2.65}$$

Expanding both sides of the derivative formula also uses

$$\int_0^1 ds\ s^m (1-s)^n = \frac{m!n!}{(m+n+1)!}. \tag{2.66}$$

For quick reference, the mathematical properties of matrix representations, operator derivatives, and nested commutator expansions referenced above are summarized in Appendices A–E below.

Appendix A. Summary of the properties of matrices

A square matrix of dimension n is a table of $n \times n$ elements a_{ij}:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

The matrix elements a_{ij} are (possibly complex) numbers. Suppose we have another $n \times n$ matrix B with elements b_{ij}.

Definition: The sum of two matrices A and B is a matrix C with matrix elements given by

$$c_{ij} = a_{ij} + b_{ij}$$

Their difference C' is given by the matrix elements $c'_{ij} = a_{ij} - b_{ij}$. We write:

$$C = A + B, \quad \text{or} \quad C' = A - B. \tag{2.67}$$

From Eq. 2.67 it follows that the summation is commutative: $A + B = B + A$.

Definition: The product of the matrices A and B is a matrix C with matrix elements c_{ij} given by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}. \tag{2.68}$$

We write:

$$C = AB. \tag{2.69}$$

Explanation: Eq. 2.68 defines the matrix elements c_{ij} if a_{ij} and b_{ij} are given. Since a_{ij} and b_{ij} are ordinary numbers, it does not matter in which order they are written. This does not hold, however, for products of matrices, where in general $C = AB \neq BA$. Note that $C' = BA$ is a matrix whose elements are given by

$$c'_{ij} = \sum_k b_{ik} a_{kj} \tag{2.70}$$

which differs from Eq. 2.68. We say that matrix multiplication is non commutative. It is associative: $A(BC) = (AB)C$.

Definition: The commutator of two matrices A and B is given by

$$[A, B] \equiv AB - BA. \tag{2.71}$$

We say that A and B commute if the commutator is equal to zero.

Definition: The product of a (possibly complex) number x with a matrix A is a matrix B with elements:

$$b_{ij} = x a_{ij} \tag{2.72}$$

Notation:

$$B = xA \tag{2.73}$$

From Eq. 2.72, it follows that $B^2 = x^2 A^2$, since

$$\sum_k b_{ik} b_{kj} = x^2 \sum_k a_{ik} a_{kj}. \tag{2.74}$$

Definition: The unit matrix 1 is defined as

$$1 = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

and gives $1A = A1 = A$ for every matrix A.

For every $n \times n$ matrix we can define its determinant, being a (possibly complex) number. It is defined recursively by taking out one of its rows, say the i^{th} row:

$$\mathrm{Det}(A) = \sum_{k=1}^{n} a_{ik} (-1)^{k+i} \mathrm{Det}(A(i,k)), \tag{2.75}$$

where the $(n-1) \times (n-1)$ matrix $A(i,k)$ is obtained from A by removing the i^{th} row and the k^{th} column. The determinant of a 0×0 matrix is 1.

Many properties of determinants are easy to derive: if $C = AB$ then

$$\mathrm{Det}(C) = \mathrm{Det}(AB) = \mathrm{Det}(A)\mathrm{Det}(B). \tag{2.76}$$

If $C = xA$ with x a number and A an $n \times n$ matrix,

$$\mathrm{Det}(C) = \mathrm{Det}(xA) = x^n \mathrm{Det}(A). \tag{2.77}$$

Furthermore: the inverse of a matrix A is a matrix B such that $BA = AB = 1$.

Notation: $B = A^{-1}$. The matrix elements of B are (note: $A(j,i)$, not $A(i,j)$)

$$b_{ij} = \frac{(-1)^{i+j} \mathrm{Det}(A(j,i))}{\mathrm{Det}(A)}. \tag{2.78}$$

From this, it follows that a matrix A has an inverse if $\mathrm{Det}(A) \neq 0$.

Another definition of the determinant makes use of the $\epsilon_{i_1 \cdots i_n}$ symbols, defined by $\epsilon_{i_1 \cdots i_n} = 0$ if two or more of the indices are equal, $\epsilon_{i_1 \cdots i_n} = \pm 1$ if in $i_1 \cdots i_n$ no two indices are equal, in which case we have $\epsilon_{i_1 \cdots i_n} = 1$ if $i_1 \cdots i_n$ are an even permutation of $123 \cdots n$, and $\epsilon_{i_1 \cdots i_n} = -1$ if $i_1 \cdots i_n$ are an odd permutation of $123 \cdots n$. In particular, $\epsilon_{123 \cdots n} = +1$.

The determinant of a matrix A is then given by

$$A_{i_1 j_1} A_{i_2 j_2} \cdots A_{i_n j_n} \epsilon_{j_1 \cdots j_n} = \mathrm{Det}(A) \epsilon_{i_1 \cdots i_n} \tag{2.79}$$

Finally, for a given matrix A, the transpose $\tilde{A}$ of a matrix A is defined by the matrix elements

$$\tilde{a}_{ij} = a_{ji} \tag{2.80}$$

and if A^{-1} is the inverse of a matrix A, one has

$$\widetilde{AB} = \tilde{B}\tilde{A}, \quad (AB)^{-1} = B^{-1}A^{-1}. \tag{2.81}$$

The first equation follows from the definition 2.68 of the matrix product, and the second formula follows from $1 = (AB)(AB)^{-1} = ABB^{-1}A^{-1} = 1$.

Appendix B. Differentiation of matrices

Matrix elements can be functions of one or more variables. These functions may be differentiable. In this case, one can define the derivative of a matrix.

Definition: The derivative of a matrix A is a matrix whose elements are the derivatives of the matrix elements of A. Thus, the matrix A' with

$$A'(x) = \frac{dA(x)}{dx} \tag{2.82}$$

has the matrix elements

$$a'_{ij}(x) = \frac{da_{ij}(x)}{dx} \tag{2.83}$$

In general, the derivative of a matrix will not commute with the matrix itself. Apart from this fact, one may use all formulae that are familiar from differentiation theory, and they follow from the definition 2.82. For example,

$$\frac{d(A+B)}{dx} = \frac{dA}{dx} + \frac{dB}{dx} \tag{2.84}$$

and

$$\frac{d(AB)}{dx} = \frac{dA}{dx}B + A\frac{dB}{dx} \tag{2.85}$$

The latter result follows from the chain rule

$$\frac{d}{dx}\sum_k a_{ik}b_{kj} = \sum_k a'_{ik}b_{kj} + \sum_k a_{ik}b'_{kj} \tag{2.86}$$

Similarly,

$$\frac{dA^n}{dx} = A^{n-1}A' + A^{n-2}A'A + \cdots + A'A^{n-1}. \tag{2.87}$$

Note that this is not always equal to $nA^{n-1}A'$, because A and A' do not necessarily commute.

The derivative of the inverse of a matrix A can be found the following way. Since $A^{-1}A = 1$, we have

$$0 = \frac{d}{dx}(A^{-1}A) = \frac{dA^{-1}}{dx}A + A^{-1}\frac{dA}{dx} \tag{2.88}$$

and consequently,

$$\frac{dA^{-1}}{dx} = -A^{-1}\frac{dA}{dx}A^{-1} \tag{2.89}$$

Appendix C. Functions of matrices

Functions based on additions and multiplications can readily be defined for matrices. But many more functions are used. In particular, the function e^x is of importance. There are two possible definitions.

Let a matrix A be given. The matrix e^A is then defined by

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!}A^k \tag{2.90}$$

where $A^0 = 1$. A second definition is

$$e^A = \lim_{m\to\infty}\left(1 + \frac{1}{m}A\right)^m \tag{2.91}$$

The proof that the two definitions 2.90 and 2.91 coincide is just like it goes for ordinary exponentials, by applying the binomial expansion of Eq. 2.91.

Eq. 2.91 allows us to derive an important relation:

$$\mathrm{Det}(e^A) = \lim_{m\to\infty}\left\{\mathrm{Det}\left(1+\frac{1}{m}A\right)\right\}^m \tag{2.92}$$

We compute the determinant between brackets by ignoring contributions of order $1/m^2$ or smaller. For instance, first consider a 2×2 matrix. There:

$$1+\frac{1}{m}A = \begin{pmatrix} 1+\frac{a_{11}}{m} & \frac{a_{12}}{m} \\ \frac{a_{21}}{m} & 1+\frac{a_{22}}{m} \end{pmatrix}$$

If, in the calculation of the determinant, we ignore the contributions of order $1/m^2$ and higher, the off-diagonal elements can be ignored (in a determinant, there are no terms containing only one off-diagonal element as a factor). In this approximation, the determinant is simply the product of the diagonal elements:

$$\mathrm{Det}\left(1+\frac{1}{m}A\right) \approx \left(1+\frac{a_{11}}{m}\right)\left(1+\frac{a_{22}}{m}\right) \approx 1+\frac{1}{m}(a_{11}+a_{22}) + O\left(\frac{1}{m^2}\right).$$

In general, one finds:

$$\mathrm{Det}\left(1+\frac{1}{m}A\right) = 1+\frac{1}{m}\mathrm{Tr}(A) + O\left(\frac{1}{m^2}\right) \tag{2.93}$$

where $\mathrm{Tr}(A)$ is the trace of the matrix A, that is, the sum of its diagonal elements:

$$\mathrm{Tr}(A) = \sum_k a_{kk} \tag{2.94}$$

Substituting this in Eq. 2.92, gives

$$\mathrm{Det}(e^A) = \lim_{m\to\infty}\left\{1+\frac{1}{m}\mathrm{Tr}(A)+O\left(\frac{1}{m^2}\right)\right\}^m = e^{\mathrm{Tr}(A)}, \tag{2.95}$$

and thus we arrive at:

$$\mathrm{Det}(e^A) = e^{\mathrm{Tr}(A)} \tag{2.96}$$

Appendix D. The Campbell-Baker-Hausdorff formula

Let there be given two matrices A and B. The CBH formula tells us that there exists a matrix C such that

$$e^A e^B = e^C \tag{2.97}$$

where $C = A+B+[A,B]+$ repeated commutators of A and B.

Repeated commutators are expressions of the following type:

$$[[A,B],B] = [A,B]B - B[A,B] = ABB - BAB - BAB + BBA,$$

$$[[A,[A,B]],B], \quad [[[A,B],B],B], \quad \text{etc.}$$

The numeric factors that come with the coefficients of these repeated commutators are not so easy to find. For our purposes, it suffices to establish that C indeed is of the form 2.97. We shall prove this and also derive the coefficients of the next order, containing either commutators with two B's and one A, or two A's and one B.

To begin with, we introduce a variable, and consider

$$e^{xA}e^{xB} = e^{C(x)} \tag{2.98}$$

So, the x dependence at the l.h.s. of 2.98 is simple and explicit. However, $C(x)$ will be a complicated function of x. In fact, the series expansion of $C(x)$ with respect to x will at the same time be the expansion of C in powers of A and B. Therefore, it will be useful to try to identify the function $C(x)$. This we do by differentiating Eq. 2.98 left and right with respect to x. This gives us an equation from which $C(x)$ can be solved, yielding a power series. First, we need to prove some more identities.

Consider

$$H(y) = e^{-yF}Ge^{yF} \tag{2.99}$$

where y is a variable and F and G are matrices not depending on y. H can be written in the form of a power expansion:

$$H = H_0 + yH_1 + \frac{y^2}{2!}H_2 + \frac{y^3}{3!}H_3 + \cdots \tag{2.100}$$

Here, the factors $2!, 3!, \ldots$ have been included for later convenience. We shall compute the matrices H_n. Differentiation of Eq. 2.99 to y yields:

$$\frac{dH(y)}{dy} = \left(\frac{de^{-yF}}{dy}\right)Ge^{yF} + e^{-yF}G\frac{de^{yF}}{dy} \tag{2.101}$$

Using

$$\frac{d}{dy}e^{yF} = Fe^{yF} = e^{yF}F \tag{2.102}$$

we find

$$\frac{dH(y)}{dy} = -Fe^{-yF}Ge^{yF} + e^{-yF}Ge^{yF}F = [H, F] \tag{2.103}$$

and substituting this in the power series 2.100,

$$H_1 + yH_2 + \frac{y^2}{2!}H_3 + \cdots = [H_0, F] + y[H_1, F] + \frac{y^2}{2!}[H_2, F] + \cdots \tag{2.104}$$

Since $H_0 = G$, identifying the coefficients for equal powers of y gives us:

$$H_n = [H_{n-1}, F] = [[[\cdots[G, F], F], F]\cdots F]. \tag{2.105}$$

Substituting this in Eq. 2.100 gives for $y = 1$:

$$e^{-F}Ge^{F} = G + [G, F] + \frac{1}{2!}[[G, F], F] + \frac{1}{3!}[[[G, F], F], F] + \ldots \tag{2.106}$$

Let us introduce a shorthand notation:

$$H_n = \{G, F^n\}, \tag{2.107}$$

and, even shorter, we rewrite Eq. 2.106 as

$$e^{-F}Ge^{F} = \{G, e^F\}, \tag{2.108}$$

with which we mean that e^F must first be replaced by its power series expansion.

Let now F be some function of a variable x. Consider

$$e^{-yF(x)}\frac{d}{dx}e^{yF(x)} \tag{2.109}$$

In a way similar to the above, one derives

$$e^{-F(x)}\frac{d}{dx}e^{F(x)} = F' + \frac{1}{2!}[F',F] + \frac{1}{3!}[[F',F],F] + \cdots = \left\{F', \frac{e^F - 1}{F}\right\} \tag{2.110}$$

In fact, Eq. 2.110 is what one gets if in Eq. 2.106, G is replaced by $\frac{d}{dx}$, because in that case,

$$[G,F] = \frac{d}{dx}F - F\frac{d}{dx} \tag{2.111}$$

Here, it was assumed that d/dx acts to the right on whatever else one has to the right of the commutator $[G,F]$. According to the chain rule,

$$\frac{d}{dx}F = \frac{dF}{dx} + F\frac{d}{dx} = F' + F\frac{d}{dx} \tag{2.112}$$

so that

$$[G,F] = F'. \tag{2.113}$$

From Eq. 2.110, it follows that

$$\frac{d}{dx}e^{F(x)} = e^{F(x)}\left\{F', \frac{e^F - 1}{F}\right\} \tag{2.114}$$

We now return to Eq. 2.98. Differentiation with respect to x gives, in view of Eq. 2.114:

$$e^{xA}Ae^{xB} + e^{xA}Be^{xB} = e^{C(x)}\left\{C', \frac{e^C - 1}{C}\right\} \tag{2.115}$$

or

$$e^{xA}e^{xB}\left(e^{-xB}Ae^{xB} + e^{-xB}Be^{xB}\right) = e^{C(x)}\left\{C', \frac{e^C - 1}{C}\right\}. \tag{2.116}$$

The factors outside the brackets are the same left and right, in view of Eq. 2.98. Because $Be^{xB} = e^{xB}B$, the second term within the brackets is nothing but B. To the first term, we can apply Eq. 2.108. Thus, we obtain:

$$\left\{C', \frac{e^C - 1}{C}\right\} = \{A, e^{xB}\} + B \tag{2.117}$$

From this equation, C can be solved as a power expansion. The strength of our abstract notation shows if we temporarily call the r.h.s. of this equation H:

$$\begin{aligned} C' &= \left\{H, \frac{C}{e^C - 1}\right\} = \left\{H, 1 - \frac{C}{2} + \frac{C^2}{12} + \cdots\right\} \\ &= H - \frac{1}{2}[H,C] + \frac{1}{12}[[H,C],C] + \cdots, \end{aligned} \tag{2.118}$$

where we used the power expansion of $\frac{C}{e^C-1} = \frac{1}{1+\frac{1}{2}C+\frac{1}{6}C^2+\cdots} = 1 - \frac{1}{2}C + \frac{1}{12}C^2 \ldots$.

Let us write

$$C = C_0 + xC_1 + \frac{x^2}{2!}C_2 + \frac{x^3}{3!}C_3 + \cdots, \tag{2.119}$$

$$C' = C_1 + xC_2 + \frac{x^2}{2!}C_3 + \cdots, \tag{2.120}$$

$$H = H_0 + xH_1 + \frac{x^2}{2!}H_2 = A + B + x[A,B] + \frac{x^2}{2!}[[A,B],B] + \cdots. \tag{2.121}$$

We can then compare equal powers of x in Eqs. 2.118 and 2.121. First of all:

$$C_0 = C(0) = 0. \tag{2.122}$$

The terms independent of x give

$$C_1 = H_0 = A + B \tag{2.123}$$

and the terms linear in x:

$$C_2 = H_1 - \frac{1}{2}[H_0, C_1] = [A, B]. \tag{2.124}$$

Continuing this way, all other coefficients C_n can be found:

$$\frac{1}{2}C_3 = \frac{1}{2}H_2 - \frac{1}{4}[H_0, C_2] - \frac{1}{2}[H_1, C_1] \tag{2.125}$$

$$\begin{aligned} C_3 &= [[A, B], B] - \frac{1}{2}[A + B, [A, B]] - [[A, B], A + B] \\ &= \frac{1}{2}[A, [A, B]] + \frac{1}{2}[[A, B], B], \end{aligned} \tag{2.126}$$

(since $[B, [A, B]] = -[[A, B], B]$).

All further coefficients C_i obtained this way, can always be written in terms of multiple commutators containing preceding C_j, $j < i$. Since these C_j only consist of multiple commutators, so do C_i, by induction.

Substituting Eqs. 2.122, 2.123, 2.124, and 2.126 in the series expansion 2.121, yields, for $x = 1$:

$$e^A e^B = e^C \tag{2.127}$$

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[[A, B], B] + \cdots \tag{2.128}$$

It is a nice challenge to further streamline the method described here for deriving the series of commutators for C.

Appendix E. Complex inner product, unitary and hermitian matrices

Let an n-dimensional linear space be given with a basis $e_1, \ldots, e_n$, which is orthogonal when regarding some well defined inner product. Every vector in this space can be written as a linear combination of the $e_1, \ldots, e_n$. In general, the coefficients are complex numbers.

Consider now two vectors α and β,

$$\alpha = \alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n. \tag{2.129}$$

$$\beta = \beta_1 e_1 + \beta_2 e_2 + \cdots + \beta_n e_n. \tag{2.130}$$

The coefficients α_i and β_i are just numbers. The inner product (α, β) is a complex number given by

$$(\alpha, \beta) = \alpha_1^* \beta_1 + \alpha_2^* \beta_2 + \cdots + \alpha_n^* \beta_n. \tag{2.131}$$

Notice that we have

$$(\beta, \alpha) = (\alpha, \beta)^*. \tag{2.132}$$

One has the following axioms for the inner product:

$$(\alpha, \alpha) \geq 0, \quad \text{the equal sign only if } \alpha = 0, \tag{2.133}$$

$$(\alpha, \beta + \gamma) = (\alpha, \beta) + (\alpha, \gamma), \tag{2.134}$$

$$(\alpha, x\beta) = x(\alpha, \beta), \quad x = \text{number}, \tag{2.135}$$

$$(x\alpha, \beta) = x^*(\alpha, \beta). \tag{2.136}$$

Here, $\beta + \gamma$ is a vector with components $\beta_i + \gamma_i$, and $x\beta$ is a vector with components $x\beta_i$.

One can turn this argument around: if the above properties hold, then Eq. 2.131 implies the orthogonality of the basis vectors:

$$(e_i, e_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{2.137}$$

Let now be given a matrix A. This matrix defines a mapping of the space into itself:

$$\alpha' = A\alpha, \tag{2.138}$$

which is symbolic for

$$\alpha'_i = \sum_k a_{ik}\alpha_k \tag{2.139}$$

So, for every vector α, the matrix A defines an image α'. Which are the conditions to be imposed on A such that the inner product is invariant, in the sense that $(\alpha, \beta) = (\alpha', \beta')$ for any pair α, β?

By expansion in components, one easily derives:

$$\begin{aligned}(\alpha', \beta') &= \sum_k \alpha'^*_k \beta'_k = \sum_{klm} (a_{kl}\alpha_l)^* a_{km}\beta_m \\ &= \sum_{k,l,m} \alpha^*_l a^*_{kl} a_{km}\beta_m\end{aligned} \tag{2.140}$$

If now $\tilde{A}$ is the transpose of A then

$$(\tilde{a})_{lk} = a_{kl} \tag{2.141}$$

and if $A^\dagger$ is the conjugate transpose of A:

$$(a^\dagger)_{lk} = a^*_{kl} \tag{2.142}$$

and if $B = A^\dagger A$, then evidently

$$\begin{aligned}b_{lm} &= \sum_k a^\dagger_{lk} a_{km} \\ &= \sum_k a^*_{kl} a_{km}\end{aligned} \tag{2.143}$$

So:

$$(\alpha', \beta') = \sum_{l,m} \alpha^*_l b_{lm} \beta_m \tag{2.144}$$

with $B = A^\dagger A$. Let now β'' be the vector obtained from β by applying B to it:

$$\beta'' = B\beta \tag{2.145}$$

that is, according to Eq. 2.139

$$\beta''_l = \sum_m b_{lm}\beta_m \tag{2.146}$$

and Eq. 2.144 is simply the inner product (α, β''), so $(\alpha', \beta') = (\alpha, \beta'')$. This must be equal to (α, β) for all choices of α and β. This can only be the case if $\beta'' = \beta$ for all β, and therefore, B must be the unit matrix.

We conclude that the inner product is invariant under the mapping with a matrix A provided one has

$$A^\dagger A = 1, \tag{2.147}$$

or equivalently, $A^\dagger = A^{-1}$. A matrix obeying this condition is called unitary. In real spaces, one has $A^\dagger = \tilde{A}$, and so $A^{-1} = \tilde{A}$. Such a matrix is orthogonal.

A brief symbolic display of the above derivation reads

$$(\alpha', \beta') = (A\alpha, A\beta) = (\alpha, A^\dagger A\beta). \tag{2.148}$$

Note that for inner products, we have

$$(A\alpha, \beta) = (\alpha, A^\dagger \beta), \tag{2.149}$$

which holds for any matrix A.

An eigenvector of a matrix A is a vector with the property

$$A\alpha = \lambda\alpha, \tag{2.150}$$

where λ is a real or complex number; i.e.

$$\sum_k a_{ik}\alpha_k = \lambda\alpha_i \tag{2.151}$$

λ is the eigenvalue of A associated to the eigenvector α. Which are the conditions on A that guarantee all λ to be real?

Consider the inner product $(\alpha, A\alpha) = \lambda(\alpha, \alpha)$. Now, (α, α) is real and larger than zero if α is not zero. Real λ apparently implies that $(\alpha, A\alpha)$ is real. That means

$$(\alpha, A\alpha) = (\alpha, A\alpha)^* = (A\alpha, \alpha) = (\alpha, A^\dagger\alpha). \tag{2.152}$$

This clearly holds if $A^\dagger = A$. A matrix obeying $A^\dagger = A$ is called hermitian. Therefore:

$$A = \text{hermitian, i.e., } A^\dagger = A \implies \text{only real eigenvalues.} \tag{2.153}$$

The converse is not necessarily true: there are many non-hermitian matrices with only real eigenvalues.

Problems

2.A. Find all components of $e^{i\alpha A}$ where

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

2.B. If $[A, B] = B$, calculate

$$e^{i\alpha A} B e^{-i\alpha A}.$$

2.C. Carry out the expansion of δ_c in (2.11) and (2.12) to third order in α and β.

Chapter 3

SU(2)

The algebra of $SU(2)$ is the angular-momentum algebra, written here with indices $1, 2, 3$ rather than x, y, z:

$$[J_j, J_k] = i\epsilon_{jk\ell}J_\ell. \tag{3.1}$$

It is the simplest compact Lie algebra. The analysis below avoids using J_aJ_a as the main tool, because the same method generalizes directly to larger compact Lie algebras.

3.1 J_3 eigenstates

Let the representation space be finite dimensional, of dimension N, and suppose it carries an irreducible representation of the algebra. We diagonalize one generator, chosen to be J_3. Let j be the largest eigenvalue of J_3. For the corresponding highest states,

$$J_3 \left|j, \alpha\right\rangle = j \left|j, \alpha\right\rangle, \tag{3.2}$$

where α labels possible degeneracy of the highest eigenvalue. The highest states may be chosen orthonormally:

$$\langle j, \alpha | j, \beta\rangle = \delta_{\alpha\beta}. \tag{3.3}$$

3.2 Raising and lowering operators

Define

$$J^\pm = \frac{J_1 \pm iJ_2}{\sqrt{2}}. \tag{3.4}$$

The commutation relations imply

$$[J_3, J^\pm] = \pm J^\pm, \tag{3.5}$$

$$[J^+, J^-] = J_3. \tag{3.6}$$

Thus, if

$$J_3 \left|m\right\rangle = m \left|m\right\rangle, \tag{3.7}$$

then

$$J_3J^\pm \left|m\right\rangle = J^\pm J_3 \left|m\right\rangle \pm J^\pm \left|m\right\rangle = (m \pm 1)J^\pm \left|m\right\rangle. \tag{3.8}$$

So J^+ raises the J_3 eigenvalue and J^- lowers it.

> The purpose of raising and lowering operator is to change the J_3 eigenvalue rather than changing the invariant subspace they belong to!

Since j is the highest eigenvalue,

$$J^+ |j, \alpha\rangle = 0 \qquad \forall \alpha. \tag{3.9}$$

Define the first lowered state by

$$J^- |j, \alpha\rangle \equiv N_j(\alpha) |j-1, \alpha\rangle . \tag{3.10}$$

The normalization and orthogonality follow from

$$\begin{aligned} N_j(\beta)^* N_j(\alpha) \langle j-1, \beta | j-1, \alpha\rangle &= \langle j, \beta | J^+ J^- |j, \alpha\rangle \\ &= \langle j, \beta | [J^+, J^-] |j, \alpha\rangle \\ &= \langle j, \beta | J_3 |j, \alpha\rangle \\ &= j \langle j, \beta | j, \alpha\rangle = j\delta_{\alpha\beta}. \end{aligned} \tag{3.11}$$

Hence we may choose

$$N_j(\alpha) = \sqrt{j} \equiv N_j. \tag{3.12}$$

Then

$$\begin{aligned} J^+ |j-1, \alpha\rangle &= \frac{1}{N_j} J^+ J^- |j, \alpha\rangle \\ &= \frac{1}{N_j} [J^+, J^-] |j, \alpha\rangle \\ &= \frac{j}{N_j} |j, \alpha\rangle = N_j |j, \alpha\rangle . \end{aligned} \tag{3.13}$$

Repeating the same construction gives orthonormal states satisfying

$$\begin{aligned} J^- |j-1, \alpha\rangle &= N_{j-1} |j-2, \alpha\rangle , \\ J^+ |j-2, \alpha\rangle &= N_{j-1} |j-1, \alpha\rangle . \end{aligned} \tag{3.14}$$

Continuing downward, one obtains a tower

$$\begin{aligned} J^- |j-k, \alpha\rangle &= N_{j-k} |j-k-1, \alpha\rangle , \\ J^+ |j-k-1, \alpha\rangle &= N_{j-k} |j-k, \alpha\rangle . \end{aligned} \tag{3.15}$$

The constants N_{j-k} may be chosen real. From the algebra,

$$\begin{aligned} N_{j-k}^2 &= \langle j-k, \alpha | J^+ J^- |j-k, \alpha\rangle \\ &= \langle j-k, \alpha | [J^+, J^-] |j-k, \alpha\rangle + \langle j-k, \alpha | J^- J^+ |j-k, \alpha\rangle \\ &= N_{j-k+1}^2 + j - k. \end{aligned} \tag{3.16}$$

Solving the recursion, beginning with $N_j^2 = j$, gives

$$\begin{aligned} N_j^2 &= j, otag \qquad &(3.17) \\ N_{j-1}^2 - N_j^2 &= j-1, otag &(3.18) \\ &\vdots \\ N_{j-k}^2 - N_{j-k+1}^2 &= j-k, otag &(3.19) \end{aligned}$$

$$\begin{aligned} N_{j-k}^2 &= (k+1)j - \frac{k(k+1)}{2} otag \qquad &(3.20) \\ &= \frac{1}{2}(k+1)(2j-k). &(3.21) \end{aligned}$$

Equivalently, putting $k = j - m$,

$$N_m = \frac{1}{\sqrt{2}} \sqrt{(j+m)(j-m+1)}. \tag{3.22}$$

Because the representation is finite dimensional, the lowering process stops. For some non-negative integer ℓ,

$$J^- \left| j - \ell, \alpha \right\rangle = 0. \tag{3.23}$$

This requires

$$N_{j-\ell} = \frac{1}{\sqrt{2}} \sqrt{(2j-\ell)(\ell+1)} = 0. \tag{3.24}$$

Since $\ell + 1 \neq 0$,

$$\ell = 2j. \tag{3.25}$$

Thus

$$j = \frac{\ell}{2} \qquad \text{for some integer } \ell. \tag{3.26}$$

The label α can now be removed in an irreducible representation: each value of α would generate an invariant tower, so irreducibility allows only one such tower.

$$m = j - l = j - 2j = j$$

is the lowest weight state in that representation.

3.3 The standard notation

The standard basis for the spin-j irreducible representation is

$$\left| \overset{\text{label of the irreducible representation}}{j} \,,\, \underset{J_3 \text{ eigenvalue of the state}}{m} \right\rangle . \tag{3.27}$$

Each representation has a unique heighest weight state from which we can obtain all the other elements belonging to the same representation by operating with lowering operator.The generator matrix elements are

$$\begin{aligned}
\left\langle j, m' \right| J_3 \left| j, m \right\rangle &= m \delta_{m'm}, \\
\left\langle j, m' \right| J^+ \left| j, m \right\rangle &= \sqrt{\frac{(j+m+1)(j-m)}{2}}\, \delta_{m',m+1}, \\
\left\langle j, m' \right| J^- \left| j, m \right\rangle &= \sqrt{\frac{(j+m)(j-m+1)}{2}}\, \delta_{m',m-1}.
\end{aligned} \tag{3.28}$$

These define the spin-j representation matrices:

$$[J_a^j]_{k\ell} = \left\langle j, j+1-k \right| J_a \left| j, j+1-\ell \right\rangle . \tag{3.29}$$

Equivalently, using m values directly as row and column labels,

$$[J_a^j]_{m'm} = \left\langle j, m' \right| J_a \left| j, m \right\rangle . \tag{3.30}$$

For $j = 1/2$ one obtains

$$\begin{aligned}
J_1^{1/2} &= \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{2}\sigma_1, \\
J_2^{1/2} &= \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{1}{2}\sigma_2, \\
J_3^{1/2} &= \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{2}\sigma_3.
\end{aligned} \tag{3.31}$$

Here

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{3.32}$$

with

$$\sigma_a \sigma_b = \delta_{ab} + i\epsilon_{abc}\sigma_c. \tag{3.33}$$

Exponentiating the spin-1/2 generators gives finite $SU(2)$ matrices of the form

$$e^{i\vec{\alpha}\cdot\vec{\sigma}/2}. \tag{3.34}$$

For the spin-1 representation,

$$\begin{aligned}
J_1^1 &= \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\
J_2^1 &= \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \\
J_3^1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.
\end{aligned} \tag{3.35}$$

For the spin-3/2 representation,

$$\begin{aligned}
J_1^{3/2} &= \begin{pmatrix} 0 & \sqrt{\frac{3}{2}} & 0 & 0 \\ \sqrt{\frac{3}{2}} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{\frac{3}{2}} \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 \end{pmatrix}, \\
J_2^{3/2} &= \begin{pmatrix} 0 & -i\sqrt{\frac{3}{2}} & 0 & 0 \\ i\sqrt{\frac{3}{2}} & 0 & -2i & 0 \\ 0 & 2i & 0 & -i\sqrt{\frac{3}{2}} \\ 0 & 0 & i\sqrt{\frac{3}{2}} & 0 \end{pmatrix}, \\
J_3^{3/2} &= \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix}.
\end{aligned} \tag{3.36}$$

The above construction is the highest-weight construction. A reducible finite dimensional representation can be block-diagonalized by repeatedly selecting the highest remaining J_3 eigenvalue and generating its full tower:

1. Diagonalize J_3.
2. Find the states with highest J_3 value j.
3. For each highest state, generate the corresponding spin-j tower by applying J^-.
4. Remove the invariant subspace already put into canonical form.
5. Repeat on the orthogonal complement.

$$\text{highest-weight reduction algorithm.} \tag{3.37}$$

The resulting basis has the form

$$|j, m, \alpha\rangle \,, \tag{3.38}$$

where α labels whatever additional observables are needed. These states satisfy

$$\big\langle j', m', \alpha' \big| j, m, \alpha \big\rangle = \delta_{m'm}\delta_{j'j}\delta_{\alpha'\alpha}. \tag{3.39}$$

Schur's lemma is reflected in the identity

$$\begin{aligned} \big\langle j', m', \alpha' \big| \, J_a \, |j, m, \alpha\rangle &= [J_a^{j'}]_{m'm''} \big\langle j', m'', \alpha' \big| j, m, \alpha \big\rangle \\ &= \big\langle j', m', \alpha' \big| j, m'', \alpha \big\rangle \, [J_a^{j}]_{m''m}. \end{aligned} \tag{3.40}$$

This says that the overlap matrix commutes with the irreducible action; hence it vanishes between inequivalent irreducibles and is proportional to the identity inside an equivalent one.

3.4 Tensor products

Tensor products appear when two independent degrees of freedom transform under the group. Write product states as

$$|i, x\rangle \equiv |i\rangle \, |x\rangle \,. \tag{3.41}$$

If $|i\rangle$ transforms in representation D_1 and $|x\rangle$ in representation D_2, then

$$\begin{aligned} D(g) \, |i, x\rangle &= |j, y\rangle \, [D_{1\otimes 2}(g)]_{jyix} \\ &= |j\rangle \, |y\rangle \, [D_1(g)]_{ji} [D_2(g)]_{yx} \\ &= \big(|j\rangle \, [D_1(g)]_{ji}\big)\big(|y\rangle \, [D_2(g)]_{yx}\big). \end{aligned} \tag{3.42}$$

Near the identity,

$$\begin{aligned} (1 + i\alpha_a J_a) \, |i, x\rangle &= |j, y\rangle \, \langle j, y| \, (1 + i\alpha_a J_a) \, |i, x\rangle \\ &= |j, y\rangle \, \big(\delta_{ji}\delta_{yx} + i\alpha_a [J_a^{1\otimes 2}(g)]_{jyix}\big) \\ &= |j, y\rangle \, \big(\delta_{ji} + i\alpha_a [J_a^1]_{ji}\big) \, \big(\delta_{yx} + i\alpha_a [J_a^2]_{yx}\big) \,. \end{aligned} \tag{3.43}$$

Equating terms linear in α_a gives

$$[J_a^{1\otimes 2}(g)]_{jyix} = [J_a^1]_{ji}\delta_{yx} + \delta_{ji}[J_a^2]_{yx}. \tag{3.44}$$

In compressed notation,

$$J_a^{1\otimes 2} = J_a^1 + J_a^2. \tag{3.45}$$

Equivalently, as an action on product states,

$$J_a\big(|j\rangle \, |x\rangle\big) = (J_a \, |j\rangle) \, |x\rangle + |j\rangle \, (J_a \, |x\rangle). \tag{3.46}$$

3.5 J_3 values add

Since J_3 is diagonal in the chosen basis, J_3 eigenvalues add in a tensor product:

$$J_3\Big(|j_1, m_1\rangle\,|j_2, m_2\rangle\Big) = (m_1 + m_2)\Big(|j_1, m_1\rangle\,|j_2, m_2\rangle\Big). \tag{3.47}$$

For example, consider the tensor product of spin 1/2 and spin 1. The unique highest-weight state is

$$|3/2, 3/2\rangle = |1/2, 1/2\rangle\,|1, 1\rangle\,. \tag{3.48}$$

Applying J^- gives

$$\begin{aligned} J^-\,|3/2, 3/2\rangle &= J^-\Big(|1/2, 1/2\rangle\,|1, 1\rangle\Big) \\ &= \sqrt{\frac{3}{2}}\,|3/2, 1/2\rangle \\ &= \sqrt{\frac{1}{2}}\,|1/2, -1/2\rangle\,|1, 1\rangle + |1/2, 1/2\rangle\,|1, 0\rangle\,. \end{aligned} \tag{3.49}$$

Hence

$$|3/2, 1/2\rangle = \sqrt{\frac{1}{3}}\,|1/2, -1/2\rangle\,|1, 1\rangle + \sqrt{\frac{2}{3}}\,|1/2, 1/2\rangle\,|1, 0\rangle\,. \tag{3.50}$$

Continuing,

$$|3/2, -1/2\rangle = \sqrt{\frac{2}{3}}\,|1/2, -1/2\rangle\,|1, 0\rangle + \sqrt{\frac{1}{3}}\,|1/2, 1/2\rangle\,|1, -1\rangle\,, \tag{3.51}$$

$$|3/2, -3/2\rangle = |1/2, -1/2\rangle\,|1, -1\rangle\,. \tag{3.52}$$

The remaining states, orthogonal to the spin-3/2 states, are

$$\sqrt{\frac{2}{3}}\,|1/2, -1/2\rangle\,|1, 1\rangle - \sqrt{\frac{1}{3}}\,|1/2, 1/2\rangle\,|1, 0\rangle\,, \tag{3.53}$$

which is orthogonal to (3.50) and by using (3.47) has J_3 eigenvalue:

$$\begin{aligned} &J_3\left(\sqrt{\frac{2}{3}}\,|1/2, -1/2\rangle\,|1, 1\rangle - \sqrt{\frac{1}{3}}\,|1/2, 1/2\rangle\,|1, 0\rangle\right) \\ &\qquad = \frac{1}{2}\left(\sqrt{\frac{2}{3}}\,|1/2, -1/2\rangle\,|1, 1\rangle - \sqrt{\frac{1}{3}}\,|1/2, 1/2\rangle\,|1, 0\rangle\right) \end{aligned} \tag{3.54}$$

and the following is orthogonal to (3.51)

$$\sqrt{\frac{1}{3}}\,|1/2, -1/2\rangle\,|1, 0\rangle - \sqrt{\frac{2}{3}}\,|1/2, 1/2\rangle\,|1, -1\rangle\,. \tag{3.55}$$

Applying the highest-weight construction to this reduced space gives

$$\begin{aligned} |1/2, 1/2\rangle &= \sqrt{\frac{2}{3}}\,|1/2, -1/2\rangle\,|1, 1\rangle - \sqrt{\frac{1}{3}}\,|1/2, 1/2\rangle\,|1, 0\rangle\,, \\ |1/2, -1/2\rangle &= \sqrt{\frac{1}{3}}\,|1/2, -1/2\rangle\,|1, 0\rangle - \sqrt{\frac{2}{3}}\,|1/2, 1/2\rangle\,|1, -1\rangle\,. \end{aligned} \tag{3.56}$$

Note that orthogonal states to (3.53) and (3.55) live in the same space, therefore the process terminates. Thus

$$\{1/2\} \otimes \{1\} = \{3/2\} \oplus \{1/2\}.$$

Problems

3.A. Use highest-weight decomposition to show

$$\{j\} \otimes \{s\} = \bigoplus_{\ell=|s-j|}^{s+j} \{\ell\},$$

where $\{k\}$ denotes the spin-k irreducible representation. Track the counting of states at each stage.

3.B. Calculate

$$e^{i\vec{r}\cdot\vec{\sigma}},$$

where $\vec{\sigma}$ are the Pauli matrices. Hint: write $\vec{r} = |\vec{r}|\hat{r}$.

3.C. Show explicitly that the spin-1 representation obtained from the highest-weight construction with $j = 1$ is equivalent to the adjoint representation with $f_{abc} = \epsilon_{abc}$, by finding the similarity transformation.

3.D. Let $[\sigma_a]_{ij}$ and $[\eta_a]_{xy}$ be Pauli matrices acting on two different two-dimensional spaces. In the tensor-product basis

$$\begin{aligned} |1\rangle &= |i=1\rangle\,|x=1\rangle\,, & \qquad |2\rangle &= |i=1\rangle\,|x=2\rangle\,, \\ |3\rangle &= |i=2\rangle\,|x=1\rangle\,, & \qquad |4\rangle &= |i=2\rangle\,|x=2\rangle\,, \end{aligned}$$

write the matrix elements of $\sigma_2 \otimes \eta_1$.

3.E. Practice the compact tensor-product notation in which identity matrices and indices are suppressed:

$$\begin{aligned} [\sigma_a]_{ij}[\eta_b]_{xy} &\quad \text{as} \quad \sigma_a\eta_b, \\ [\sigma_a]_{ij}\delta_{xy} &\quad \text{as} \quad \sigma_a, \\ \delta_{ij}[\eta_b]_{xy} &\quad \text{as} \quad \eta_b, \\ \delta_{ij}\delta_{xy} &\quad \text{as} \quad 1. \end{aligned}$$

For instance,

$$(\sigma_1)(\sigma_2\eta_1) = i\sigma_3\eta_1, \qquad (\sigma_1\eta_2)(\sigma_1\eta_3) = i\eta_1.$$

Compute

(a) $[\sigma_a, \sigma_b\eta_c]$,

(b) $\mathrm{Tr}\big(\sigma_a\{\eta_b, \sigma_c\eta_d\}\big)$,

(c) $[\sigma_1\eta_1, \sigma_2\eta_2]$,

where $\{A, B\} = AB + BA$.

Chapter 4

Tensor Operators

A tensor operator is a collection of operators whose commutator with the algebra generators transforms in the same way as an irreducible representation of the algebra. For the $SU(2)$ algebra, a spin-s tensor operator consists of components O_ℓ, with $\ell = -s, -s+1, \ldots, s$, satisfying

$$[J_a, O_\ell] = O_{\ell'} [J^s_a]_{\ell'\ell}. \tag{4.1}$$

Here J^s_a denotes the standard spin-s representation matrices constructed in Chapter 3. The repeated index ℓ' is summed.

4.1 Orbital angular momentum

For a particle in a spherically symmetric potential, and with no intrinsic spin, the generators are the orbital angular momentum operators

$$J_a = \epsilon_{abc} r_b p_c. \tag{4.2}$$

The position vector transforms under the adjoint representation:

$$\begin{aligned} [J_a, r_b] &= \epsilon_{acd}[r_c p_d, r_b] = -i\epsilon_{acd} r_c \delta_{bd} \\ &= -i f_{acb} r_c = r_c [J^{\text{adj}}_a]_{cb}. \end{aligned} \tag{4.3}$$

The adjoint representation of $SU(2)$ is equivalent to the ordinary spin-1 representation.

This is derived in section 5.1.2 of Quantum Mechanics by Ernest S. Abers or section 3.4 of Ballentine's Quantum Mechanics or section 5.1 of Quantum Mechanics by Schwabl. The way Ernest Abers has done this is by considering the transformation of basis and expectation value to deduce how the vector should transform. Given

$$|\alpha\rangle_R = \hat{R}\,|\alpha\rangle,$$

then,

$$_R\langle\alpha|\, V_i\, |\beta\rangle_R = R_{ij}\,\langle\alpha|\, V_j\, |\beta\rangle,$$

where R_{ij} acts on the expectation value.

$$\langle\alpha|\, R^\dagger V_i R\, |\beta\rangle = \langle\alpha|\, R_{ij} V_j\, |\beta\rangle.$$

Hence,

$$R^\dagger V_i R = R_{ij} V_j.$$

That is,

$$e^{i\theta\hat{n}\cdot J}\, V_i\, e^{-i\theta\hat{n}\cdot J} = \left(e^{-i\theta\hat{n}\cdot J}\right)_{ij} V_j.$$

Expanding,

$$e^{i\theta\hat{n}\cdot[J,\ \ ]}V_i = \left(e^{-i\theta\hat{n}\cdot J}\right)_{ij} V_j.$$

At first order,

$$i\theta\, \hat{n}\cdot[J,\, V_i] = (-i\theta\, \hat{n}\cdot J)_{ij} V_j.$$

Thus,

$$[J_a,\, V_i] = -(J_a)_{ij}V_j = -(-i\epsilon_{aij})V_j = i\epsilon_{aij}V_j.$$

Otherwise, we could study how the vectors transform:

$$(R_{\hat{n}}\vec{V})_i = \left\{\vec{V} - (\hat{n}\times\vec{V})\theta\right\}_i = V_i - \theta\, n_j V_k \epsilon_{ijk},$$

for an infinitesimal rotation.
Now,

$$U(R)V_iU^{-1}(R) = (R_{\hat{n}}\vec{V})_i.$$

Hence,

$$e^{-i\theta\hat{n}\cdot J}V_i e^{i\theta\hat{n}\cdot J} = (R_{\hat{n}}\vec{V})_i.$$

Expanding,

$$e^{-i\theta\hat{n}\cdot[J,\ \ ]}V_i = (R_{\hat{n}}\vec{V})_i.$$

At first order,

$$-i\theta\, \hat{n}\cdot[J, V_i] = -\theta\, n_j V_k \epsilon_{ijk}.$$

Therefore,

$$[J_a, V_i] = -i\epsilon_{iak}V_k = i\epsilon_{aik}V_k.$$

4.2 Using tensor operators

The components r_a do not yet appear in the standard spin-1 basis. In general, suppose a set of operators O_x, $x = 1, \ldots, 2s+1$, transforms under some representation D equivalent to the spin-s representation:

$$[J_a, O_x] = O_y[D_a]_{yx}. \tag{4.4}$$

Since D is equivalent to the standard spin-s representation, there is an invertible matrix S such that

$$S^{-1}D_aS = J_a^s, \tag{4.5}$$

or, in components,

$$[S^{-1}]_{\ell z}[D_a]_{zy}[S]_{y\ell'} = [J_a^s]_{\ell\ell'}. \tag{4.6}$$

Define a new operator basis by

$$O_\ell = O_y[S^{-1}]_{y\ell}, \qquad \ell = -s, -s+1, \ldots, s. \tag{4.7}$$

Then

$$\begin{aligned}
[J_a, O_\ell] &= [J_a, O_y][S^{-1}]_{y\ell} \\
&= O_z[D_a]_{zy}[S^{-1}]_{y\ell} \\
&= O_z[S^{-1}]_{z\ell'}[S]_{\ell' z'}[D_a]_{z'y}[S^{-1}]_{y\ell}
\end{aligned}$$

$$= O_{\ell'}[J^s_a]_{\ell'\ell}. \tag{4.8}$$

This is the desired tensor-operator transformation law.

Commutation relation of cartesian components (see (4.3)) is very different from the ones obeyed by spherical tensor.)

For J_3 the standard spin-s matrices are diagonal:

$$[J^s_3]_{\ell'\ell} = \ell\,\delta_{\ell\ell'}, \qquad \ell, \ell' = -s, -s+1, \ldots, s. \tag{4.9}$$

Thus the J_3 commutator reads

$$[J_3, O_\ell] = O_{\ell'}[J^s_3]_{\ell'\ell} = \ell O_\ell. \tag{4.10}$$

For the position operator, r_3 has J_3 value zero (as $\epsilon_{3a3} = 0$), so one may choose

$$r_0 = r_3. \tag{4.11}$$

The remaining spherical components follow from the raising and lowering operators:

$$[J_\pm, r_0] = r_{\pm 1} = -\frac{[r_3, J_1]}{\sqrt{2}} \mp \frac{i[r_3, J_2]}{\sqrt{2}} = \mp \frac{r_1 \pm i r_2}{\sqrt{2}}. \tag{4.12}$$

4.3 The Wigner–Eckart theorem

Now consider how the product of a tensor operator and a state transforms. For a spin-s tensor operator acting on a spin-j state,

$$\begin{aligned} J_a O_\ell \left|j, m, \alpha\right\rangle &= [J_a, O_\ell] \left|j, m, \alpha\right\rangle + O_\ell J_a \left|j, m, \alpha\right\rangle \\ &= O_{\ell'} \left|j, m, \alpha\right\rangle [J^s_a]_{\ell'\ell} + O_\ell \left|j, m', \alpha\right\rangle [J^j_a]_{m'm}. \end{aligned} \tag{4.13}$$

Thus $O_\ell \left|j, m, \alpha\right\rangle$ transforms like the tensor product $s \otimes j$. For J_3 this becomes especially transparent:

$$J_3 O_\ell \left|j, m, \alpha\right\rangle = (\ell + m) O_\ell \left|j, m, \alpha\right\rangle. \tag{4.14}$$

The J_3 eigenvalue of the product is the sum of the J_3 eigenvalues of the operator component and the state.

The product representation decomposes exactly like an ordinary tensor product of states. The highest-weight construction gives

$$\sum_\ell O_\ell \left|j, M-\ell, \alpha\right\rangle (s, j, \ell, M-\ell \mid J, M) = k_J \left|J, M\right\rangle. \tag{4.15}$$

The same Clebsch–Gordan coefficients appear in the corresponding tensor product of kets,

$$\sum_\ell \left|s, \ell\right\rangle \left|j, M-\ell\right\rangle (s, j, \ell, M-\ell \mid J, M) = \left|J, M\right\rangle. \tag{4.16}$$

Inverting the decomposition gives

$$O_\ell \left|j, m, \alpha\right\rangle = \sum_{J=|j-s|}^{j+s} (J, \ell+m \mid s, j, \ell, m) k_J \left|J, \ell+m\right\rangle. \tag{4.17}$$

The state $k_J\,|J,\ell+m\rangle$ may be expanded in a Hilbert-space basis as

$$k_J\,|J,\ell+m\rangle = \sum_\beta k_{\alpha\beta}\,|J,\ell+m,\beta\rangle\,. \tag{4.18}$$

The constants $k_{\alpha\beta}$ are reduced matrix elements:

$$k_{\alpha\beta} = (J,\beta\|O^s\|j,\alpha). \tag{4.19}$$

Putting these facts together gives the Wigner–Eckart theorem:

$$\left\langle J,m',\beta\right| O_\ell\,|j,m,\alpha\rangle = \delta_{m',\ell+m}(J,\ell+m \mid s,j,\ell,m)(J,\beta\|O^s\|j,\alpha). \tag{4.20}$$

Once one nonzero reduced matrix element is known, the remaining matrix elements within the same irreducible multiplets are determined by the Clebsch–Gordan coefficients.

4.4 Example

Suppose

$$\langle 1/2,1/2,\alpha|\,r_3\,|1/2,1/2,\beta\rangle = A. \tag{4.21}$$

We want to find

$$\langle 1/2,1/2,\alpha|\,r_1\,|1/2,-1/2,\beta\rangle = ? \tag{4.22}$$

Since $r_0 = r_3$,

$$\langle 1/2,1/2,\alpha|\,r_0\,|1/2,1/2,\beta\rangle = A. \tag{4.23}$$

From the spherical-basis relation,

$$r_1 = \frac{1}{\sqrt{2}}(-r_{+1}+r_{-1}). \tag{4.24}$$

Therefore

$$\begin{aligned}\langle 1/2,1/2,\alpha|\,r_1\,|1/2,-1/2,\beta\rangle &= \frac{1}{\sqrt{2}}\,\langle 1/2,1/2,\alpha|\,(-r_{+1}+r_{-1})\,|1/2,-1/2,\beta\rangle \\ &= -\frac{1}{\sqrt{2}}\,\langle 1/2,1/2,\alpha|\,r_{+1}\,|1/2,-1/2,\beta\rangle\,. \end{aligned} \tag{4.25}$$

The highest-weight state in the spin-3/2 part of $1\otimes 1/2$ is

$$|3/2,3/2\rangle = r_{+1}\,|1/2,1/2,\beta\rangle\,. \tag{4.26}$$

Applying the lowering operator gives

$$|3/2,1/2\rangle = \sqrt{\frac{2}{3}}\,r_0\,|1/2,1/2,\beta\rangle + \sqrt{\frac{1}{3}}\,r_{+1}\,|1/2,-1/2,\beta\rangle\,. \tag{4.27}$$

A spin-3/2 state has zero matrix element with a spin-1/2 bra, so

$$\begin{aligned}0 &= \langle 1/2,1/2,\alpha|\,3/2,1/2\rangle \\ &= \sqrt{\frac{2}{3}}\,\langle 1/2,1/2,\alpha|\,r_0\,|1/2,1/2,\beta\rangle + \sqrt{\frac{1}{3}}\,\langle 1/2,1/2,\alpha|\,r_{+1}\,|1/2,-1/2,\beta\rangle\,. \end{aligned} \tag{4.28}$$

Thus

$$\langle 1/2,1/2,\alpha|\,r_{+1}\,|1/2,-1/2,\beta\rangle = -\sqrt{2}\,\langle 1/2,1/2,\alpha|\,r_0\,|1/2,1/2,\beta\rangle = -\sqrt{2}\,A. \tag{4.29}$$

Hence

$$\langle 1/2, 1/2, \alpha| \, r_1 \, |1/2, -1/2, \beta\rangle = A. \tag{4.30}$$

The orthogonal spin-1/2 combination in the product space is

$$|1/2, 1/2\rangle = \sqrt{\frac{1}{3}} \, r_0 \, |1/2, 1/2, \alpha\rangle - \sqrt{\frac{2}{3}} \, r_{+1} \, |1/2, -1/2, \alpha\rangle \, . \tag{4.31}$$

Here one must be careful: unlike ordinary tensor-product states, the objects

$$r_0 \, |1/2, 1/2, \alpha\rangle \, , \qquad r_{+1} \, |1/2, -1/2, \alpha\rangle \tag{4.32}$$

do not come with known norms from symmetry alone. Nevertheless, the algebra implies

$$J_+ \, |1/2, 1/2\rangle = 0, \tag{4.33}$$

so the state is identified as the highest-weight state of a spin-1/2 representation.

Another way to see the result is to study the matrix elements

$$\langle 1/2, m, \alpha| \, r_a \, \left|1/2, m', \beta\right\rangle . \tag{4.34}$$

The Wigner–Eckart theorem says all these matrix elements are proportional to one reduced matrix element. Since J_a has the same commutation relations as r_a, the matrix elements of r_a are proportional to those of J_a when the latter are nonzero:

$$\langle 1/2, m, \alpha| \, J_a \, \left|1/2, m', \beta\right\rangle = \delta_{\alpha\beta} \frac{1}{2} [\sigma_a]_{mm'}. \tag{4.35}$$

Thus

$$\langle 1/2, m, \alpha| \, r_a \, \left|1/2, m', \beta\right\rangle \propto [\sigma_a]_{mm'}. \tag{4.36}$$

4.5 * Making tensor operators

Often one encounters operators transforming as a reducible representation:

$$[J_a, O_x] = O_y [D_a]_{yx}, \tag{4.37}$$

where D is reducible. The goal is to form linear combinations that are irreducible tensor operators. First find combinations with definite J_3 value:

$$[J_3, O_j] = j O_j. \tag{4.38}$$

The highest-weight construction is then applied directly to operators. At lower levels one must find combinations annihilated by J_+:

$$[J_3, O_{j-1,\alpha}] = (j-1) O_{j-1,\alpha}, \qquad [J_+, O_{j-1,\alpha}] = 0. \tag{4.39}$$

The point is that for operators there is no natural scalar product, so one cannot simply use orthogonality to remove already-identified irreducible components.

Consider seven operators $a_{\pm 1}$, $b_{\pm 1}$, a_0, b_0, and c_0 with commutation relations

$$\begin{gathered} [J_3, a_{+1}] = a_{+1}, \qquad [J_3, b_{+1}] = b_{+1}, \\ [J_3, a_0] = [J_3, b_0] = [J_3, c_0] = 0. \end{gathered} \tag{4.40}$$

The raising commutators are

$$\begin{gathered} [J_+, a_0] = a_{+1}, \qquad [J_+, a_{-1}] = c_0, \\ [J_+, b_0] = b_{+1}, \qquad [J_+, c_0] = a_{+1} - b_{+1}, \\ [J_+, b_{-1}] = \frac{1}{2}(a_0 + b_0 - 3c_0). \end{gathered} \tag{4.41}$$

The lowering commutators are

$$\begin{gathered}[J_-, a_{+1}] = \frac{1}{2}(a_0 + b_0 + c_0), \qquad [J_-, b_{+1}] = \frac{1}{2}(a_0 + b_0 - c_0), \\ [J_-, a_0] = 2a_{-1} + b_{-1}, \qquad [J_-, b_0] = a_{-1} + b_{-1}, \\ [J_-, c_0] = a_{-1}, \qquad [J_-, a_{-1}] = [J_-, b_{-1}] = 0. \end{gathered} \tag{4.42}$$

The highest-weight components are chosen as

$$A_{+1} = a_{+1}, \qquad B_{+1} = b_{+1}. \tag{4.43}$$

Lowering gives the zero components

$$A_0 = \frac{1}{2}(a_0 + b_0 + c_0), \qquad B_0 = \frac{1}{2}(a_0 + b_0 - c_0), \tag{4.44}$$

and the -1 components

$$A_{-1} = a_{-1} + \frac{1}{2}b_{-1}, \qquad B_{-1} = \frac{1}{2}b_{-1}. \tag{4.45}$$

There remains one spin-0 operator, the combination with vanishing commutator with both J_+ and J_-:

$$C_0 = a_0 - b_0 - c_0. \tag{4.46}$$

4.6 Products of operators

Products of tensor operators are important because they transform under tensor product representations. If $O^{s_1}_{m_1}$ and $O^{s_2}_{m_2}$ are tensor operators of spins s_1 and s_2, then

$$\begin{aligned}[J_a, O^{s_1}_{m_1} O^{s_2}_{m_2}] &= [J_a, O^{s_1}_{m_1}] O^{s_2}_{m_2} + O^{s_1}_{m_1} [J_a, O^{s_2}_{m_2}] \\ &= O^{s_1}_{m_1'} O^{s_2}_{m_2} [J^{s_1}_a]_{m_1' m_1} + O^{s_1}_{m_1} O^{s_2}_{m_2'} [J^{s_2}_a]_{m_2' m_2}. \end{aligned} \tag{4.47}$$

Thus products may be decomposed into irreducible tensor operators by the same highest-weight procedure used for tensor products of states. For J_3,

$$[J_3, O^{s_1}_{m_1} O^{s_2}_{m_2}] = (m_1 + m_2) O^{s_1}_{m_1} O^{s_2}_{m_2}. \tag{4.48}$$

Problems

4.A. Consider an operator O_x, $x = 1, 2$, transforming according to the spin-1/2 representation:

$$[J_a, O_x] = O_y \left[\frac{\sigma_a}{2}\right]_{yx}. \tag{4.49}$$

Given

$$\langle 3/2, -1/2, \alpha | \, O_1 \, |1, -1, \beta\rangle = A, \tag{4.50}$$

find

$$\langle 3/2, -3/2, \alpha | \, O_2 \, |1, -1, \beta\rangle. \tag{4.51}$$

4.B. The operator $(r_{+1})^2$ satisfies

$$[J_3, (r_{+1})^2] = 2(r_{+1})^2, \qquad [J_+, (r_{+1})^2] = 0. \tag{4.52}$$

It is therefore the O_{+2} component of a spin-2 tensor operator. Construct the other components O_m. Relate this construction to the spherical harmonics $Y_{\ell m}(\theta, \phi)$ by taking

$$r_1 = \sin\theta\cos\phi, \qquad r_2 = \sin\theta\sin\phi, \qquad r_3 = \cos\theta. \tag{4.53}$$

Generalize the construction to arbitrary ℓ.

4.C. Find

$$e^{ia_a X_a^1}, \tag{4.54}$$

where the X_a^1 are the spin-1 matrices. A useful trick is to write

$$X_a^1 a_a = A \hat{a}_a X_a^1, \tag{4.55}$$

where

$$A = \sqrt{a_a a_a}, \qquad \hat{a}_a = \frac{a_a}{A}. \tag{4.56}$$

Since $\hat{a}_a X_a^1$ has eigenvalues $+1, -1, 0$, its square is a projection operator and

$$(\hat{a}_a X_a^1)^3 = \hat{a}_a X_a^1. \tag{4.57}$$

Use this to manipulate the exponential and obtain an explicit expression for $e^{ia_a X_a^1}$.

Chapter 5

Isospin

Isospin was introduced in nuclear physics to treat the proton and neutron as members of a single two-component object, the nucleon doublet,

$$N \equiv \begin{pmatrix} p \\ n \end{pmatrix}. \tag{5.1}$$

This notation later became useful because the strong interaction is nearly independent of whether the nucleon is a proton or a neutron. The original historical motivation was not quite right dynamically, but the group-theoretic idea survived.

5.1 Charge independence

The nuclear force is approximately charge independent: the strong force between pp, pn, and nn pairs is approximately the same. This is naturally encoded by treating p and n as the two T_3 components of an $SU(2)$ doublet, in close analogy with ordinary spin-1/2. The name "isospin" is historical: what matters is that the transformations mix states with the same baryon number, not isotopes in the literal chemical sense.

5.2 Creation operators

For particle physics, it is useful to describe isospin using creation and annihilation operators. For the nucleon doublet,

$$|p, \alpha\rangle = a^\dagger_{N,\frac{1}{2},\alpha} |0\rangle, \qquad |n, \alpha\rangle = a^\dagger_{N,-\frac{1}{2},\alpha} |0\rangle. \tag{5.2}$$

Here α denotes all additional quantum numbers of the one-particle state. The nucleon creation operators are

$$a^\dagger_{N,\pm\frac{1}{2},\alpha}. \tag{5.3}$$

Their adjoints are annihilation operators,

$$a_{N,\pm\frac{1}{2},\alpha}. \tag{5.4}$$

They annihilate the vacuum,

$$a_{N,\pm\frac{1}{2},\alpha} |0\rangle = 0. \tag{5.5}$$

Because protons and neutrons are fermions, the nucleon creation and annihilation operators obey anticommutation relations:

$$\begin{aligned} \left\{a_{N,m,\alpha}, a^\dagger_{N,m',\beta}\right\} &= \delta_{mm'}\delta_{\alpha\beta}, \\ \left\{a^\dagger_{N,m,\alpha}, a^\dagger_{N,m',\beta}\right\} &= \left\{a_{N,m,\alpha}, a_{N,m',\beta}\right\} = 0. \end{aligned} \tag{5.6}$$

A state of n protons is made by applying n proton creation operators to $|0\rangle$:

$$\underbrace{a^\dagger_{N,\frac{1}{2},\alpha_1} \cdots a^\dagger_{N,\frac{1}{2},\alpha_n}}_{n \text{ proton creation operators}} |0\rangle \propto |n \text{ protons}; \alpha_1, \ldots, \alpha_n\rangle . \tag{5.7}$$

More generally, an n-nucleon state is

$$\underbrace{a^\dagger_{N,m_1,\alpha_1} \cdots a^\dagger_{N,m_n,\alpha_n}}_{n \text{ nucleon creation operators}} |0\rangle \propto |n \text{ nucleon}; m_1, \alpha_1; \ldots ; m_n, \alpha_n\rangle . \tag{5.8}$$

The anticommutation relation makes the state antisymmetric under exchange of both labels belonging to two nucleons:

$$\begin{aligned} &|n \text{ nucleon}; m_1, \alpha_1; m_2, \alpha_2; \ldots ; m_n, \alpha_n\rangle \\ &\qquad = - |n \text{ nucleon}; m_2, \alpha_2; m_1, \alpha_1; \ldots ; m_n, \alpha_n\rangle . \end{aligned} \tag{5.9}$$

This is the generalized exclusion principle in this notation.

5.3 Number operators

Number operators are obtained by combining creation and annihilation operators, with summation over repeated labels understood:

$$\begin{aligned} a^\dagger_{N,+\frac{1}{2},\alpha} a_{N,+\frac{1}{2},\alpha} &\qquad \text{counts protons,} \\ a^\dagger_{N,-\frac{1}{2},\alpha} a_{N,-\frac{1}{2},\alpha} &\qquad \text{counts neutrons,} \\ a^\dagger_{N,m,\alpha} a_{N,m,\alpha} &\qquad \text{counts nucleons.} \end{aligned} \tag{5.10}$$

For a generic creation-annihilation pair one has

$$\left[a^\dagger a, a^\dagger\right] = a^\dagger. \tag{5.11}$$

Thus the operators in (5.10), summed over all α, count the total number of protons, neutrons, or nucleons.

5.4 Isospin generators

On one-particle states the isospin generators act like spin-1/2 generators:

$$T_a |m, \alpha\rangle = |m', \alpha\rangle \, [J_a^{1/2}]_{m'm} = \frac{1}{2} |m', \alpha\rangle \, [\sigma_a]_{m'm}. \tag{5.12}$$

Equivalently,

$$T_a a^\dagger_{N,m,\alpha} |0\rangle = a^\dagger_{N,m',\alpha} |0\rangle \, [J_a^{1/2}]_{m'm} = \frac{1}{2} a^\dagger_{N,m',\alpha} |0\rangle \, [\sigma_a]_{m'm}. \tag{5.13}$$

The vacuum transforms trivially:

$$T_a |0\rangle = 0. \tag{5.14}$$

Therefore the creation operators must transform as spin-1/2 tensor operators under isospin:

$$\left[T_a, a^\dagger_{N,m,\alpha}\right] = a^\dagger_{N,m',\alpha} [J_a^{1/2}]_{m'm} = \frac{1}{2} a^\dagger_{N,m',\alpha} [\sigma_a]_{m'm}. \tag{5.15}$$

A generator with the desired action is

$$\begin{aligned} T_a &= a^\dagger_{N,m',\alpha}[J^{1/2}_a]_{m'm}a_{N,m,\alpha} + \cdots \\ &= \frac{1}{2}a^\dagger_{N,m',\alpha}[\sigma_a]_{m'm}a_{N,m,\alpha} + \cdots \\ &= \frac{1}{2}a^\dagger_{N,\alpha}\sigma_a a_{N,\alpha} + \cdots . \end{aligned} \tag{5.16}$$

The ellipsis denotes terms commuting with the nucleon creation and annihilation operators, and also annihilating the vacuum. The commutator check is

$$\begin{aligned} \left[T_a, a^\dagger_{N,m,\alpha}\right] &= \left[a^\dagger_{N,m',\beta}[J^{1/2}_a]_{m'm''}a_{N,m'',\beta}, a^\dagger_{N,m,\alpha}\right] \\ &= a^\dagger_{N,m',\beta}[J^{1/2}_a]_{m'm''}\left\{a_{N,m'',\beta}, a^\dagger_{N,m,\alpha}\right\} \\ &\quad - \left\{a^\dagger_{N,m',\beta}, a^\dagger_{N,m,\alpha}\right\}[J^{1/2}_a]_{m'm''}a_{N,m'',\beta} \\ &= a^\dagger_{N,m',\alpha}[J^{1/2}_a]_{m'm}. \end{aligned} \tag{5.17}$$

Thus multiparticle states transform as tensor products, because they are built by applying tensor-operator creation operators to the vacuum.

5.5 Symmetry of tensor products

For two identical spin-1/2 representations, the spin-1 piece is symmetric under exchange, while the spin-0 piece is antisymmetric. Denote the two spin-1/2 states by

$$|1/2, \pm 1/2, \alpha\rangle . \tag{5.18}$$

The highest-weight triplet state is symmetric:

$$|1,1\rangle = |1/2,1/2,\alpha\rangle\,|1/2,1/2,\beta\rangle = |1/2,1/2,\beta\rangle\,|1/2,1/2,\alpha\rangle . \tag{5.19}$$

Lowering gives the rest of the spin-1 multiplet:

$$\begin{aligned} |1,0\rangle &= \frac{1}{\sqrt{2}}\Big(\,|1/2,-1/2,\alpha\rangle\,|1/2,1/2,\beta\rangle + |1/2,1/2,\alpha\rangle\,|1/2,-1/2,\beta\rangle\,\Big), \\ |1,-1\rangle &= |1/2,-1/2,\alpha\rangle\,|1/2,-1/2,\beta\rangle . \end{aligned} \tag{5.20}$$

The orthogonal spin-0 state is antisymmetric:

$$|0,0\rangle = \frac{1}{\sqrt{2}}\Big(\,|1/2,-1/2,\alpha\rangle\,|1/2,1/2,\beta\rangle - |1/2,1/2,\alpha\rangle\,|1/2,-1/2,\beta\rangle\,\Big). \tag{5.21}$$

5.6 The deuteron

Nucleons carry both spin 1/2 and isospin 1/2. In a two-nucleon s-wave, the spatial wavefunction is symmetric, while the fermionic creation operators make the full two-particle state antisymmetric under simultaneous exchange of the spin and isospin labels. Therefore the spin and isospin symmetry types must compensate: spin-symmetric goes with isospin-antisymmetric, and spin-antisymmetric goes with isospin-symmetric. Hence the allowed s-wave two-nucleon states are

$$(T = 1, S = 0) \qquad \text{or} \qquad (T = 0, S = 1).$$

The deuteron is the $T = 0$, $S = 1$ case.

5.7 Superselection rules

The ordinary Pauli exclusion principle says that the creation operator for a fermion in a fixed state satisfies

$$\left(a_\alpha^\dagger\right)^2 = 0. \tag{5.22}$$

For another state of the same particle, the superposed creation operator must also square to zero:

$$\left(a_\alpha^\dagger + a_\beta^\dagger\right)^2 = 0. \tag{5.23}$$

Thus

$$\left\{a_\alpha^\dagger, a_\beta^\dagger\right\} = 0. \tag{5.24}$$

For different particles, such superpositions are physically suspect because states with different exactly conserved quantum numbers lie in different superselection sectors. Nevertheless, in the isospin formalism the anticommutation of the proton and neutron creation operators follows from the requirement that the creation operators transform as tensor operators.

Write the baryon creation operators as $a_{\pm\pm}^\dagger$, dropping the N. The first sign is T_3, the second sign is J_3. Since $(a_{++}^\dagger)^2 = 0$, there is no state with $T_3 = 1$ and $J_3 = 1$. For example,

$$\left[T^-, \left(a_{++}^\dagger\right)^2\right] = \left\{a_{-+}^\dagger, a_{++}^\dagger\right\} = 0. \tag{5.25}$$

Applying both lowering operators gives

$$\left[J^-, \left[T^-, \left(a_{++}^\dagger\right)^2\right]\right] = \left\{a_{--}^\dagger, a_{++}^\dagger\right\} + \left\{a_{-+}^\dagger, a_{+-}^\dagger\right\} = 0. \tag{5.26}$$

The two terms create physically distinguishable states, so they must vanish separately. In particular,

$$\left\{a_{--}^\dagger, a_{++}^\dagger\right\} |0\rangle = 0, \tag{5.27}$$

and

$$\left\{a_{-+}^\dagger, a_{+-}^\dagger\right\} |0\rangle = 0. \tag{5.28}$$

Consequently the spin-0, isospin-0 combination also vanishes:

$$\left\{a_{--}^\dagger, a_{++}^\dagger\right\} |0\rangle - \left\{a_{-+}^\dagger, a_{+-}^\dagger\right\} |0\rangle = 0. \tag{5.29}$$

5.8 Other particles

The pions are spinless bosons with charges $Q = +1, 0, -1$ and $T_3 = Q$, forming an isospin triplet. Their creation and annihilation operators are

$$a_{\pi,m,\alpha}^\dagger, \qquad a_{\pi,m,\alpha}, \qquad m = -1, 0, 1. \tag{5.30}$$

They obey bosonic commutation relations:

$$\begin{aligned} \left[a_{\pi,m,\alpha}, a_{\pi,m',\beta}^\dagger\right] &= \delta_{mm'}\delta_{\alpha\beta}, \\ \left[a_{\pi,m,\alpha}^\dagger, a_{\pi,m',\beta}^\dagger\right] &= \left[a_{\pi,m,\alpha}, a_{\pi,m',\beta}\right] = 0. \end{aligned} \tag{5.31}$$

The pion contribution to the isospin generators is

$$T_a = a_{\pi,m,\alpha}^\dagger [J_a^1]_{mm'} a_{\pi,m',\alpha} + \cdots . \tag{5.32}$$

For all particle species carrying isospin, the full isospin generators have the form

$$T_a = \sum_{\substack{\text{particles } x \\ \text{states } \alpha \\ T_3 \text{ values } m,m'}} a^\dagger_{x,m,\alpha}[J^{j_x}_a]_{mm'} a_{x,m',\alpha}. \tag{5.33}$$

The general canonical relations are

$$\begin{aligned} \left[a_{x,m,\alpha}, a^\dagger_{x',m',\beta}\right]_\pm &= \delta_{mm'}\delta_{\alpha\beta}\delta_{xx'}, \\ \left[a^\dagger_{x,m,\alpha}, a^\dagger_{x',m',\beta}\right]_\pm &= \left[a_{x,m,\alpha}, a_{x',m',\beta}\right]_\pm = 0. \end{aligned} \tag{5.34}$$

Here $[A, B]_+$ is the anticommutator and $[A, B]_-$ is the commutator; use the anticommutator only when both particles are fermions, otherwise use the commutator. The symbol j_x denotes the isospin of the x-type particles.

5.9 Approximate isospin symmetry

Isospin is only approximate. The Hamiltonian is split into an isospin-invariant part and an isospin-breaking perturbation,

$$H = H_0 + \Delta H. \tag{5.35}$$

Here H_0 commutes with the isospin generators, while ΔH does not. The traditional language is that strong interactions respect isospin, while electromagnetic and weak interactions break it. In modern terms this division is somewhat simplified, since quark-mass differences also break isospin.

5.10 Perturbation theory

In applications one first classifies states using the H_0 eigenstates, which also carry definite isospin. Then ΔH is treated perturbatively when needed. For example, pion-nucleon scattering involves an isospin-1 pion and an isospin-1/2 nucleon, so the total isospin channels are

$$1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}.$$

Thus, in the isospin-symmetric approximation, the scattering is described by only two independent isospin amplitudes.

Problems

5.A. Suppose that in some process, a pair of pions is produced in a state with zero relative orbital angular momentum. What total isospin values are possible for this state?

5.B. Show that the operators defined in (5.33) obey the commutation relations of the isospin generators.

5.C. The particles Δ^{++}, Δ^+, Δ^0, and Δ^- form an isospin-3/2 multiplet with $T_3 = 3/2, 1/2, -1/2, -3/2$, respectively, and baryon number 1. They are produced by strong interactions in pion-nucleon collisions. Compare the probability of producing Δ^{++} in

$$\pi^+ p \to \Delta^{++}$$

with the probability of producing Δ^0 in

$$\pi^- p \to \Delta^0.$$

Chapter 6

Roots and Weights

The purpose of this chapter is to generalize the familiar analysis of the $SU(2)$ algebra to an arbitrary simple Lie algebra. The strategy is the same as in quantum mechanics: choose as many mutually commuting Hermitian operators as possible, diagonalize them, and use their eigenvalues as labels. For a general Lie algebra these operators are the Cartan generators. The remaining generators then behave like analogues of the $SU(2)$ raising and lowering operators.

6.1 Weights

A maximal commuting set of Hermitian generators is called a *Cartan subalgebra*. In an irreducible representation D, write its Hermitian Cartan generators as H_i, $i = 1, \ldots, m$. They obey

$$H_i = H_i^\dagger, \qquad [H_i, H_j] = 0. \tag{6.1}$$

Since the Cartan generators form a vector space, one can choose a basis with trace normalization

$$\mathrm{Tr}(H_i H_j) = k_D \delta_{ij}, \qquad i, j = 1, \ldots, m. \tag{6.2}$$

The number m is the *rank* of the algebra.

Because the H_i commute, they can be diagonalized simultaneously. A state in the representation D may then be denoted $|\mu, x, D\rangle$, with

$$H_i |\mu, x, D\rangle = \mu_i |\mu, x, D\rangle. \tag{6.3}$$

The eigenvalues μ_i are the *weights*; the vector μ is the weight vector. We use the Euclidean notation

$$\alpha \cdot \mu \equiv \alpha_i \mu_i, \qquad \alpha^2 \equiv \alpha_i \alpha_i. \tag{6.4}$$

6.2 More on the adjoint representation

The adjoint representation is special because its states can be identified with the generators themselves. The adjoint state corresponding to a generator X_a is written

$$|X_a) \, . \tag{6.5}$$

Linear combinations of generators correspond to linear combinations of adjoint states:

$$\alpha \, |X_a) + \beta \, |X_b) = |\alpha X_a + \beta X_b) \, . \tag{6.6}$$

A useful scalar product on this space is

$$(X_a | \, X_b\rangle = \lambda^{-1} \, \mathrm{Tr}\Big(X_a^\dagger X_b\Big), \tag{6.7}$$

where λ is the adjoint-representation trace constant.

Using the adjoint action, the action of a generator on an adjoint state is

$$\begin{aligned} X_a\,|X_b) &= |X_c)\,(X_c|\,X_a|X_b\rangle = |X_c)\,[T_a]_{cb} = -\mathrm{i}f_{acb}\,|X_c) \\ &= \mathrm{i}f_{abc}\,|X_c) = |\mathrm{i}f_{abc}X_c) = |[X_a, X_b])\,. \end{aligned} \tag{6.8}$$

The dagger in the scalar product matters because the natural Cartan/root basis will use complex combinations of Hermitian generators.

6.3 Roots

The roots are precisely the nonzero weights of the adjoint representation. For Cartan generators,

$$H_i\,|H_j) = |[H_i, H_j]) = 0. \tag{6.9}$$

With the normalization above, the Cartan states are orthonormal:

$$(H_i|\,H_j\rangle = \lambda^{-1}\,\mathrm{Tr}(H_iH_j) = \delta_{ij}. \tag{6.10}$$

The remaining adjoint states have nonzero weight vectors α:

$$H_i\,|E_\alpha) = \alpha_i\,|E_\alpha)\,. \tag{6.11}$$

Equivalently, the corresponding generators obey

$$[H_i, E_\alpha] = \alpha_i E_\alpha. \tag{6.12}$$

Taking the Hermitian adjoint gives

$$[H_i, E_\alpha^\dagger] = -\alpha_i E_\alpha^\dagger. \tag{6.13}$$

Thus the natural convention is

$$E_\alpha^\dagger = E_{-\alpha}. \tag{6.14}$$

States with different weights are orthogonal. The root generators can be normalized so that

$$(E_\alpha|\,E_\beta\rangle = \lambda^{-1}\,\mathrm{Tr}\Big(E_\alpha^\dagger E_\beta\Big) = \delta_{\alpha\beta}\left(= \prod_i \delta_{\alpha_i\beta_i}\right). \tag{6.15}$$

The components α_i are called roots, and α is the root vector.

6.4 Raising and lowering

The operators $E_{\pm\alpha}$ move weights by $\pm\alpha$. For any representation D,

$$\begin{aligned} H_iE_{\pm\alpha}|\mu, D\rangle &= [H_i, E_{\pm\alpha}]|\mu, D\rangle + E_{\pm\alpha}H_i|\mu, D\rangle \\ &= (\mu \pm \alpha)_i E_{\pm\alpha}|\mu, D\rangle. \end{aligned} \tag{6.16}$$

At this stage there is no preferred notion of positive direction, so "raising" and "lowering" are relative words.

For the adjoint representation, $E_\alpha\,|E_{-\alpha})$ has zero weight and therefore must be a linear combination of Cartan states:

$$E_\alpha\,|E_{-\alpha}) = \beta_i\,|H_i) = |\beta_i H_i) = |\beta\cdot H) = |[E_\alpha, E_{-\alpha}])\,. \tag{6.17}$$

The coefficients are computed as

$$\begin{aligned}\beta_i &= (H_i|\, E_\alpha|E_{-\alpha}\rangle \\ &= \lambda^{-1}\,\mathrm{Tr}\big(H_i[E_\alpha, E_{-\alpha}]\big) \\ &= \lambda^{-1}\,\mathrm{Tr}\big(E_{-\alpha}[H_i, E_\alpha]\big) \\ &= \lambda^{-1}\alpha_i\,\mathrm{Tr}(E_{-\alpha}E_\alpha) \\ &= \alpha_i.\end{aligned} \tag{6.18}$$

Therefore

$$[E_\alpha, E_{-\alpha}] = \alpha\cdot H. \tag{6.19}$$

This is the direct analogue of $[J^+, J^-] = J_3$ in $SU(2)$.

6.5 Lots of $SU(2)$s

Each pair of roots $\pm\alpha$ defines an $SU(2)$ subalgebra. Define

$$E^\pm \equiv |\alpha|^{-1}E_{\pm\alpha}, \qquad E_3 \equiv |\alpha|^{-2}\alpha\cdot H. \tag{6.20}$$

Then

$$\begin{aligned}[E_3, E^\pm] &= |\alpha|^{-3}[\alpha\cdot H, E_{\pm\alpha}] \\ &= |\alpha|^{-3}\alpha\cdot(\pm\alpha)E_{\pm\alpha} = \pm|\alpha|^{-1}E_{\pm\alpha} = \pm E^\pm,\end{aligned} \tag{6.21}$$

while

$$\begin{aligned}[E^+, E^-] &= |\alpha|^{-2}[E_\alpha, E_{-\alpha}] \\ &= |\alpha|^{-2}\alpha\cdot H = E_3.\end{aligned} \tag{6.22}$$

Now suppose there were two independent generators with the same root, E_α and E'_α. Choose them orthogonal in the adjoint scalar product:

$$(E_\alpha|\, E'_\alpha\rangle = \lambda^{-1}\,\mathrm{Tr}\Big(E_\alpha^\dagger E'_\alpha\Big) = \lambda^{-1}\,\mathrm{Tr}\big(E_{-\alpha}E'_\alpha\big) = 0. \tag{6.23}$$

Consider $E^-\,|E'_\alpha)$. It has zero weight, hence only Cartan components are possible. But for every i,

$$\begin{aligned}(H_i|\, E^-|E'_\alpha\rangle &= \lambda^{-1}\,\mathrm{Tr}\big(H_i[E^-, E'_\alpha]\big) \\ &= -\lambda^{-1}\,\mathrm{Tr}\big(E^-[H_i, E'_\alpha]\big) \\ &= -\alpha_i\lambda^{-1}\,\mathrm{Tr}\big(E'_\alpha E^-\big) = 0.\end{aligned} \tag{6.24}$$

Therefore

$$E^-\,\big|E'_\alpha\big) = 0. \tag{6.25}$$

On the other hand,

$$E_3\,\big|E'_\alpha\big) = |\alpha|^{-2}\alpha\cdot H\,\big|E'_\alpha\big) = \big|E'_\alpha\big)\,. \tag{6.26}$$

Equations (6.25) and (6.26) contradict the $SU(2)$ representation structure: a lowest J_3 state cannot have positive J_3. Hence a nonzero root α labels a unique generator E_α. The same $SU(2)$ argument also rules out nonzero multiples of a root, except for the opposite root $-\alpha$.

6.6 Watch carefully - this is important!

For any weight μ of a representation D, the E_3 eigenvalue in the $SU(2)$ subalgebra associated with α is

$$E_3|\mu, x, D\rangle = \frac{\alpha \cdot \mu}{\alpha^2}|\mu, x, D\rangle. \tag{6.27}$$

Since $SU(2)$ eigenvalues are integral or half-integral,

$$\frac{2\alpha \cdot \mu}{\alpha^2} \quad \text{is an integer.} \tag{6.28}$$

Decompose a state $|\mu, x, D\rangle$ into irreducible multiplets of this $SU(2)$. If the highest spin that appears is j, then for some non-negative integer p,

$$(E^+)^p|\mu, x, D\rangle \neq 0, \tag{6.29}$$

while

$$(E^+)^{p+1}|\mu, x, D\rangle = 0. \tag{6.30}$$

The weight of the state in (6.29) is $\mu + p\alpha$, and its E_3 value is

$$\frac{\alpha \cdot (\mu + p\alpha)}{\alpha^2} = \frac{\alpha \cdot \mu}{\alpha^2} + p = j. \tag{6.31}$$

Similarly, for some non-negative integer q,

$$(E^-)^q|\mu, x, D\rangle \neq 0, \tag{6.32}$$

while

$$(E^-)^{q+1}|\mu, x, D\rangle = 0. \tag{6.33}$$

The lowest state has weight $\mu - q\alpha$, with

$$\frac{\alpha \cdot (\mu - q\alpha)}{\alpha^2} = \frac{\alpha \cdot \mu}{\alpha^2} - q = -j. \tag{6.34}$$

Adding (6.31) and (6.34) gives

$$\frac{2\alpha \cdot \mu}{\alpha^2} + p - q = 0, \tag{6.35}$$

or equivalently

$$\frac{\alpha \cdot \mu}{\alpha^2} = -\frac{1}{2}(p - q). \tag{6.36}$$

This is the master formula.

Apply the master formula to a pair of distinct roots α and β. Using the $SU(2)$ generated by E_α,

$$\frac{\alpha \cdot \beta}{\alpha^2} = -\frac{1}{2}(p - q). \tag{6.37}$$

Using the $SU(2)$ generated by E_β,

$$\frac{\beta \cdot \alpha}{\beta^2} = -\frac{1}{2}(p' - q'). \tag{6.38}$$

Multiplying gives the angle constraint

$$\cos^2 \theta_{\alpha\beta} = \frac{(\alpha \cdot \beta)^2}{\alpha^2 \beta^2} = \frac{(p - q)(p' - q')}{4}. \tag{6.39}$$

Since $(p-q)(p'-q')$ is a non-negative integer, the possible nontrivial angles between roots are summarized by

$$\begin{array}{c|c} (p-q)(p'-q') & \theta_{\alpha\beta} \\ \hline 0 & 90^\circ \\ 1 & 60^\circ \text{ or } 120^\circ \\ 2 & 45^\circ \text{ or } 135^\circ \\ 3 & 30^\circ \text{ or } 150^\circ \end{array} \tag{6.40}$$

The remaining value $(p-q)(p'-q') = 4$ would give 0° or 180°: the first is excluded by uniqueness of roots, and the second is just the opposite-root pair $\alpha, -\alpha$ in the same $SU(2)$ subalgebra.

Problems

6.A. Show that $[E_\alpha, E_\beta]$ must be proportional to $E_{\alpha+\beta}$. What happens if $\alpha+\beta$ is not a root?

6.B. Suppose the raising/lowering operators of a Lie algebra satisfy

$$[E_\alpha, E_\beta] = N E_{\alpha+\beta} \tag{5.1}$$

for some nonzero N. Calculate

$$[E_\alpha, E_{-\alpha-\beta}]. \tag{5.2}$$

6.C. Consider the simple Lie algebra formed by the ten matrices

$$\sigma_a, \qquad \sigma_a\tau_1, \qquad \sigma_a\tau_3, \qquad \tau_2, \tag{5.3}$$

for $a = 1, 2, 3$, where σ_a and τ_a are Pauli matrices in orthogonal spaces. Take

$$H_1 = \sigma_3, \qquad H_2 = \sigma_3\tau_3 \tag{5.4}$$

as the Cartan subalgebra. Find:

1. the weights of the four-dimensional representation generated by these matrices;
2. the weights of the adjoint representation.

The problem becomes easier after seeing the explicit $SU(3)$ example in the next chapter.

Chapter 7

$SU(3)$

After $SU(2)$, the next central example for particle physics is $SU(3)$: the special unitary group of 3×3 matrices. The word "special" means that the determinant is fixed to be one. This chapter works out the standard matrix realization, its Cartan generators, and the corresponding weights and roots.

7.1 The Gell-Mann matrices

The Lie algebra of $SU(3)$ is generated by Hermitian traceless 3×3 matrices. Exponentiating Hermitian generators gives unitary matrices,

$$U(\alpha) = e^{\mathrm{i}\alpha_a X_a}. \tag{7.1}$$

To see the determinant condition, diagonalize the Hermitian combination $\alpha_a X_a$:

$$V \alpha_a X_a V^{-1} = D, \tag{7.2}$$

where D is diagonal. Then

$$\det(U(\alpha)) = \det\left(e^{\mathrm{i}D}\right) = \prod_j e^{\mathrm{i}[D]_{jj}} = e^{\mathrm{i}\,\mathrm{Tr}\,D} = e^{\mathrm{i}\,\mathrm{Tr}\,\alpha_a X_a}. \tag{7.3}$$

Thus traceless generators give determinant one.

The conventional physics basis is given by the Gell-Mann matrices:

$$\begin{aligned}
\lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\
\lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \\
\lambda_5 &= \begin{pmatrix} 0 & 0 & -\mathrm{i} \\ 0 & 0 & 0 \\ \mathrm{i} & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\
\lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.
\end{aligned} \tag{7.4}$$

The first three contain the Pauli matrices in the upper-left 2×2 block:

$$\lambda_a = \begin{pmatrix} \sigma_a & 0 \\ 0 & 0 \end{pmatrix}, \qquad a = 1, 2, 3. \tag{7.5}$$

The corresponding $SU(3)$ generators are normalized as

$$T_a = \frac{1}{2}\lambda_a, \tag{7.6}$$

and satisfy

$$\mathrm{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}. \tag{7.7}$$

The generators T_1, T_2, T_3 close on an $SU(2)$ subalgebra. A convenient Cartan subalgebra for $SU(3)$ is chosen as

$$H_1 = T_3, \qquad H_2 = T_8. \tag{7.8}$$

7.2 Weights and roots of $SU(3)$

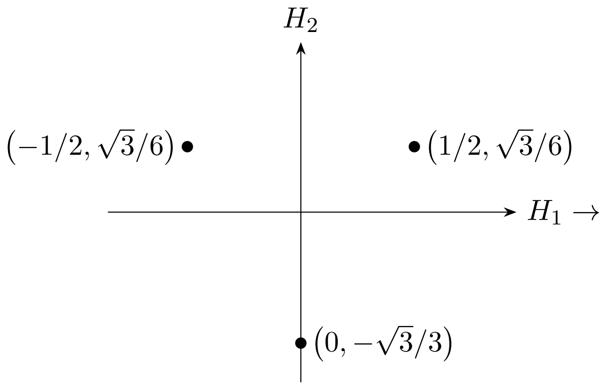
In the defining three-dimensional representation, the Cartan generators are already diagonal:

$$T_3 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad T_8 = \frac{\sqrt{3}}{6}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{7.9}$$

Thus the basis eigenvectors and their weight vectors are

$$\begin{gathered} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \longrightarrow \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \qquad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \longrightarrow \left(-\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \\ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \longrightarrow \left(0, -\frac{\sqrt{3}}{3}\right). \end{gathered} \tag{7.10}$$

These weights form an equilateral triangle in the (H_1, H_2) plane:



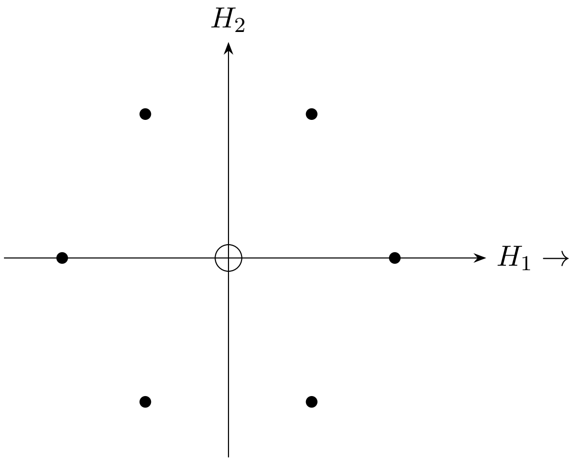
(7.11)

The roots are obtained from differences of weights. The corresponding off-diagonal generators are

$$\begin{aligned} \frac{1}{\sqrt{2}}(T_1 \pm \mathrm{i}T_2) &= E_{\pm 1,0}, \\ \frac{1}{\sqrt{2}}(T_4 \pm \mathrm{i}T_5) &= E_{\pm 1/2, \pm\sqrt{3}/2}, \\ \frac{1}{\sqrt{2}}(T_6 \pm \mathrm{i}T_7) &= E_{\mp 1/2, \pm\sqrt{3}/2}, \end{aligned} \tag{7.12}$$

where the signs in each line are correlated as shown.

The six roots form a regular hexagon, while the two Cartan generators sit at the origin:



(7.13)

Problems

7.A. Compute the structure constants f_{147} and f_{458} for $SU(3)$.

7.B. Verify that T_1, T_2, T_3 generate an $SU(2)$ subalgebra of $SU(3)$. Decompose the defining three-dimensional representation, and then the adjoint representation, into irreducible representations of this subalgebra.

7.C. Verify that $\lambda_2, \lambda_5, \lambda_7$ generate an $SU(2)$ subalgebra of $SU(3)$. Decompose the defining and adjoint representations under this subalgebra.

Chapter 8

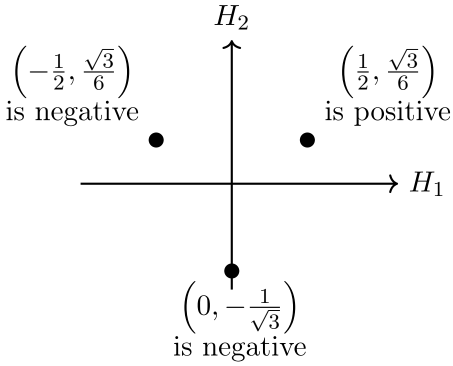
Simple Roots

8.1 Positive weights

To extend the analogy with $SU(2)$ to a general simple Lie algebra, one first needs a notion of positivity for weights. Then one can speak meaningfully about raising and lowering operators and about highest weights. The definition should ensure that every nonzero weight is either positive or negative, and that if μ is positive then $-\mu$ is negative, and conversely.

Choose an arbitrary basis in the Cartan subalgebra. If the components of a weight are $\mu_1, \mu_2, \ldots$, then we call the weight *positive* if its first nonzero component is positive, and *negative* if its first nonzero component is negative.

For $SU(3)$, the defining representation has weight diagram

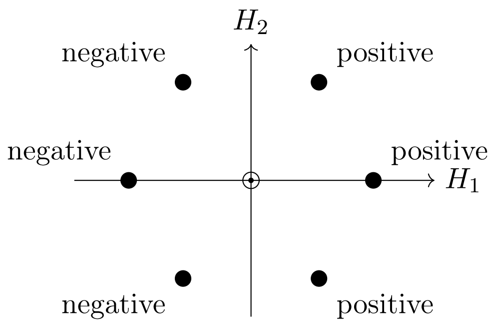


(8.1)

The weight $(0, -1/\sqrt{3})$ is negative because its first component vanishes, so the sign is determined by the second component. This induces the ordering

$$\mu > \nu \quad \text{if } \mu - \nu \text{ is positive.} \tag{8.2}$$

This allows one to speak about the highest weight in a representation. In the adjoint representation, positive roots correspond to raising operators and negative roots to lowering operators. For $SU(3)$, the positive roots lie on the right and the negative roots on the left:



(8.3)

8.2 Simple roots

A *simple root* is a positive root that cannot be written as a sum of two positive roots. The geometry of the simple roots determines the algebra.

If α and β are different simple roots, then $\alpha - \beta$ is not a root. Therefore

$$E_{-\alpha}\,|E_\beta\rangle = E_{-\beta}\,|E_\alpha\rangle = 0. \tag{8.4}$$

Using the master formula (6.36),

$$\frac{\alpha\cdot\beta}{\alpha^2} = -\frac{1}{2}(p-q), \qquad \frac{\beta\cdot\alpha}{\beta^2} = -\frac{1}{2}(p'-q'),$$

with $q = q' = 0$, one gets

$$\frac{\alpha\cdot\beta}{\alpha^2} = -\frac{p}{2}, \qquad \frac{\beta\cdot\alpha}{\beta^2} = -\frac{p'}{2}. \tag{8.5}$$

Hence

$$\cos\theta_{\alpha\beta} = -\frac{\sqrt{pp'}}{2}, \qquad \frac{\beta^2}{\alpha^2} = \frac{p}{p'}. \tag{8.6}$$

Thus the angle between simple roots obeys

$$\frac{\pi}{2} \le \theta < \pi. \tag{8.7}$$

Simple roots are linearly independent. Consider

$$\gamma = \sum_\alpha x_\alpha \alpha. \tag{8.8}$$

If the coefficients have mixed signs, write

$$\gamma = \mu - \nu, \tag{8.9}$$

with

$$\mu = \sum_{x_\alpha > 0} x_\alpha \alpha, \qquad \nu = -\sum_{x_\alpha < 0} x_\alpha \alpha. \tag{8.10}$$

Then

$$(\mu-\nu)^2 = \mu^2 + \nu^2 - 2(\mu\cdot\nu) \ge \mu^2 + \nu^2 > 0. \tag{8.11}$$

So no nontrivial linear combination vanishes.

Any positive root can be written as

$$\phi = \sum_\alpha k_\alpha \alpha, \qquad k_\alpha \in \mathbb{Z}_{\ge 0}. \tag{8.12}$$

If a vector ξ is orthogonal to all simple roots, then it is orthogonal to all roots, and hence

$$[\xi\cdot H, E_\phi] = 0 \qquad \text{for all roots } \phi. \tag{8.13}$$

Since the algebra is simple, this is impossible unless the simple roots span the Cartan subalgebra.

Thus every positive root has the form

$$\phi_k = \sum_\alpha k_\alpha \alpha, \tag{8.14}$$

where

$$k = \sum_\alpha k_\alpha. \tag{8.15}$$

Acting on roots of level $k = \ell$ with E_α gives candidates at level $\ell + 1$:

$$E_\alpha \left|\phi_\ell\right\rangle . \tag{8.16}$$

The master formula gives

$$\frac{2\alpha \cdot \phi_\ell}{\alpha^2} = -(p - q). \tag{8.17}$$

For $\ell = 1$, with $\phi_1 = \beta$ a simple root, this becomes

$$\frac{2\alpha \cdot \phi_1}{\alpha^2} = \frac{2\alpha \cdot \beta}{\alpha^2} = -p. \tag{8.18}$$

If $\alpha \cdot \beta = 0$, then $p = 0$ and $\alpha + \beta$ is not a root; otherwise $p > 0$ and $\alpha + \beta$ is a root. A hypothetical missed positive root would satisfy

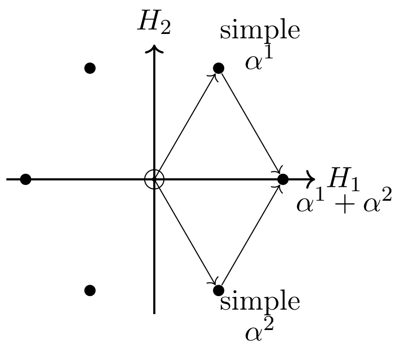
$$\phi_{\ell+1}^2 = \sum_\alpha k_\alpha \, \alpha \cdot \phi_{\ell+1} \leq 0, \tag{8.19}$$

which is impossible.

For $SU(3)$, the simple roots can be chosen as

$$\alpha^1 = (1/2, \sqrt{3}/2), \qquad \alpha^2 = (1/2, -\sqrt{3}/2). \tag{8.20}$$

Their sum is the remaining positive root:



(8.21)

And

$$\alpha^{1\,2} = \alpha^{2\,2} = 1, \qquad \alpha^1 \cdot \alpha^2 = -\frac{1}{2}. \tag{8.22}$$

Hence

$$\frac{2\alpha^1 \cdot \alpha^2}{\alpha^{1\,2}} = \frac{2\alpha^1 \cdot \alpha^2}{\alpha^{2\,2}} = -1. \tag{8.23}$$

So $p = 1$ for α^1 acting on $\left|\alpha^2\right\rangle$, and vice versa; therefore $\alpha^1 + \alpha^2$ is a root, but $2\alpha^1 + \alpha^2$ and $\alpha^1 + 2\alpha^2$ are not.

8.3 Constructing the algebra

From (6.31) and (6.34),

$$\frac{\alpha \cdot \mu}{\alpha^2} + p = j, \qquad \frac{\alpha \cdot \mu}{\alpha^2} - q = -j. \tag{8.24}$$

The $SU(2)$ generators associated with α are

$$E^\pm = |\alpha|^{-1} E_{\pm\alpha}, \qquad E_3 = |\alpha|^{-2} \, \alpha \cdot H. \tag{8.25}$$

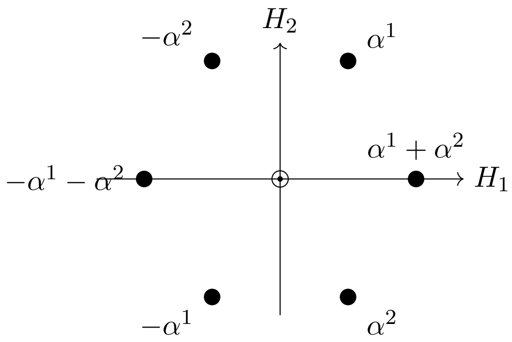
Subtracting the two relations in (8.24) gives

$$p + q = 2j. \tag{8.26}$$

If β is a root, then under the $SU(2)$ generated by α,

$$|\beta\rangle = \left|j, \alpha \cdot \beta/\alpha^2\right\rangle . \tag{8.27}$$

For $SU(3)$ the root diagram is



(8.28)

with

$$\alpha^{1\,2} = \alpha^{2\,2} = 1, \qquad \frac{\alpha^1 \cdot \alpha^2}{\alpha^{1\,2}} = \frac{\alpha^1 \cdot \alpha^2}{\alpha^{2\,2}} = -\frac{1}{2}.$$

Then

$$\begin{aligned} E^+ \left|E_{\alpha^2}\right\rangle &= |\alpha^1|^{-1} E_{\alpha^1} \left|E_{\alpha^2}\right\rangle = E_{\alpha^1} \left|E_{\alpha^2}\right\rangle = \left|[E_{\alpha^1}, E_{\alpha^2}]\right\rangle \\ &= E_{\alpha^1} \left|1/2, -1/2\right\rangle = \frac{1}{\sqrt{2}} \left|1/2, 1/2\right\rangle = \frac{1}{\sqrt{2}} \eta \left|E_{\alpha^1+\alpha^2}\right\rangle , \end{aligned} \tag{8.29}$$

so, choosing $\eta = 1$,

$$\left|E_{\alpha^1+\alpha^2}\right\rangle = \sqrt{2}\, \left|[E_{\alpha^1}, E_{\alpha^2}]\right\rangle , \tag{8.30}$$

and therefore

$$E_{\alpha^1+\alpha^2} = \sqrt{2}\, [E_{\alpha^1}, E_{\alpha^2}]. \tag{8.31}$$

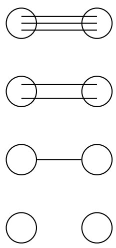
As examples,

$$\begin{aligned} [E_{-\alpha^1}, E_{\alpha^1+\alpha^2}] &= \sqrt{2}[E_{-\alpha^1}, [E_{\alpha^1}, E_{\alpha^2}]] = \sqrt{2}[[E_{-\alpha^1}, E_{\alpha^1}], E_{\alpha^2}] \\ &= \sqrt{2}[-\alpha^1 \cdot H, E_{\alpha^2}] = -\sqrt{2}\, \alpha^1 \cdot \alpha^2\, E_{\alpha^2} = \frac{1}{\sqrt{2}} E_{\alpha^2}, \end{aligned} \tag{8.32}$$

$$\begin{aligned} [E_{-\alpha^2}, E_{\alpha^1+\alpha^2}] &= \sqrt{2}[E_{-\alpha^2}, [E_{\alpha^1}, E_{\alpha^2}]] = \sqrt{2}[E_{\alpha^1}, [E_{-\alpha^2}, E_{\alpha^2}]] \\ &= \sqrt{2}[E_{\alpha^1}, -\alpha^2 \cdot H] = \sqrt{2}[\alpha^2 \cdot H, E_{\alpha^1}] = \sqrt{2}\, \alpha^1 \cdot \alpha^2\, E_{\alpha^1} = -\frac{1}{\sqrt{2}} E_{\alpha^1}. \end{aligned} \tag{8.33}$$

8.4 Dynkin diagrams

A Dynkin diagram encodes the angles between simple roots. Each simple root is drawn as an open circle, and the number of links is determined by the angle:



if the angle is 150°

if the angle is 135°

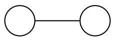
if the angle is 120°

if the angle is 90°

(8.34)

Thus

 is the diagram for $SU(2)$

 is the diagram for $SU(3)$ (8.35)

8.5 Example: G_2

Take simple roots

$$\alpha^1 = (0,1), \qquad \alpha^2 = (\sqrt{3}/2, -3/2). \tag{8.36}$$

Then

$$\alpha^{1\,2} = 1, \qquad \alpha^{2\,2} = 3, \qquad \alpha^1 \cdot \alpha^2 = -\frac{3}{2}, \qquad \frac{2\alpha^1 \cdot \alpha^2}{\alpha^{1\,2}} = -3, \qquad \frac{2\alpha^1 \cdot \alpha^2}{\alpha^{2\,2}} = -1. \tag{8.37}$$

Hence

$$\cos\theta_{\alpha^1\alpha^2} = -\frac{\sqrt{3}}{2}, \qquad \theta_{\alpha^1\alpha^2} = 150°. \tag{8.38}$$

This is the Dynkin diagram of G_2.

8.6 The roots of G_2

For E_{α^1} acting on $|\alpha^2\rangle$ one has $p = 3$; for E_{α^2} acting on $|\alpha^1\rangle$ one has $p = 1$. Therefore

$$\alpha^1 + \alpha^2, \qquad 2\alpha^1 + \alpha^2, \qquad 3\alpha^1 + \alpha^2 \tag{8.39}$$

are roots, while

$$\alpha^1 + 2\alpha^2, \qquad 4\alpha^1 + \alpha^2 \tag{8.40}$$

are not. In terms of the ϕ_k of (8.14),

$$\phi_2 = \alpha^1 + \alpha^2, \qquad \phi_3 = 2\alpha^1 + \alpha^2, \qquad \phi_4 = 3\alpha^1 + \alpha^2. \tag{8.41}$$

To test for $2\alpha^1 + 2\alpha^2$,

$$\frac{2\alpha^2 \cdot (2\alpha^1 + \alpha^2)}{\alpha^{2\,2}} = \frac{-6+6}{3} = 0 = -(p-q). \tag{8.42}$$

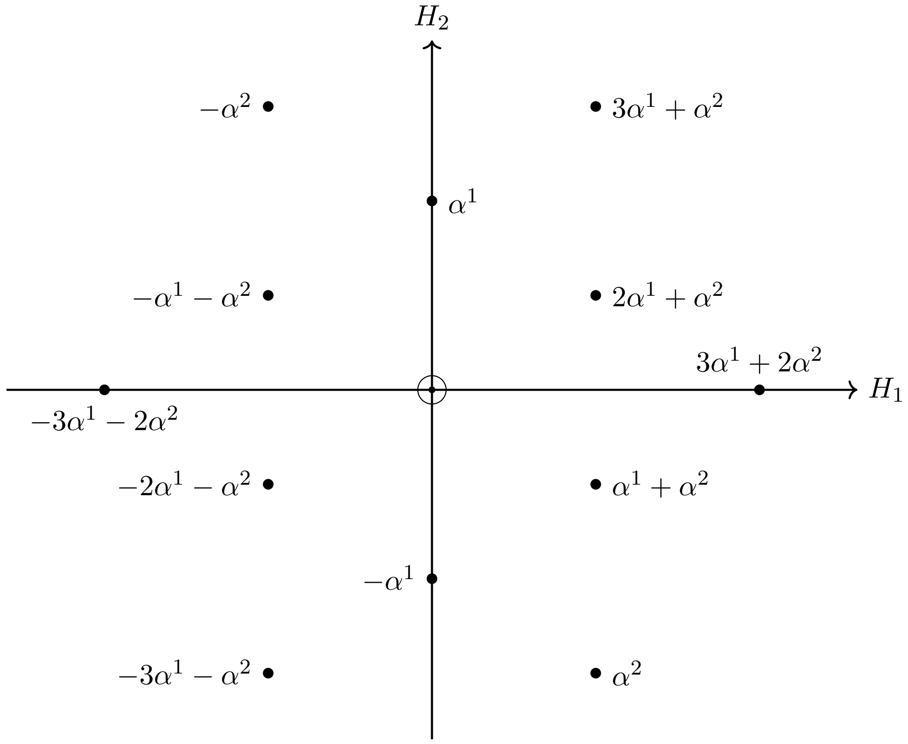
Since $q = 0$, one gets $p = 0$, so $2\alpha^1 + 2\alpha^2$ is not a root. Next,

$$\frac{2\alpha^2 \cdot (3\alpha^1 + \alpha^2)}{\alpha^{2\,2}} = \frac{-9+6}{3} = -1 = -(p-q). \tag{8.43}$$

Thus $p = 1$ and $\phi_5 = 3\alpha^1 + 2\alpha^2$ is a root. Finally,

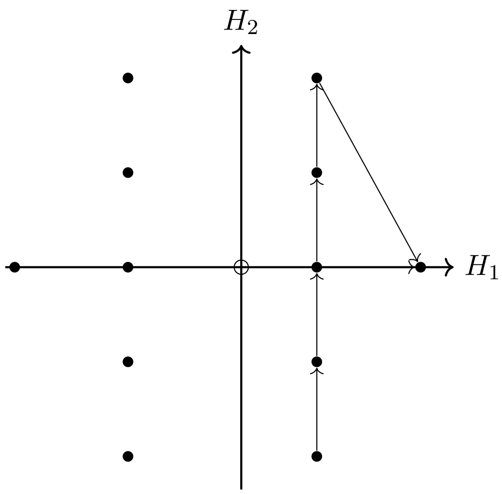
$$\frac{2\alpha^1 \cdot (3\alpha^1 + 2\alpha^2)}{\alpha^{1\,2}} = \frac{6-6}{1} = 0 = -(p-q). \tag{8.44}$$

so there is no further positive root. The root diagram is:



(8.45)

The same information can be summarized by the directed construction diagram:



(8.46)

8.7 The Cartan matrix

For a positive root ϕ, the integers p^i and q^i associated with the action of a simple root α^i can be encoded through the quantity $q^i - p^i$. Since

$$2E_3\,|\mu\rangle = \frac{2H\cdot\alpha^i}{\alpha^{i\,2}}\,|\mu\rangle = \frac{2\mu\cdot\alpha^i}{\alpha^{i\,2}}\,|\mu\rangle = (q^i-p^i)\,|\mu\rangle\,, \tag{8.47}$$

we obtain

$$q^i - p^i = \frac{2\phi\cdot\alpha^i}{\alpha^{i\,2}} = \sum_j k_j\frac{2\alpha^j\cdot\alpha^i}{\alpha^{i\,2}} = \sum_j k_j A_{ji}. \tag{8.48}$$

Here the Cartan matrix is

$$A_{ji} \equiv \frac{2\alpha^j \cdot \alpha^i}{\alpha^{i\,2}}. \tag{8.49}$$

For $SU(3)$,

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{8.50}$$

For G_2,

$$A = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}. \tag{8.51}$$

8.8 Finding all the roots

When one goes from ϕ to $\phi + \alpha^j$, the coefficient k_j changes by one and therefore

$$k_j \to k_j + 1, \qquad q^i - p^i \to q^i - p^i + A_{ji}. \tag{8.52}$$

This leads to a graphical algorithm.

For $SU(3)$, one starts with the simple roots at level $k = 1$:

$$\begin{array}{llll} k=1 & \boxed{2\ -1} & \boxed{-1\ 2} & \alpha^1,\ \alpha^2 \end{array} \tag{8.53}$$

Then place the Cartan generators at $k = 0$:

$$\begin{array}{lcccl} k=1 & \boxed{2\ -1} & & \boxed{-1\ 2} & \alpha^1,\ \alpha^2 \\ \\ k=0 & & \boxed{0\ 0} & & H_i \end{array} \tag{8.54}$$

Now the q values are known:

$$\begin{array}{lcccl} q= & 2\quad 0 & & 0\quad 2 & \\ k=1 & \boxed{2\ -1} & & \boxed{-1\ 2} & \alpha^1,\ \alpha^2 \\ \\ k=0 & & \boxed{0\ 0} & & H_i \end{array} \tag{8.55}$$

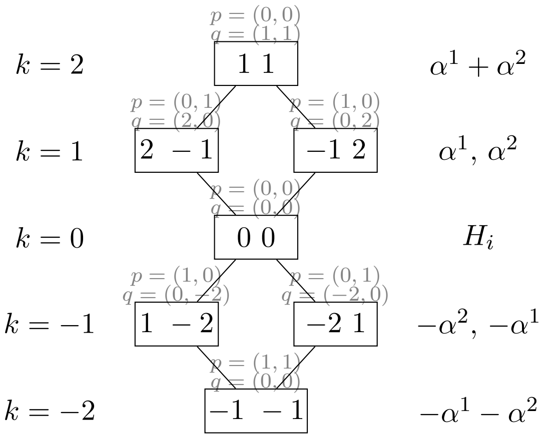
Hence the p values are

$$\begin{array}{lcccl} p= & 0\quad 1 & & 1\quad 0 & \\ k=1 & \boxed{2\ -1} & & \boxed{-1\ 2} & \alpha^1,\ \alpha^2 \\ \\ k=0 & & \boxed{0\ 0} & & H_i \end{array} \tag{8.56}$$

Now, as requested, the p and q values are placed above *every* box in the higher-level diagrams as well:

$$\begin{array}{lcccl} & & \begin{smallmatrix} p=(0,0) \\ q=(1,1) \end{smallmatrix} & & \\ k=2 & & \boxed{1\quad 1} & & \alpha^1+\alpha^2 \\ & \begin{smallmatrix} p=(0,1) \\ q=(2,0) \end{smallmatrix} & & \begin{smallmatrix} p=(1,0) \\ q=(0,2) \end{smallmatrix} & \\ k=1 & \boxed{2\ \ -1} & & \boxed{-1\ \ 2} & \alpha^1,\ \alpha^2 \\ & & \begin{smallmatrix} p=(1,1) \\ q=(1,1) \end{smallmatrix} & & \\ k=0 & & \boxed{0\quad 0} & & H_i \end{array} \tag{8.57}$$

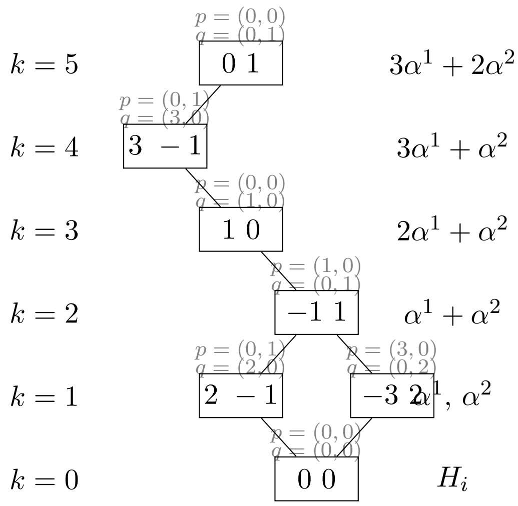
Continuing downward gives



(8.58)

8.9 The $SU(2)$s

The same information may be read directly in terms of the $SU(2)$ multiplets. For G_2 the positive-root construction is:



(8.59)

8.10 Constructing the G_2 algebra

The relevant raising operators are

$$E_1^+ = E_{\alpha^1}, \qquad E_2^+ = \frac{1}{\sqrt{3}} E_{\alpha^2}. \tag{8.60}$$

Start from $|E_{\alpha^2}\rangle$, viewed as the lowest weight state in a spin-3/2 representation of the α^1 $SU(2)$:

$$|E_{\alpha^2}\rangle = |3/2, -3/2, 1\rangle\,. \tag{8.61}$$

Applying the α^1 raising operator once gives

$$|[E_{\alpha^1}, E_{\alpha^2}]\rangle = |\alpha^1| E_1^+ \,|E_{\alpha^2}\rangle = \sqrt{3}\;|3/2, -1/2, 1\rangle = \sqrt{2}\;|E_{\alpha^1+\alpha^2}\rangle\,. \tag{8.62}$$

A second application gives

$$|[E_{\alpha^1}, [E_{\alpha^1}, E_{\alpha^2}]]\rangle = 2\sqrt{3}\;|3/2, 1/2, 1\rangle = \sqrt{6}\;|E_{2\alpha^1+\alpha^2}\rangle\,. \tag{8.63}$$

A third one yields

$$|[E_{\alpha^1},[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]]]\rangle = 3\,|E_{3\alpha^1+\alpha^2}\rangle = 3\,|1/2,-1/2,2\rangle\,. \tag{8.64}$$

Now act with the α^2 raising operator:

$$|[E_{\alpha^2},[E_{\alpha^1},[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]]]]\rangle = 3\sqrt{2}\,E_2^+\,|1/2,-1/2,2\rangle = \sqrt{6}\,\,|E_{3\alpha^1+2\alpha^2}\rangle\,. \tag{8.65}$$

Therefore the positive-root generators are

$$E_{\alpha^1+\alpha^2} = \frac{1}{\sqrt{2}}[E_{\alpha^1},E_{\alpha^2}], \qquad E_{2\alpha^1+\alpha^2} = \frac{1}{\sqrt{6}}[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]],$$

$$E_{3\alpha^1+\alpha^2} = \frac{1}{3}[E_{\alpha^1},[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]]], \qquad E_{3\alpha^1+2\alpha^2} = \frac{1}{\sqrt{6}}[E_{\alpha^2},[E_{\alpha^1},[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]]]]. \tag{8.66}$$

As a check,

$$[E_{-\alpha^1},E_{2\alpha^1+\alpha^2}] = \frac{1}{\sqrt{6}}[E_{-\alpha^1},[E_{\alpha^1},[E_{\alpha^1},E_{\alpha^2}]]] = \frac{2}{\sqrt{6}}[E_{\alpha^1},E_{\alpha^2}] = \sqrt{3}\,E_{\alpha^1+\alpha^2}. \tag{8.67}$$

8.11 Another example: the algebra C_3

Consider the Dynkin diagram

$$\circ - \circ \Rightarrow \circ \tag{8.68}$$

with Cartan matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}. \tag{8.70}$$

The positive roots may be organized by Dynkin coefficients as

$$(2,-1,0),\quad (-1,2,-2),\quad (0,-1,2),\quad (1,1,-2),\quad (-1,1,0),\quad (1,0,0),\quad (0,1,-2),\quad (1,-1,2),\quad (0,0,2) \tag{8.71}$$

This is the algebra C_3.

8.12 Fundamental weights

Let the simple roots be α^j, $j=1,\ldots,m$. A highest weight μ satisfies $\mu+\phi$ not a weight for any positive root ϕ. It is sufficient that $\mu+\alpha^j$ not be a weight for any j, because then

$$E_{\alpha^j}\,|\mu\rangle = 0 \qquad \text{for all } j. \tag{8.72}$$

Then the master formula implies

$$\frac{2\alpha^i\cdot\mu}{\alpha^{i\,2}} = \ell_i, \tag{8.73}$$

where the ℓ_i are nonnegative integers, the Dynkin coefficients. The fundamental weights μ_k are defined by

$$\frac{2\alpha^i\cdot\mu_k}{\alpha^{i\,2}} = \delta_{ik}. \tag{8.74}$$

Any highest weight can be written uniquely as

$$\mu = \sum_{j=1}^{m}\ell_j\mu_j. \tag{8.75}$$

For a general weight μ, the Dynkin coefficients are simply

$$\ell_i = q_i - p_i. \tag{8.76}$$

8.13 The trace of a generator

Theorem 8.9. The trace of any generator in any representation of a compact simple Lie group is zero.

Proof. It is enough to work in the standard weight basis. Every generator is a linear combination of Cartan generators and raising/lowering operators. Raising and lowering operators have vanishing trace because they have no diagonal matrix elements. The Cartan generators are linear combinations of $\alpha \cdot H$ for simple roots α, and each such operator is proportional to the E_3 of an $SU(2)$ subalgebra. Every $SU(2)$ representation is symmetric about zero, so the trace of E_3 vanishes. Hence the trace of every generator vanishes. □

Problems

8.A. Find the simple roots, fundamental weights, and Dynkin diagram for the algebra discussed in problem (6.C).

8.B. Consider the algebra generated by σ_a and $\sigma_a \tau_1$, where σ_a and τ_a are independent Pauli matrices. Show that this algebra generates a group which is semisimple but not simple. Nevertheless, simple roots can still be defined. What does the Dynkin diagram look like?

8.C. Consider the algebra corresponding to the Dynkin diagram

$$\circ\!\!\Longrightarrow\!\!\circ - \circ$$

where the lengths are similar to those in the C_3 example but differ in the relative normalization. Find the Cartan matrix and the Dynkin coefficients of all positive roots, using the diagrammatic construction of this chapter.

Chapter 9

More $SU(3)$

In this chapter we continue with $SU(3)$ and use it as the testing ground for general facts about irreducible representations of Lie algebras.

9.1 Fundamental representations of $SU(3)$

Label the two simple roots of $SU(3)$ by

$$\alpha^1 = (1/2, \sqrt{3}/2), \qquad \alpha^2 = (1/2, -\sqrt{3}/2). \tag{9.1}$$

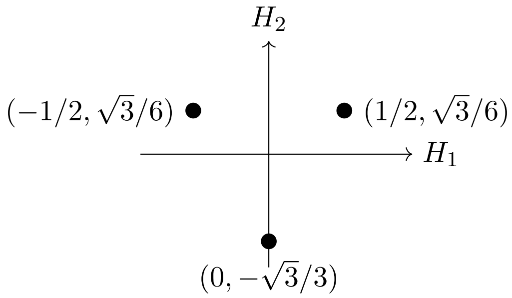
The fundamental weights μ^i are found from the condition that $\mu^i \cdot \alpha^j = 0$ when $i \neq j$. Writing

$$\mu^i = (a_i, b_i), \qquad \mu^i \cdot \alpha^j = (a_i \mp \sqrt{3} b_i)/2, \qquad \Rightarrow a_i = \pm\sqrt{3} b_i, \tag{9.2}$$

and using the normalization condition (8.74), one gets

$$\mu^1 = (1/2, \sqrt{3}/6), \qquad \mu^2 = (1/2, -\sqrt{3}/6). \tag{9.3}$$

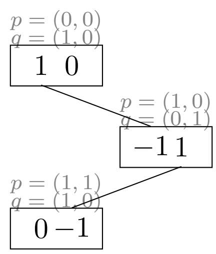
The first fundamental weight μ^1 is the highest weight of the defining representation:



The simple roots in the $q-p$ notation are

$$\boxed{2\ \ -1} \quad \boxed{-1\ 2} \tag{9.4}$$

The highest weight $(1, 0)$ then has the lowering diagram

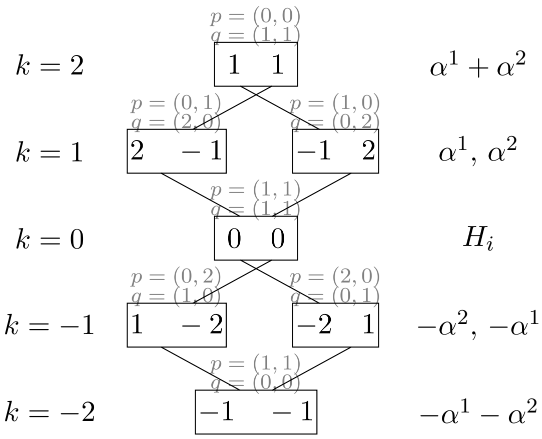


(9.5)

Thus the weights are

$$\mu^1, \qquad \mu^1 - \alpha^1, \qquad \mu^1 - \alpha^1 - \alpha^2. \tag{9.6}$$

The second fundamental representation is obtained similarly. Its $q-p$ diagram is



(9.7)

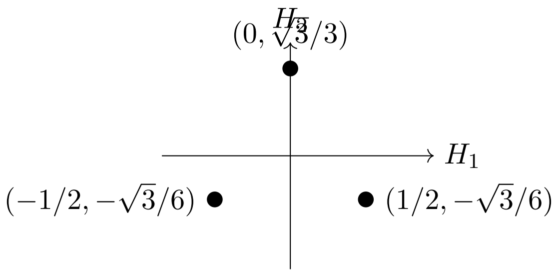
so the weights are

$$\mu^2, \qquad \mu^2 - \alpha^2, \qquad \mu^2 - \alpha^2 - \alpha^1, \tag{9.8}$$

or explicitly

$$(1/2, -\sqrt{3}/6), \qquad (0, \sqrt{3}/3), \qquad (-1/2, -\sqrt{3}/6). \tag{9.9}$$

In the H_i plane this representation is



(9.10)

9.2 Constructing the states

Let an irreducible representation have highest weight μ. Its states are generated by repeated action of root generators:

$$E_{\phi_1} E_{\phi_2} \cdots E_{\phi_k} \left|\mu\right\rangle, \tag{9.11}$$

where the ϕ_i are roots. Positive roots may be moved rightward using commutators until they annihilate the highest-weight state, so all states can be generated using simple lowering operators:

$$E_{-\alpha^{\beta_1}} E_{-\alpha^{\beta_2}} \cdots E_{-\alpha^{\beta_k}} \left|\mu\right\rangle, \tag{9.12}$$

where the α^{β_i} are simple roots.

The norm of such a state has the form

$$\left\langle\mu\right| E_{\alpha^{\beta_k}} \cdots E_{\alpha^{\beta_2}} E_{\alpha^{\beta_1}} E_{-\alpha^{\beta_1}} E_{-\alpha^{\beta_2}} \cdots E_{-\alpha^{\beta_k}} \left|\mu\right\rangle. \tag{9.13}$$

This is evaluated by commuting raising operators to the right until they hit $|\mu\rangle$. Matrix elements between states with the same set of lowering operators in different orders are similarly

$$\left\langle\mu\right| E_{\alpha^{\gamma_k}} \cdots E_{\alpha^{\gamma_2}} E_{\alpha^{\gamma_1}} E_{-\alpha^{\beta_1}} E_{-\alpha^{\beta_2}} \cdots E_{-\alpha^{\beta_k}} \left|\mu\right\rangle. \tag{9.14}$$

For two states $|A\rangle$ and $|B\rangle$ in the same weight space, linear dependence occurs when

$$\langle A|B|A|B\rangle \langle B|A|B|A\rangle = \langle A|A|A|A\rangle \langle B|B|B|B\rangle . \tag{9.15}$$

In that case one basis state may be chosen as

$$|A\rangle . \tag{9.16}$$

If instead

$$\langle A|B|A|B\rangle \langle B|A|B|A\rangle \neq \langle A|A|A|A\rangle \langle B|B|B|B\rangle , \tag{9.17}$$

then an orthonormal basis can be built by Gram-Schmidt, for example from

$$|A\rangle , \qquad \frac{|B\rangle \langle A|A|A|A\rangle - |A\rangle \langle A|B|A|B\rangle}{\sqrt{\langle A|A|A|A\rangle \left(\langle A|A|A|A\rangle \langle B|B|B|B\rangle - \langle A|B|A|B\rangle \langle B|A|B|A\rangle\right)}}. \tag{9.18}$$

9.3 The Weyl group

For every root direction there is an associated $SU(2)$, and every $SU(2)$ representation is symmetric under $E_3 \to -E_3$. If μ is a weight and $E_3 = \alpha \cdot H/\alpha^2$, then

$$\frac{2\alpha \cdot \mu}{\alpha^2} = q - p. \tag{9.19}$$

The reflected weight is

$$\mu - (q-p)\alpha,$$

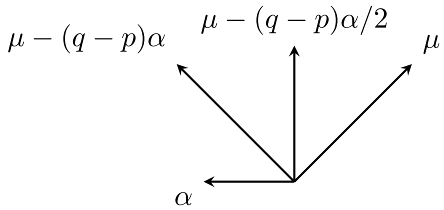
and it has the opposite E_3 value:

$$E_3 \,|\mu - (q-p)\alpha\rangle = -\frac{\alpha \cdot \mu}{\alpha^2} \,|\mu - (q-p)\alpha\rangle . \tag{9.20}$$

The midpoint between the two reflected weights lies in the hyperplane perpendicular to α:

$$\alpha \cdot \left(\mu - \frac{q-p}{2}\alpha\right) = 0. \tag{9.21}$$

The reflection is shown schematically as

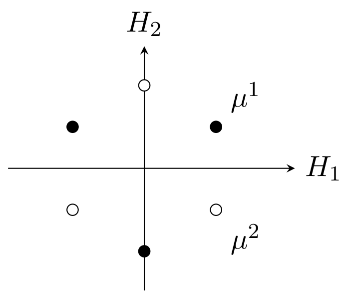


(9.22)

The transformations generated by these reflections form the Weyl group.

9.4 Complex conjugation

The weights of the second fundamental representation are the negatives of the weights of the first:



(9.23)

If T_a are the generators of a representation D, then the complex-conjugate matrices obey the same Lie algebra after inserting a minus sign:

$$\begin{aligned} ([T_a, T_b])^* &= (if_{abc}T_c)^*, \\ [T_a^*, T_b^*] &= -if_{abc}T_c^*, \\ [-T_a^*, -T_b^*] &= if_{abc}(-T_c^*). \end{aligned} \tag{9.24}$$

Thus $-T_a^*$ generate the complex conjugate representation. For $SU(3)$,

$$\overline{(1,0)} = (0,1), \qquad \overline{(0,1)} = (1,0). \tag{9.25}$$

A common shorthand is

$$(1,0) = 3, \qquad (0,1) = \bar{3}. \tag{9.26}$$

More generally,

$$\begin{array}{lll} (n,m) & \text{has highest weight} & n\mu^1 + m\mu^2, \\ (n,m) & \text{has lowest weight} & -n\mu^2 - m\mu^1, \\ \Rightarrow (m,n) & \text{has highest weight} & n\mu^2 + m\mu^1. \end{array} \tag{9.27}$$

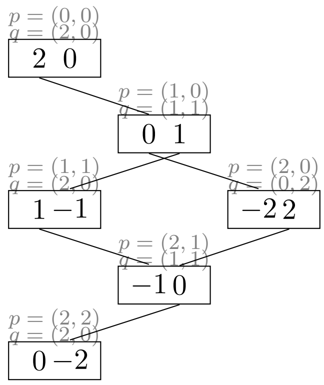
Representations (n,n) are real.

9.5 Examples of other representations

The representation $(2,0)$ has highest weight

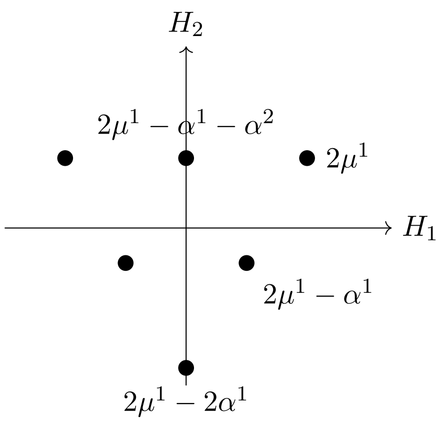
$$2\mu^1. \tag{9.28}$$

In $q-p$ notation it is



(9.29)

and in Cartan space:

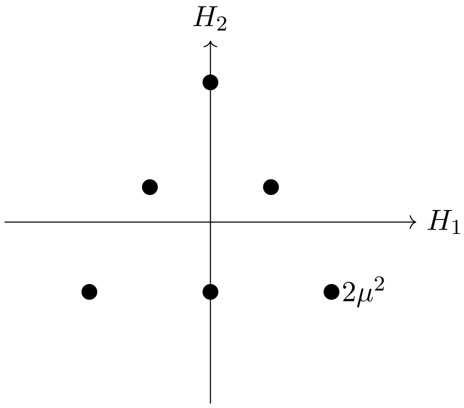


(9.30)

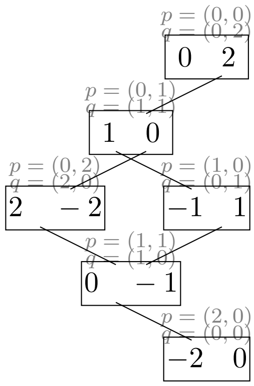
This is the six-dimensional representation, often denoted

$$(2,0) = 6.$$

Its complex conjugate $(0,2) = \bar{6}$ has the reflected diagrams:



(9.31)



(9.32)

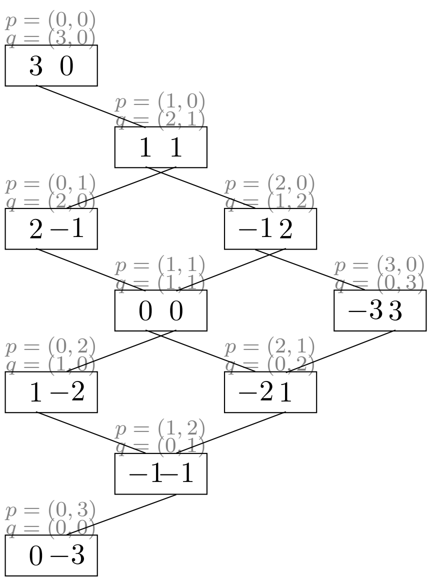
Now consider $(1,1)$. The highest weight is

$$\mu^1 + \mu^2 = \alpha^1 + \alpha^2. \tag{9.33}$$

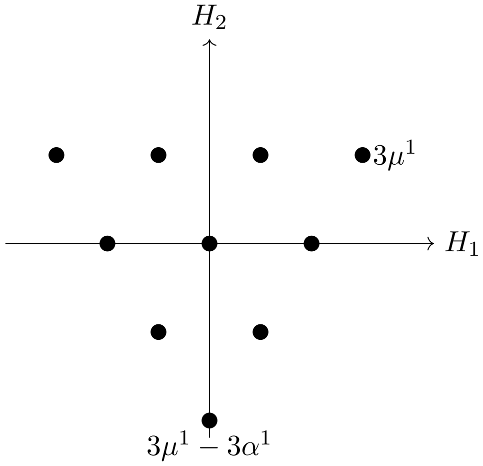
This is the adjoint representation. The zero weight is doubly degenerate. The two ways of lowering to zero give different states:

$$E_{-\alpha^1} E_{-\alpha^2} \left|\mu^1 + \mu^2\right\rangle \not\propto E_{-\alpha^2} E_{-\alpha^1} \left|\mu^1 + \mu^2\right\rangle. \tag{9.34}$$

Next, consider $(3,0)$ with highest weight $3\mu^1$



The weight diagram is



(9.36)

The potentially ambiguous central state can be reached by

$$E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle, \qquad E_{-\alpha^2}E_{-\alpha^1}E_{-\alpha^1}\left|3\mu^1\right\rangle. \tag{9.37}$$

These are linearly dependent:

$$\begin{aligned} E_{-\alpha^2}E_{-\alpha^1}E_{-\alpha^1}\left|3\mu^1\right\rangle &= [E_{-\alpha^2}, E_{-\alpha^1}]E_{-\alpha^1}\left|3\mu^1\right\rangle + E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle \\ &= E_{-\alpha^1}[E_{-\alpha^2}, E_{-\alpha^1}]\left|3\mu^1\right\rangle + E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle \\ &= E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle + E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle \\ &= 2E_{-\alpha^1}E_{-\alpha^2}E_{-\alpha^1}\left|3\mu^1\right\rangle. \end{aligned} \tag{9.38}$$

The first simplification uses

$$-2\alpha^1 - \alpha^2 \text{ is not a root,} \tag{9.39}$$

and the second uses

$$3\mu^1 - \alpha^2 \text{ is not a weight.} \tag{9.40}$$

Thus $(3,0)$ is ten-dimensional, often denoted 10, and its complex conjugate $(0,3)$ is denoted $\overline{10}$.

Problems

9.A. If $|\mu\rangle$ is the state of highest weight $\mu = \mu^1 + \mu^2$ of the adjoint representation of $SU(3)$, show that the states

$$|A\rangle = E_{-\alpha^1}E_{-\alpha^2}|\mu\rangle, \qquad |B\rangle = E_{-\alpha^2}E_{-\alpha^1}|\mu\rangle$$

are linearly independent. Hint: calculate $\langle A|A|A|A\rangle$, $\langle A|B|A|B\rangle$, and $\langle B|B|B|B\rangle$, and use the condition

$$\langle A|A|A|A\rangle\,\langle B|B|B|B\rangle = \langle A|B|A|B\rangle\,\langle B|A|B|A\rangle\,.$$

9.B. Consider the matrices in the six-dimensional tensor-product space of the $SU(3)$ λ_a matrices and the Pauli matrices σ_a:

$$\frac{1}{2}\lambda_a\sigma_2 \quad \text{for } a = 1,3,4,6,8, \qquad \frac{1}{2}\lambda_a \quad \text{for } a = 2,5,7.$$

Show that these generate a reducible representation and reduce it.

9.C. Decompose the tensor product 3×3 using highest-weight techniques.

Chapter 10

Tensor Methods

This chapter develops tensor notation for $SU(3)$ representations. The central point is that tensor components are wave functions for states in tensor-product representation spaces. Irreducible representations are obtained by imposing symmetry and tracelessness conditions on these tensors.

10.1 Lower and upper indices

The states of the defining **3** representation are labeled by lower indices:

$$\begin{aligned}\left|1/2, \sqrt{3}/6\right\rangle &= |_1\rangle \equiv \mu^1, \\ \left|-1/2, \sqrt{3}/6\right\rangle &= |_2\rangle \,, \\ \left|0, -1/\sqrt{3}\right\rangle &= |_3\rangle \,.\end{aligned} \tag{10.1}$$

Here $1, 2, 3$ label the eigenvectors of H_1, H_2 with one nonzero entry in the corresponding position. Define matrices with one upper and one lower index by

$$[T_a]^i{}_j = \frac{1}{2}[\lambda_a]_{ij}. \tag{10.2}$$

Then the triplet states transform as

$$T_a \,|_i\rangle = |_j\rangle \,[T_a]^j{}_i. \tag{10.3}$$

The summed index is one upper and one lower index.

The states of the conjugate $\overline{\mathbf{3}}$ are labeled with upper indices:

$$\begin{aligned}\left|-1/2, -\sqrt{3}/6\right\rangle &= \left|^1\right\rangle , \\ \left|1/2, -\sqrt{3}/6\right\rangle &= \left|^2\right\rangle \equiv \mu^2, \\ \left|0, 1/\sqrt{3}\right\rangle &= \left|^3\right\rangle .\end{aligned} \tag{10.4}$$

They transform as

$$T_a \left|^i\right\rangle = -[T_a]^i{}_j \left|^j\right\rangle . \tag{10.5}$$

This follows because the conjugate representation is generated by

$$[T_a]^i{}_j \longrightarrow -[T_a^*]^i{}_j = -[T_a^T]^i{}_j. \tag{10.6}$$

A general tensor-product basis state with m lower-index triplets and n upper-index antitriplets is

$$\left|\begin{smallmatrix} i_1\cdots i_m \\ j_1\cdots j_n \end{smallmatrix}\right\rangle = \left|{}^{i_1}\right\rangle \cdots \left|{}^{i_m}\right\rangle \left|{}_{j_1}\right\rangle \cdots \left|{}_{j_n}\right\rangle . \tag{10.7}$$

It transforms as (this is the $SU(3)$ version of (3.46))

$$\begin{aligned} T_a \left|\begin{smallmatrix} i_1\cdots i_m \\ j_1\cdots j_n \end{smallmatrix}\right\rangle = & \sum_{\ell=1}^{m} \left|\begin{smallmatrix} i_1\cdots i_m \\ j_1\cdots j_\ell k j_{\ell+1}\cdots j_n \end{smallmatrix}\right\rangle [T_a]^k{}_{j_\ell} \\ & - \sum_{\ell=1}^{n} \left|\begin{smallmatrix} i_1\cdots i_{\ell-1} k i_{\ell+1}\cdots i_m \\ j_1\cdots j_n \end{smallmatrix}\right\rangle [T_a]^{i_\ell}{}_k . \end{aligned} \tag{10.8}$$

The convention distinguishes the two inequivalent three-dimensional representations of $SU(3)$.

10.2 Tensor components and wave functions

An arbitrary state in the tensor product is expanded as linear combination of states in (10.7)

$$|v\rangle = v^{j_1\cdots j_n}_{i_1\cdots i_m} \left|\begin{smallmatrix} i_1\cdots i_m \\ j_1\cdots j_n \end{smallmatrix}\right\rangle . \tag{10.9}$$

The tensor components are matrix elements of the state:

$$v^{j_1\cdots j_n}_{i_1\cdots i_m} = \left\langle\begin{smallmatrix} i_1\cdots i_m \\ j_1\cdots j_n \end{smallmatrix}\middle| v\right\rangle . \tag{10.10}$$

This is the same relation as the usual wave-function relation

$$\psi(x) = \langle x|\psi\rangle . \tag{10.11}$$

The action of generators on the state can be transferred to an action on tensor components:

$$T_a\,|v\rangle = |T_a v\rangle , \tag{10.12}$$

where

$$(T_a v)^{j_1\cdots j_n}_{i_1\cdots i_m} = \sum_{\ell=1}^{m} [T_a]^{j_\ell}{}_k v^{j_1\cdots j_{\ell-1} k j_{\ell+1}\cdots j_n}_{i_1\cdots i_m} \sum_{\ell=1}^{n} [T_a]^k_{i_\ell} v^{j_1\cdots j_n}_{i_1\cdots i_{\ell-1} k i_{\ell+1}\cdots i_m} . \tag{10.13}$$

10.3 Irreducible representations and symmetry

The highest-weight state of the irreducible representation (n, m) is obtained from n copies of the $\mathbf{3}$ highest state and m copies of the $\overline{\mathbf{3}}$ highest state:

$$\left|\begin{smallmatrix} 22\cdots 2 \\ 11\cdots 1 \end{smallmatrix}\right\rangle . \tag{10.14}$$

The corresponding tensor may be written

$$(v_H)^{j_1\cdots j_n}_{i_1\cdots i_m} = N\,\delta^{j_1}_1 \cdots \delta^{j_n}_1 \delta^2_{i_1} \cdots \delta^2_{i_m} . \tag{10.15}$$

All states in (n, m) are generated by lowering operators (constructed out of T_a). The tensor components remain symmetric separately in all upper indices and in all lower indices, and they remain traceless:

$$\delta^{i_1}_{j_1} v^{j_1\cdots j_n}_{i_1\cdots i_m} = 0. \tag{10.16}$$

Any state created by lowering operators respect these properties. Thus the irreducible representation (n, m) is represented by tensors with n upper and m lower indices, symmetric separately in both index sets, and traceless under any upper-lower contraction.

10.4 Invariant tensors

The Kronecker delta is invariant:

$$[T_a]^i{}_k \delta^k{}_j - [T_a]^k{}_j \delta^i{}_k = 0. \tag{10.17}$$

The other basic invariant tensors in $SU(3)$ are the completely antisymmetric tensors ϵ^{ijk} and ϵ_{ijk}. For example,

$$[T_a \epsilon]^{ijk} = [T_a]^i{}_\ell \epsilon^{\ell jk} + [T_a]^j{}_\ell \epsilon^{i\ell k} + [T_a]^k{}_\ell \epsilon^{ij\ell}. \tag{10.18}$$

For the 123 component,

$$\begin{aligned} [T_a \epsilon]^{123} &= [T_a]^1{}_\ell \epsilon^{\ell 23} + [T_a]^2{}_\ell \epsilon^{1\ell 3} + [T_a]^3{}_\ell \epsilon^{12\ell} \\ &= ([T_a]^1{}_1 + [T_a]^2{}_2 + [T_a]^3{}_3)\epsilon^{123} = 0, \end{aligned} \tag{10.19}$$

because the generators are traceless.

10.5 Clebsch–Gordan decomposition

If u has n upper and m lower indices and v has p upper and q lower indices, then their tensor-product state is represented by the product tensor

$$(u \otimes v)^{i_1 \cdots i_n k_1 \cdots k_p}{}_{j_1 \cdots j_m \ell_1 \cdots \ell_q} = u^{i_1 \cdots i_n}{}_{j_1 \cdots j_m} v^{k_1 \cdots k_p}{}_{\ell_1 \cdots \ell_q}. \tag{10.20}$$

For two triplets, u^i and v^i,

$$u^i v^j = \frac{1}{2}(u^i v^j + u^j v^i) + \frac{1}{2}(u^i v^j - u^j v^i). \tag{10.21}$$

The symmetric term

$$\frac{1}{2}(u^i v^j + u^j v^i) \tag{10.22}$$

transforms as a **6**, while the antisymmetric term can be written with a lower index:

$$\epsilon_{kij} u^i v^j. \tag{10.23}$$

Thus

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \overline{\mathbf{3}}, \tag{10.24}$$

or in Dynkin labels,

$$(1,0) \otimes (1,0) = (2,0) \oplus (0,1). \tag{10.25}$$

For $\mathbf{3} \otimes \overline{\mathbf{3}}$,

$$u^i v_j = \left(u^i v_j - \frac{1}{3}\delta^i{}_j u^k v_k \right) + \frac{1}{3}\delta^i{}_j u^k v_k. \tag{10.26}$$

The first term is traceless and transforms as the adjoint **8**; the second is a singlet. Hence

$$\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \tag{10.27}$$

that is,

$$(1,0) \otimes (0,1) = (1,1) \oplus (0,0). \tag{10.28}$$

For $\mathbf{3} \otimes \mathbf{8}$, with u^i a triplet and $v^j{}_k$ traceless,

$$u^i v^j{}_k = \frac{1}{2}\left(u^i v^j{}_k + u^j v^i{}_k - \frac{1}{4}\delta^i{}_k u^\ell v^j{}_\ell - \frac{1}{4}\delta^j{}_k u^\ell v^i{}_\ell \right)$$

$$+\frac{1}{4}\epsilon^{ij\ell}\left(\epsilon_{\ell mn}u^m v^n{}_k + \epsilon_{kmn}u^m v^n{}_\ell\right)$$
$$+\frac{1}{4}\left(3\delta^i{}_k u^\ell v^j{}_\ell - \delta^j{}_k u^\ell v^i{}_\ell\right). \tag{10.29}$$

This gives

$$\mathbf{3}\otimes\mathbf{8} = \mathbf{15}\oplus\overline{\mathbf{6}}\oplus\mathbf{3}, \tag{10.30}$$

or

$$(1,0)\otimes(1,1) = (2,1)\oplus(0,2)\oplus(1,0). \tag{10.31}$$

$$\mathbf{3} = (1,0), \qquad \mathbf{8} = (1,1)$$

The weights of **3** are

$$(1,0), \qquad (-1,1), \qquad (0,-1)$$

The weights of **8** are

$$(1,1), \quad (-1,2), \quad (2,-1), \quad (0,0)_2, \quad (-2,1), \quad (1,-2), \quad (-1,-1)$$

where

$$(0,0)_2$$

means that the zero weight has multiplicity 2.

Add every weight of 3 to every weight of 8

$\mathbf{w_3} = (1,0)$	$\mathbf{w_3} + \mathbf{w_8}$
	$(2,1),\ (0,2),\ (3,-1),\ (1,0)_2,$ $(-1,1),\ (2,-2),\ (0,-1)$

$\mathbf{w_3} = (-1,1)$	$\mathbf{w_3} + \mathbf{w_8}$
	$(0,2),\ (-2,3),\ (1,0),\ (-1,1)_2,$ $(-3,2),\ (0,-1),\ (-2,0)$

$\mathbf{w_3} = (0,-1)$	$\mathbf{w_3} + \mathbf{w_8}$
	$(1,0),\ (-1,1),\ (2,-2),\ (0,-1)_2,$ $(-2,0),\ (1,-3),\ (-1,-2)$

where in every block

$$\mathbf{w_8} = (1,1),\ (-1,2),\ (2,-1),\ (0,0)_2,\ (-2,1),\ (1,-2),\ (-1,-1).$$

Collecting equal weights,

weight	multiplicity
$(2,1)$	1
$(-2,3)$	1
$(-3,2)$	1
$(0,2)$	2
$(-1,1)$	4
$(1,0)$	4
$(0,-1)$	4
$(-2,0)$	2
$(2,-2)$	2
$(3,-1)$	1
$(-1,-2)$	1
$(1,-3)$	1

First highest weight: $(2,1)$

The highest remaining weight is

$$(2,1)$$

Hence the first irrep is

$$\boxed{(2,1) = \mathbf{15}}$$

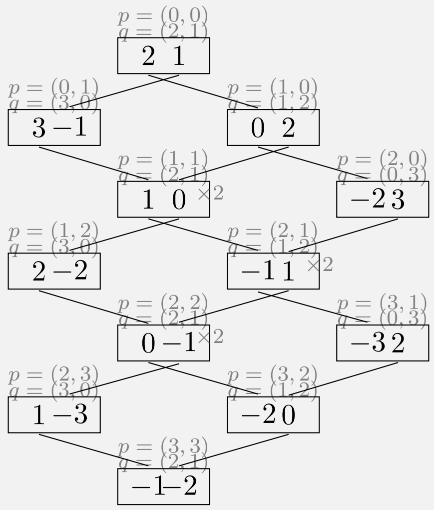
Its dimension is

$$\dim(2,1) = \frac{1}{2}(3)(2)(5) = 15$$

The weights of $(2,1)$ are

weight	multiplicity in **15**
$(2,1)$	1
$(-2,3)$	1
$(-3,2)$	1
$(0,2)$	1
$(-1,1)$	2
$(1,0)$	2
$(0,-1)$	2
$(-2,0)$	1
$(2,-2)$	1
$(3,-1)$	1
$(-1,-2)$	1
$(1,-3)$	1

A ladder diagram for $(2,1) = \mathbf{15}$, with multiplicities shown on the repeated weights, is



Subtract these multiplicities from the product:

weight	remaining multiplicity
$(0,2)$	1
$(-1,1)$	2
$(1,0)$	2
$(0,-1)$	2
$(-2,0)$	1
$(2,-2)$	1

Second highest weight: $(0,2)$

The highest remaining weight is

$$(0,2)$$

Hence

$$\boxed{(0,2) = \overline{\mathbf{6}}}$$

Its six weights are

$$(0,2), \quad (-1,1), \quad (-2,0), \quad (1,0), \quad (0,-1), \quad (2,-2)$$

Subtracting one copy of $\overline{\mathbf{6}}$ leaves

$$(-1,1), \qquad (1,0), \qquad (0,-1)$$

These are precisely the three weights of

$$\boxed{(1,0) = \mathbf{3}}$$

Therefore

$$\boxed{\mathbf{3} \otimes \mathbf{8} = \mathbf{15} \oplus \overline{\mathbf{6}} \oplus \mathbf{3}}$$

or in Dynkin labels,

$$\boxed{(1,0) \otimes (1,1) = (2,1) \oplus (0,2) \oplus (1,0)}$$

10.6 Triality

The quantity

$$n - m \pmod 3$$

is conserved in $SU(3)$ tensor products, because δ and ϵ tensors carry triality zero. This conserved quantity is called *triality*.

10.7 Matrix elements and operators

The bra corresponding to a ket tensor is

$$\langle v| = v^{*\,i_1\cdots i_m}{}_{j_1\cdots j_n}{}^{j_1\cdots j_n}\langle i_1\cdots i_m| \,. \tag{10.32}$$

For instance, a triplet bra transforms as

$$\langle i| \, T_a = [T_a]^i{}_j \, \langle j| \,. \tag{10.33}$$

It is convenient to define the tensor components of the bra with upper and lower indices interchanged:

$$v_{i_1\cdots i_n}{}^{j_1\cdots j_m} \equiv \left(v^{i_1\cdots i_m}{}_{j_1\cdots j_n}\right)^* . \tag{10.34}$$

Then

$$\langle v| = v_{i_1\cdots i_n}{}^{j_1\cdots j_m i_1\cdots i_n}\langle j_1\cdots j_m| \,. \tag{10.35}$$

A typical invariant matrix element is a full contraction, for example

$$\langle u| \, W \, |v\rangle = u_{i_1\cdots i_n}{}^{j_1\cdots j_m} W^{i_1\cdots i_n k_1\cdots k_p}{}_{j_1\cdots j_m \ell_1\cdots \ell_q} v^{\ell_1\cdots \ell_q}{}_{k_1\cdots k_p} . \tag{10.36}$$

10.8 Normalization

A normalized state satisfies

$$v_{i_1\cdots i_m}{}^{j_1\cdots j_n} v^{i_1\cdots i_m}{}_{j_1\cdots j_n} = 1. \tag{10.37}$$

For example, a tensor of the form

$$v^i{}_j = N\left(3\delta^i{}_1\delta^1{}_j - \delta^i{}_j\right) \tag{10.38}$$

since $\delta^i{}_j = \delta^i{}_1\delta^1{}_j + \delta^i{}_2\delta^2{}_j + \delta^i{}_3\delta^3{}_j$, it has norm

$$\begin{aligned}
\langle v|v\rangle &= \sum_{j,k}\left|N\left(2\delta^j_1\delta^1_k - \delta^j_2\delta^2_k - \delta^j_3\delta^3_k\right)\right|^2 \\
&= |N|^2\sum_{j=1}^{3}\sum_{k=1}^{3}\left(2\delta^j_1\delta^1_k - \delta^j_2\delta^2_k - \delta^j_3\delta^3_k\right)^2 \\
&= |N|^2\sum_{j=1}^{3}\sum_{k=1}^{3}\left(4\delta^j_1\delta^1_k\delta^j_1\delta^1_k + \delta^j_2\delta^2_k\delta^j_2\delta^2_k + \delta^j_3\delta^3_k\delta^j_3\delta^3_k - 4\delta^j_1\delta^1_k\delta^j_2\delta^2_k - 4\delta^j_1\delta^1_k\delta^j_3\delta^3_k + 2\delta^j_2\delta^2_k\delta^j_3\delta^3_k\right) \\
&= |N|^2\sum_{j=1}^{3}\sum_{k=1}^{3}\left(4\delta^j_1\delta^1_k + \delta^j_2\delta^2_k + \delta^j_3\delta^3_k\right) \quad (\text{since } \delta^j_a\delta^a_k\delta^j_b\delta^b_k = 0 \text{ for } a \neq b) \\
&= |N|^2\left(\sum_{j,k}4\delta^j_1\delta^1_k + \sum_{j,k}\delta^j_2\delta^2_k + \sum_{j,k}\delta^j_3\delta^3_k\right) \\
&= |N|^2\left(4\overbrace{(1\cdot 1)}^{j=k=1} + 1\overbrace{(1\cdot 1)}^{j=k=2} + 1\overbrace{(1\cdot 1)}^{j=k=3}\right) \\
&= |N|^2(4+1+1) \\
&= 6|N|^2
\end{aligned} \tag{10.39}$$

10.9 Tensor operators

If a set of operators transforms as (n, m), a general tensor operator can be written

$$W = W^{i_1\cdots i_n}{}_{j_1\cdots j_m} O_{i_1\cdots i_n}{}^{j_1\cdots j_m}. \tag{10.40}$$

Under commutation with generators, the coefficients transform according to the same tensor rule (10.13).

10.10 The dimension of (n, m)

A completely symmetric object with n indices, each running from 1 to 3, has

$$\binom{n+2}{2} = \frac{(n+2)!}{n!2!} = \frac{(n+2)(n+1)}{2} \tag{10.41}$$

independent components. Before imposing tracelessness,

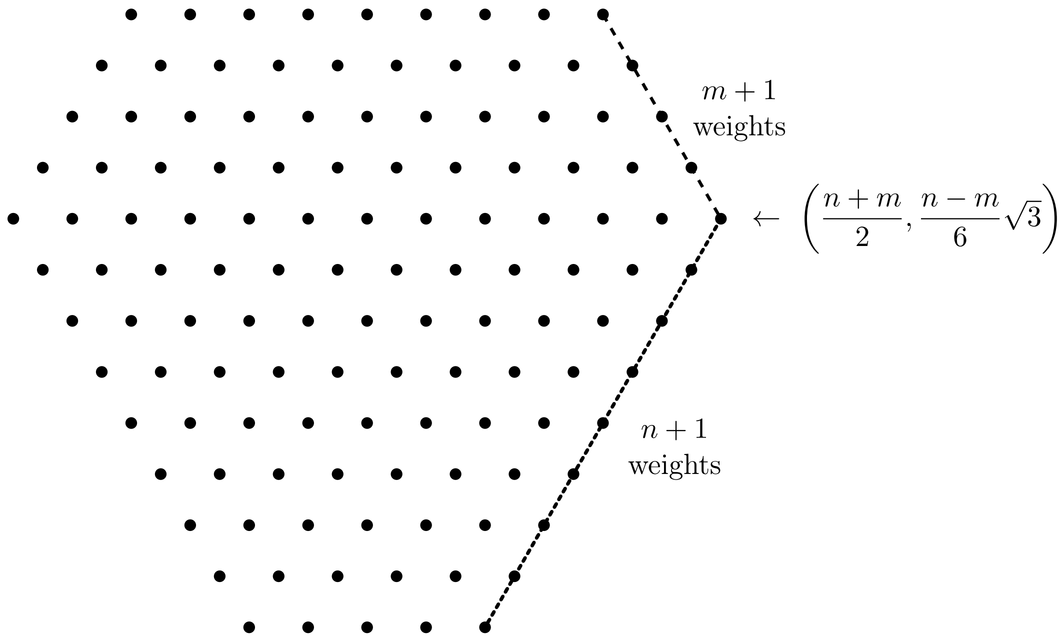
$$B(n, m) = \frac{(n+2)(n+1)}{2}\frac{(m+2)(m+1)}{2}. \tag{10.42}$$

The trace is symmetric in $n-1$ upper and $m-1$ lower indices, so

$$\begin{aligned} D(n,m) &= B(n,m) - B(n-1,m-1) \\ &= \frac{(n+2)(n+1)(m+2)(m+1)}{4} - \frac{(n+1)n(m+1)m}{4} \\ &= \frac{(n+1)(m+1)}{4}\big[(n+2)(m+2) - nm\big] \\ &= \frac{(n+1)(m+1)(n+m+2)}{2}. \end{aligned} \tag{10.43}$$

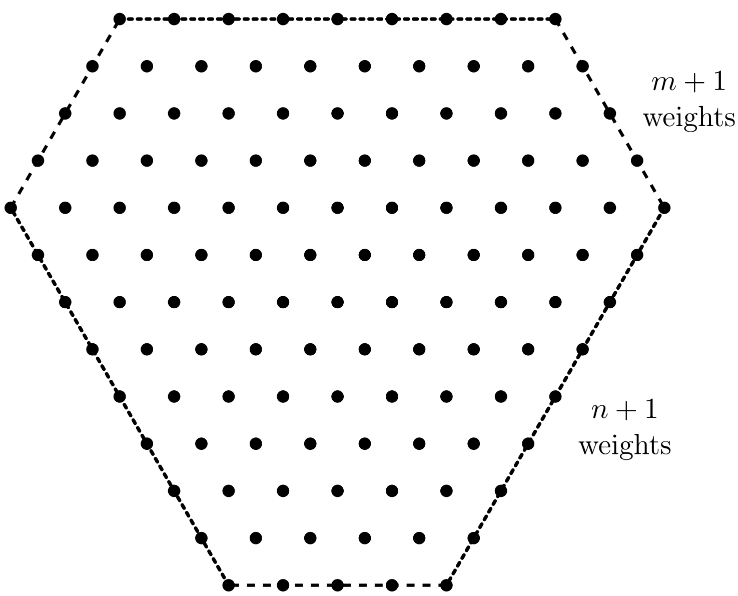
10.11 *The weights of (n, m)

The weights of a general $SU(3)$ representation (n, m) form a hexagonal pattern, degenerating to a triangle when $n = 0$ or $m = 0$. For example, for $n = 8, m = 4$:



(10.44)

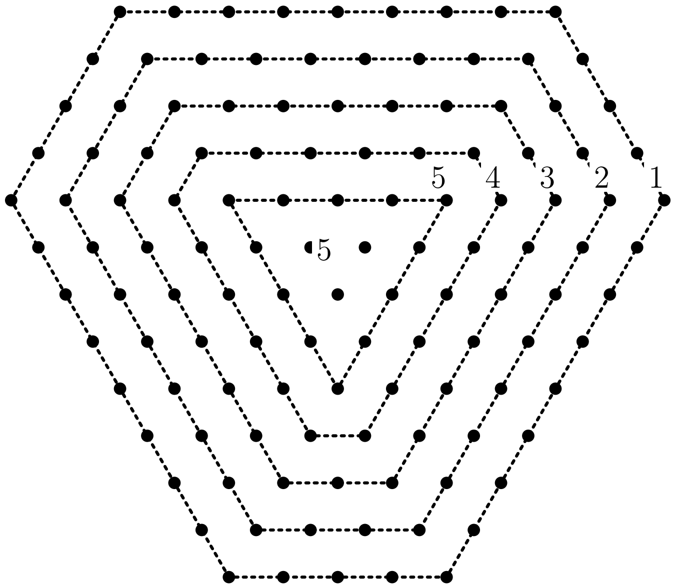
The highest weight is at one corner. One side is generated by repeated action of $E_{-\alpha^1}$ and the adjacent side by repeated action of $E_{-\alpha^2}$. Weyl reflections make the corresponding opposite sides equivalent:



(10.45)

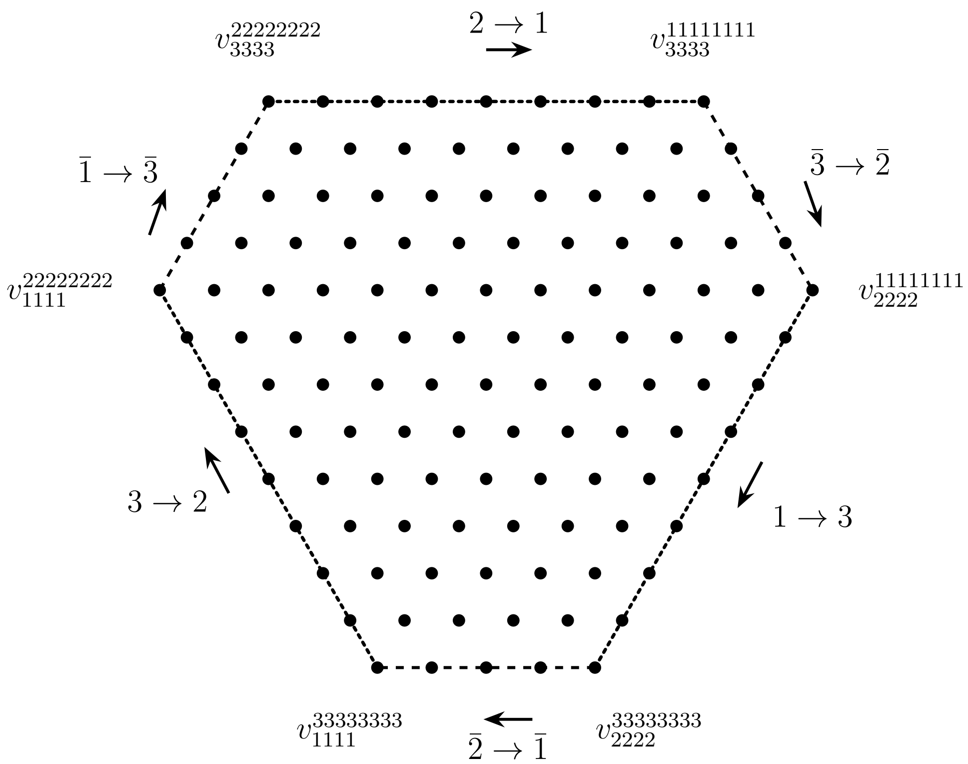
Theorem 10.10. Moving from the outer layer inward, the degeneracy of the weight space increases by one at each layer until a triangular layer is reached. Thereafter the degeneracy stays constant.

For $n = 8, m = 4$, the layer degeneracies are schematically



(10.46)

The outer layer can be described by how indices change around the diagram:



[H] (10.47)

Let $z(j)$ denote the number of upper indices equal to j, and $\bar{z}(j)$ the number of lower indices equal to j. Then the differences

$$z(1) - \bar{z}(1), \qquad z(2) - \bar{z}(2), \qquad z(3) - \bar{z}(3) \tag{10.48}$$

are fixed by the weight and by $n - m$. The components of the weight are

$$\mu_1 = \frac{1}{2}[z(1) - \bar{z}(1)] - rac12[z(2) - \bar{z}(2)], \tag{10.49}$$

$$\mu_2 = \frac{\sqrt{3}}{6}[z(1) - \bar{z}(1)] + \frac{\sqrt{3}}{6}[z(2) - \bar{z}(2)] - \frac{\sqrt{3}}{3}[z(3) - \bar{z}(3)], \tag{10.50}$$

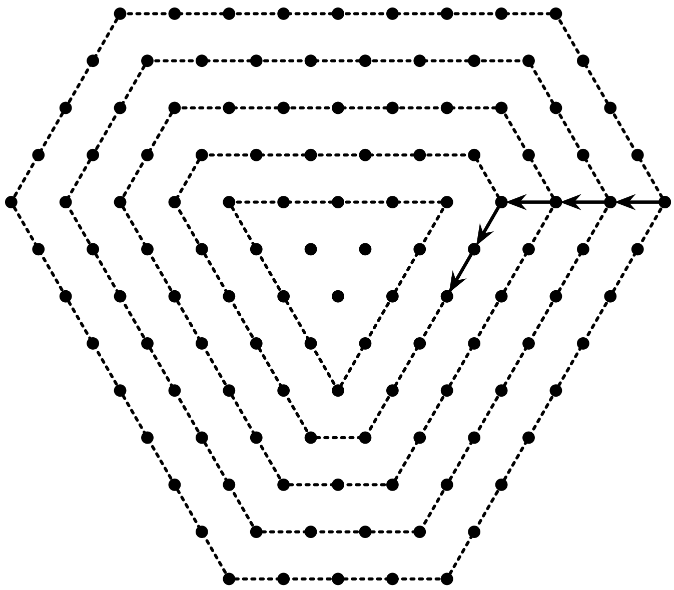
and

$$n - m = [z(1) - \bar{z}(1)] + [z(2) - \bar{z}(2)] + [z(3) - \bar{z}(3)]. \tag{10.51}$$

Solving,

$$\begin{aligned} z(1) - \bar{z}(1) &= \frac{1}{6}[6\mu_1 + 2\sqrt{3}\mu_2 + 2(n-m)], \\ z(2) - \bar{z}(2) &= \frac{1}{6}[-6\mu_1 + 2\sqrt{3}\mu_2 + 2(n-m)], \\ z(3) - \bar{z}(3) &= \frac{1}{3}[(n-m) - 2\sqrt{3}\mu_2]. \end{aligned} \tag{10.52}$$

For a point k layers inward and j steps downward along that layer,



(10.53)

one obtains

$$\begin{aligned} z(1) - \bar{z}(1) &= n - k - j, \\ z(2) - \bar{z}(2) &= -(m - k), \\ z(3) - \bar{z}(3) &= j. \end{aligned} \tag{10.54}$$

Hence a tensor component can be written with

$$\begin{gathered} n-k-j \text{ 1s}, \qquad m-k \text{ 2s}, \qquad j \text{ 3s}, \\ \text{and } k\ \ell\bar{\ell} \text{ pairs, where } \ell = 1,2,3. \end{gathered} \tag{10.55}$$

The number of such components is

$$\binom{k+2}{2}. \tag{10.56}$$

The trace contributes the same count with $k \to k-1$, so the traceless tensor has

$$\binom{k+2}{2} - \binom{k+1}{2} = \frac{(k+2)(k+1)}{2} - \frac{(k+1)k}{2} = k+1 \tag{10.57}$$

independent components at that weight. For $k > m$, the form changes to

$$\begin{gathered} n-k-j \text{ 1s}, \qquad k-m \text{ 2s}, \qquad j \text{ 3s}, \\ \text{and } m\ \ell\bar{\ell} \text{ pairs, where } \ell = 1,2,3. \end{gathered} \tag{10.58}$$

so the degeneracy remains $m+1$ thereafter. This proves theorem 10.10.

10.12 Generalization of Wigner–Eckart

Consider a matrix element of a tensor operator between tensor states:

$$\langle u | W | v \rangle . \tag{10.59}$$

Let the tensor components of v be

$$v^{i_1 \cdots i_n}{}_{j_1 \cdots j_m}, \tag{10.60}$$

those of W be

$$W^{k_1 \cdots k_p}{}_{\ell_1 \cdots \ell_q}, \tag{10.61}$$

and those of u be

$$u_{r_1 \cdots r_s}{}^{t_1 \cdots t_u}. \tag{10.62}$$

The full matrix element must be an invariant tensor:

$$u\, W\, v \quad \text{with all indices contracted using } \delta \text{ and } \epsilon. \tag{10.63}$$

For the case in which u, W, v are all adjoints, treat them as traceless matrices:

$$u^i{}_j, W^i{}_j, v^i{}_j. \tag{10.64}$$

There are two independent contractions,

$$\mathrm{Tr}(uWv), \qquad \mathrm{Tr}(uvW). \tag{10.65}$$

Thus

$$\langle u | W | v \rangle = \lambda_1 \,\mathrm{Tr}(uWv) + \lambda_2 \,\mathrm{Tr}(uvW). \tag{10.66}$$

For u and v in the **10** and W in the **8**, the tensor components are

$$v^{ijk}, \tag{10.67}$$

$$W^i{}_k, \tag{10.68}$$

and

$$u_{ijk}. \tag{10.69}$$

The only independent contraction is

$$u_{ijk} W^i{}_\ell v^{\ell jk}. \tag{10.70}$$

Therefore

$$\langle u | W | v \rangle = \lambda\, u_{ijk} W^i{}_\ell v^{\ell jk}. \tag{10.71}$$

10.13 *Tensors for $SU(2)$

For $SU(2)$ there is one fundamental doublet, and irreducible tensors are completely symmetric tensors with upper indices. A spin-s representation corresponds to a symmetric tensor with $2s$ indices. Define

$$v_{s,m}^{i_1 \cdots i_{2s}} = 1 \quad \text{if exactly } s+m \text{ indices are 1}, \quad 0 \text{ otherwise}. \tag{10.72}$$

There are

$$\binom{2s}{s+m} = \frac{(2s)!}{(s+m)!(s-m)!} \tag{10.73}$$

such nonzero components. For example,

$$v_{1,0}^{ij} = \delta_1^i \delta_2^j + \delta_2^i \delta_1^j. \tag{10.74}$$

The normalized state is

$$|s, m\rangle = \binom{2s}{s+m}^{-1/2} |v_{s,m}\rangle . \tag{10.75}$$

10.14 *Clebsch–Gordan coefficients from tensors

Consider the tensor product

$$v^{i_1\cdots i_{2s_1}}_{s_1,m_1} v^{j_1\cdots j_{2s_2}}_{s_2,m_2}. \tag{10.76}$$

The corresponding normalized tensor-product state is

$$|s_1, s_2, m_1, m_2\rangle = \binom{2s_1}{s_1+m_1}^{-1/2} \binom{2s_2}{s_2+m_2}^{-1/2} |v_{s_1,m_1} v_{s_2,m_2}\rangle . \tag{10.77}$$

The spin-s_1+s_2 state is represented by

$$|s_1+s_2, m_1+m_2\rangle = \binom{2s_1+2s_2}{s_1+s_2+m_1+m_2}^{-1/2} |v_{s_1+s_2,m_1+m_2}\rangle . \tag{10.78}$$

The Clebsch–Gordan coefficient is

$$\langle s_1+s_2, m_1+m_2 | s_1, s_2, m_1, m_2\rangle . \tag{10.79}$$

It is given by the tensor contraction

$$v^{i_1\cdots i_{2s_1} j_1\cdots j_{2s_2}}_{s_1+s_2,m_1+m_2} v^{i_1\cdots i_{2s_1}}_{s_1,m_1} v^{j_1\cdots j_{2s_2}}_{s_2,m_2}. \tag{10.80}$$

The unnormalized contraction is

$$\binom{2s_1}{s_1+m_1}\binom{2s_2}{s_2+m_2}. \tag{10.81}$$

Thus

$$\langle s_1+s_2, m_1+m_2 | s_1, s_2, m_1, m_2\rangle = \left[\frac{\binom{2s_1}{s_1+m_1}\binom{2s_2}{s_2+m_2}}{\binom{2s_1+2s_2}{s_1+s_2+m_1+m_2}}\right]^{1/2}. \tag{10.82}$$

10.15 *Spin s_1+s_2-1

To obtain the spin s_1+s_2-1 piece, insert one invariant tensor ϵ^{ij}:

$$\epsilon^{i_1 j_1} v^{i_2\cdots i_{2s_1} j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}. \tag{10.83}$$

Symmetrizing separately in the i and j indices gives

$$V^{i_1\cdots i_{2s_1} j_1\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2;2s_1,2s_2}. \tag{10.84}$$

If N normalizes this tensor, then

$$\begin{aligned}&\langle s_1+s_2-1, m_1+m_2 | s_1, s_2, m_1, m_2\rangle \\ &= N\binom{2s_1}{s_1+m_1}^{-1/2}\binom{2s_2}{s_2+m_2}^{-1/2} V^{i_1\cdots i_{2s_1} j_1\cdots j_{2s_2}} v^{i_1\cdots i_{2s_1}}_{s_1,m_1} v^{j_1\cdots j_{2s_2}}_{s_2,m_2}.\end{aligned} \tag{10.85}$$

Using the symmetrized form,

$$\begin{aligned}&\langle s_1+s_2-1, m_1+m_2 | s_1, s_2, m_1, m_2\rangle \\ &= N\binom{2s_1}{s_1+m_1}^{-1/2}\binom{2s_2}{s_2+m_2}^{-1/2} 4s_1 s_2 \\ &\quad\times\left[\binom{2s_1-1}{s_1+m_1-1}\binom{2s_2-1}{s_2+m_2} - \binom{2s_1-1}{s_1+m_1}\binom{2s_2-1}{s_2+m_2-1}\right].\end{aligned} \tag{10.86}$$

The last line comes from expanding

$$\sum_{i_2\cdots,j_2\cdots}\big(v^{2i_2\cdots i_{2s_1}1j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}v^{2i_2\cdots i_{2s_1}}_{s_1,m_1}v^{1j_2\cdots j_{2s_2}}_{s_2,m_2} - v^{1i_2\cdots i_{2s_1}2j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}v^{1i_2\cdots i_{2s_1}}_{s_1,m_1}v^{2j_2\cdots j_{2s_2}}_{s_2,m_2}\big). \tag{10.87}$$

The normalization satisfies

$$\begin{aligned}\frac{1}{N^2} &= \sum_{i_1\cdots,j_1\cdots}\left|V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2;2s_1,2s_2}\right|^2\\ &= 4s_1s_2\sum\epsilon^{i_1j_1}v^{i_2\cdots i_{2s_1}j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}.\end{aligned} \tag{10.88}$$

The contraction obeys

$$\sum_{i_1,j_1}\epsilon^{i_1j_1}V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}} = C\,v^{i_2\cdots i_{2s_1}j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}, \tag{10.89}$$

and hence

$$\frac{1}{N^2} = 4s_1s_2C\sum\left|v^{i_2\cdots i_{2s_1}j_2\cdots j_{2s_2}}_{s_1+s_2-1,m_1+m_2}\right|^2. \tag{10.90}$$

The direct contraction gives

$$\sum_{i_1,j_1}|\epsilon^{i_1j_1}|^2 = 2, \tag{10.91}$$

whereas

$$\sum_{i_1}\epsilon^{i_1j_1}\epsilon^{i_1j_m} = \delta^{j_1j_m}. \tag{10.92}$$

Thus $C = 2s_1+2s_2$, and

$$\frac{1}{N^2} = 4s_1s_2(2s_1+2s_2)\binom{2s_1+2s_2-2}{s_1+s_2+m_1+m_2-1}. \tag{10.93}$$

Combining the pieces yields

$$\begin{aligned}&\langle s_1+s_2-1, m_1+m_2|s_1,s_2,m_1,m_2\rangle\\ &= \left[\binom{2s_1}{s_1+m_1}\binom{2s_2}{s_2+m_2}\right]^{-1/2}\left[\frac{4s_1s_2}{(2s_1+2s_2)\binom{2s_1+2s_2-2}{s_1+s_2+m_1+m_2-1}}\right]^{1/2}\\ &\quad\times\left[\binom{2s_1-1}{s_1+m_1-1}\binom{2s_2-1}{s_2+m_2} - \binom{2s_1-1}{s_1+m_1}\binom{2s_2-1}{s_2+m_2-1}\right].\end{aligned} \tag{10.94}$$

10.16 *Spin s_1+s_2-k

For the general component of spin s_1+s_2-k, insert k epsilon tensors:

$$\epsilon^{i_1j_1}\epsilon^{i_2j_2}\cdots\epsilon^{i_kj_k}v^{i_{k+1}\cdots i_{2s_1}j_{k+1}\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2}. \tag{10.95}$$

After separate symmetrization in the i and j indices,

$$V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2;2s_1,2s_2}. \tag{10.96}$$

The number of symmetrized terms is

$$\mu = \frac{(2s_1)!}{(2s_1-k)!}\frac{(2s_2)!}{(2s_2-k)!}\frac{1}{k!}. \tag{10.97}$$

The Clebsch–Gordan coefficient is

$$\begin{aligned}&\langle s_1+s_2-k, m_1+m_2|s_1,s_2,m_1,m_2\rangle\\&= N\binom{2s_1}{s_1+m_1}^{-1/2}\binom{2s_2}{s_2+m_2}^{-1/2}\sum V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}} v_{s_1,m_1}^{i_1\cdots i_{2s_1}} v_{s_2,m_2}^{j_1\cdots j_{2s_2}}.\end{aligned}\tag{10.98}$$

Expanding the epsilons gives

$$\begin{aligned}&\langle s_1+s_2-k, m_1+m_2|s_1,s_2,m_1,m_2\rangle\\&= N\binom{2s_1}{s_1+m_1}^{-1/2}\binom{2s_2}{s_2+m_2}^{-1/2}\mu\sum_{\ell=0}^{k}(-1)^\ell\binom{k}{\ell}\binom{2s_1-k}{s_1+m_1-k+\ell}\binom{2s_2-k}{s_2+m_2-\ell}.\end{aligned}\tag{10.99}$$

For the normalization,

$$\begin{aligned}\frac{1}{N^2} &= \sum\left|V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2;2s_1,2s_2}\right|^2\\&= \mu\sum\epsilon^{i_1j_1}\cdots\epsilon^{i_kj_k}v^{i_{k+1}\cdots i_{2s_1}j_{k+1}\cdots j_{2s_2}}V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}.\end{aligned}\tag{10.100}$$

At each contraction stage,

$$\begin{aligned}&\sum_{i_1,j_1}\epsilon^{i_1j_1}V^{i_1\cdots i_{2s_1}j_1\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2;2s_1,2s_2}\\&= C\,V^{i_2\cdots i_{2s_1}j_2\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2;2s_1-1,2s_2-1}.\end{aligned}\tag{10.101}$$

A representative term is

$$\epsilon^{i_2j_2}\cdots\epsilon^{i_kj_k}v^{i_{k+1}\cdots i_{2s_1}j_{k+1}\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2}.\tag{10.102}$$

The term

$$\epsilon^{i_1j_1}\epsilon^{i_2j_2}\cdots\epsilon^{i_kj_k}v^{\cdots}\tag{10.103}$$

contributes 2. Terms of the form

$$\epsilon^{i_1j_m}\epsilon^{i_2j_2}\cdots\widehat{\epsilon^{i_mj_m}}\cdots\epsilon^{i_kj_k}v^{\cdots j_1\cdots}\tag{10.104}$$

contribute 1, and terms of the form

$$\epsilon^{i_mj_1}\epsilon^{i_2j_2}\cdots\widehat{\epsilon^{i_mj_m}}\cdots\epsilon^{i_kj_k}v^{\cdots i_1\cdots}\tag{10.105}$$

also contribute 1. Terms exchanging one of the existing epsilon pairs contribute

$$1 \quad \text{for } 2\le m\le k.\tag{10.106}$$

Thus

$$\sum_{i_1,j_1}\epsilon^{i_1j_1}V^{i_1\cdots j_1\cdots}_{s_1+s_2-k,m_1+m_2;2s_1,2s_2} = (2s_1+2s_2-k+1)V_{s_1+s_2-k,m_1+m_2;2s_1-1,2s_2-1}.\tag{10.107}$$

Iterating,

$$\sum_{i_{p+1},j_{p+1}}\epsilon^{i_{p+1}j_{p+1}}V_{s_1+s_2-k,m_1+m_2;2s_1-p,2s_2-p} = C_pV_{s_1+s_2-k,m_1+m_2;2s_1-p-1,2s_2-p-1},\tag{10.108}$$

with terminal object

$$V^{i_{k+1}\cdots i_{2s_1}j_{k+1}\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2;2s_1-k,2s_2-k} = v^{i_{k+1}\cdots i_{2s_1}j_{k+1}\cdots j_{2s_2}}_{s_1+s_2-k,m_1+m_2}.\tag{10.109}$$

A representative contribution at the pth step is

$$\epsilon^{i_{p+2}j_{p+2}} \cdots \epsilon^{i_k j_k} v^{i_{k+1}\cdots i_{2s_1} j_{k+1}\cdots j_{2s_2}}. \tag{10.110}$$

The direct pair contributes

$$2, \tag{10.111}$$

terms of type

$$\epsilon^{i_{p+1}j_m} \cdots \tag{10.112}$$

and

$$\epsilon^{i_m j_{p+1}} \cdots \tag{10.113}$$

contribute 1, and the remaining epsilon-pair exchanges give

$$1 \quad \text{for } p+2 \le m \le k. \tag{10.114}$$

Therefore

$$C_p = 2s_1 + 2s_2 - k - p + 1. \tag{10.115}$$

The normalization sum is then

$$\begin{aligned}
\sum |V|^2 &= \left(\prod_{p=0}^{k-1} C_p \right) \sum |v_{s_1+s_2-k, m_1+m_2}|^2 \\
&= \frac{(2s_1+2s_2-k+1)!}{(2s_1+2s_2-2k+1)!} \binom{2s_1+2s_2-2k}{s_1+s_2-k+m_1+m_2}.
\end{aligned} \tag{10.116}$$

Thus

$$\frac{1}{N^2} = \mu \frac{(2s_1+2s_2-k+1)!}{(2s_1+2s_2-2k+1)!} \binom{2s_1+2s_2-2k}{s_1+s_2-k+m_1+m_2}. \tag{10.117}$$

Putting all pieces together,

$$\begin{aligned}
&\langle s_1+s_2-k, m_1+m_2 | s_1, s_2, m_1, m_2 \rangle \\
&= [inom2s_1+2s_2-2ks_1+s_2+m_1+m_2-kinom2s_1s_1+m_1inom2s_2s_2+m_2]^{-1/2} \\
&\quad \times \left(\frac{(2s_1)!(2s_2)!}{k!(2s_1-k)!(2s_2-k)!} \right)^{1/2} \left(\frac{(2s_1+2s_2-2k+1)!}{(2s_1+2s_2-k+1)!} \right)^{1/2} \\
&\quad \times \sum_{\ell=0}^{k} (-1)^\ell \binom{k}{\ell} \binom{2s_1-k}{s_1+m_1-k+\ell} \binom{2s_2-k}{s_2+m_2-\ell}.
\end{aligned} \tag{10.118}$$

Problems

10.A. Decompose the product of tensor components $u^i v^{jk}$, where $v^{jk} = v^{kj}$ transforms like a **6** of $SU(3)$.

10.B. Find the matrix elements $\langle u | T_a | v \rangle$, where T_a are the $SU(3)$ generators and $|u\rangle, |v\rangle$ are tensors in the adjoint representation with components $u^i{}_j$ and $v^i{}_j$. Write the result in terms of the tensor components and the λ_a matrices.

10.C. In the **6** of $SU(3)$, for each weight find the corresponding tensor component v^{ij}.

***10.D.** Use the $SU(2)$ tensor methods described in this chapter to redo problem V.C. Also check equation (10.86) by using the highest-weight procedure to find the state $|3/2, 1/2\rangle$ in the tensor product of spin-3/2 and spin-1 states, and compare with the tensor result for

$$\langle 3/2, 1, 3/2, 1/2 | 1, 3/2, 0, 1/2 \rangle .$$

***10.E.** Consider πp scattering, ignoring isospin breaking. The accessible amplitudes are

$$\begin{aligned} A_{+p} &= \left\langle \pi^+ p \right| H_I \left| \pi^+ p \right\rangle , \\ A_{-p} &= \left\langle \pi^- p \right| H_I \left| \pi^- p \right\rangle , \\ A_{0n} &= \left\langle \pi^0 n \right| H_I \left| \pi^- p \right\rangle . \end{aligned} \tag{1}$$

Using $SU(2)$ tensors $\pi^{jk} = \pi^{kj}$ for pions and N^j for nucleons, the most general amplitude is

$$\langle \pi N | \, H_I \, | \pi N \rangle = A_1 \pi^{jk} N^\ell \pi_{jk} N_\ell + A_2 \pi^{jk} N^\ell \pi_{j\ell} N_k . \tag{2}$$

Use this to express the three amplitudes in terms of A_1, A_2, find their relation, and rewrite them in the $I = 3/2$ and $I = 1/2$ channel basis.

Chapter 11

Hypercharge and Strangeness

This chapter records the $SU(3)$ organization of the light hadrons, the hypercharge–strangeness bookkeeping, and the standard Gell-Mann–Okubo mass relation.

11.1 The eight-fold way

The lightest strange mesons are the K mesons, an isospin doublet with $S = 1$:

$$K^+ \quad \text{with } I_3 = \frac{1}{2}, \qquad K^0 \quad \text{with } I_3 = -\frac{1}{2}. \tag{11.1}$$

Their antiparticles have $S = -1$:

$$K^- \quad \text{with } I_3 = -\frac{1}{2}, \qquad \overline{K}^0 \quad \text{with } I_3 = \frac{1}{2}. \tag{11.2}$$

The strange baryons with $B = 1$ and $S = -1$ include the isotriplet

$$\Sigma^+ \quad (I_3 = 1), \qquad \Sigma^0 \quad (I_3 = 0), \qquad \Sigma^- \quad (I_3 = -1), \tag{11.3}$$

and the isosinglet

$$\Lambda \quad (I_3 = 0). \tag{11.4}$$

For $S = -2$ there is the isodoublet

$$\Xi^0 \quad (I_3 = \frac{1}{2}), \qquad \Xi^- \quad (I_3 = -\frac{1}{2}). \tag{11.5}$$

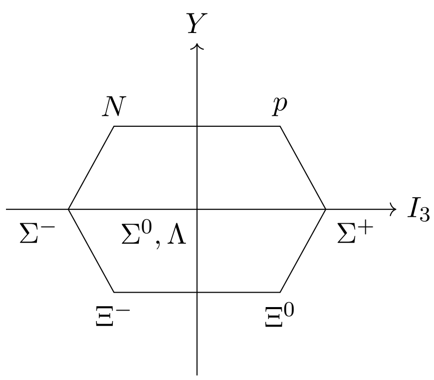
All these states satisfy

$$Q = I_3 + \frac{Y}{2}, \tag{11.6}$$

where the hypercharge is

$$Y = B + S. \tag{11.7}$$

The baryon octet is naturally represented in the (I_3, Y) plane:

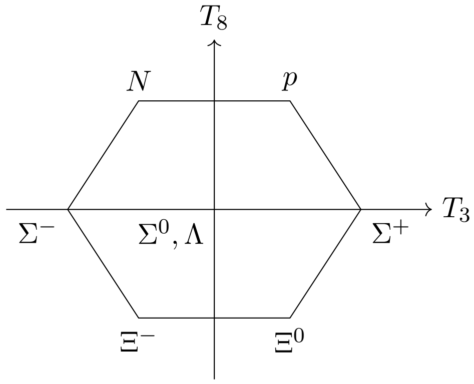


(11.8)

To make the hexagon regular, one chooses

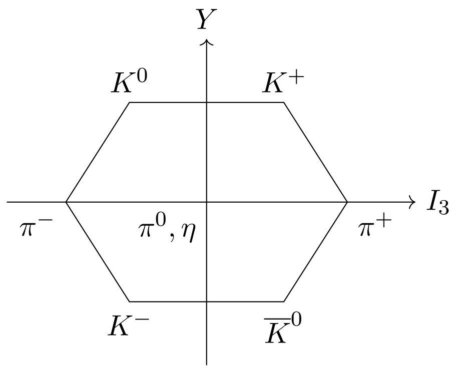
$$Y = \frac{2}{\sqrt{3}} T_8. \tag{11.9}$$

Then the same octet can be drawn in the usual $SU(3)$ weight-plane normalization:



(11.10)

The light spin-zero mesons similarly fill an octet once the η is included:



(11.11)

Gell-Mann's useful organizing assumption was that the strong interactions split into

$$\begin{aligned}&\text{very strong interactions: invariant under } SU(3),\\ &\text{medium strong interactions: break } SU(3) \text{ but conserve isospin and hypercharge.}\end{aligned} \tag{11.12}$$

11.2 The Gell-Mann Okubo formula

Write the baryon octet states as

$$B^i{}_j \left|i\bar{j}\right\rangle, \tag{11.13}$$

with, for example,

$$B^1{}_3 = p. \tag{11.14}$$

A convenient octet matrix is

$$B^i{}_j = \begin{pmatrix} \frac{\Lambda}{\sqrt{6}} + \frac{\Sigma^0}{\sqrt{2}} & \Sigma^+ & p \\ \Sigma^- & \frac{\Lambda}{\sqrt{6}} - \frac{\Sigma^0}{\sqrt{2}} & N \\ \Xi^- & \Xi^0 & -\frac{2\Lambda}{\sqrt{6}} \end{pmatrix}. \tag{11.15}$$

Thus

$$B^i{}_j \left|i\bar{j}\right\rangle = p\left|p\right\rangle + N\left|N\right\rangle + \cdots. \tag{11.16}$$

For the Hamiltonian matrix element,

$$\langle B|H_S|B\rangle = \langle B|H_{VS}|B\rangle + \langle B|H_{MS}|B\rangle. \tag{11.17}$$

Assume that the $SU(3)$-breaking medium-strong Hamiltonian transforms as the eighth component of an octet:

$$H_{MS} = [T_8]^i{}_j O^j{}_i, \tag{11.18}$$

where

$$O^i{}_j \sim 8 \quad \text{under } SU(3). \tag{11.19}$$

Tensor methods reduce the matrix element to

$$\langle B|H_{MS}|B\rangle. \tag{11.20}$$

The two independent invariants are

$$[B^\dagger]^i{}_j [T_8]^j{}_k B^k{}_i = \mathrm{Tr}\Big(B^\dagger T_8 B\Big), \qquad [B^\dagger]^i{}_j B^j{}_k [T_8]^k{}_i = \mathrm{Tr}\Big(B^\dagger B T_8\Big), \tag{11.21}$$

using

$$[B^\dagger]^i{}_j = (B^j{}_i)^*. \tag{11.22}$$

Therefore

$$\begin{aligned} \langle B|H_{MS}|B\rangle &= X\,\mathrm{Tr}\Big(B^\dagger B T_8\Big) + Y\,\mathrm{Tr}\Big(B^\dagger T_8 B\Big) \\ &= \frac{X}{\sqrt{12}}\Big([B^\dagger B]^1{}_1 + [B^\dagger B]^2{}_2 - 2[B^\dagger B]^3{}_3\Big) \\ &\quad + \frac{Y}{\sqrt{12}}\Big([BB^\dagger]^1{}_1 + [BB^\dagger]^2{}_2 - 2[BB^\dagger]^3{}_3\Big). \end{aligned} \tag{11.23}$$

Explicitly,

$$\begin{aligned} [B^\dagger B]^1{}_1 + [B^\dagger B]^2{}_2 - 2[B^\dagger B]^3{}_3 &= |\Sigma|^2 + |\Xi|^2 - |\Lambda|^2 - 2|N|^2, \\ [BB^\dagger]^1{}_1 + [BB^\dagger]^2{}_2 - 2[BB^\dagger]^3{}_3 &= |\Sigma|^2 + |N|^2 - |\Lambda|^2 - 2|\Xi|^2. \end{aligned} \tag{11.24}$$

Thus

$$\begin{aligned} \langle B|H_{MS}|B\rangle = &\frac{X}{\sqrt{12}}\left(|\Sigma|^2 + |\Xi|^2 - |\Lambda|^2 - 2|N|^2\right) \\ &+ \frac{Y}{\sqrt{12}}\left(|\Sigma|^2 + |N|^2 - |\Lambda|^2 - 2|\Xi|^2\right). \end{aligned} \tag{11.25}$$

Adding the common octet mass M_0 gives

$$\begin{aligned} M_N &= M_0 - \frac{2X}{\sqrt{12}} + \frac{Y}{\sqrt{12}}, \\ M_\Sigma &= M_0 + \frac{X}{\sqrt{12}} + \frac{Y}{\sqrt{12}}, \\ M_\Lambda &= M_0 - \frac{X}{\sqrt{12}} - \frac{Y}{\sqrt{12}}, \\ M_\Xi &= M_0 + \frac{X}{\sqrt{12}} - \frac{2Y}{\sqrt{12}}. \end{aligned} \tag{11.26}$$

Eliminating M_0, X, Y gives the Gell-Mann–Okubo formula

$$\frac{3M_\Lambda + M_\Sigma}{4} = \frac{M_N + M_\Xi}{2}. \tag{11.27}$$

Equivalently,

$$M_\Lambda = \frac{2M_N + 2M_\Xi - M_\Sigma}{3}. \tag{11.28}$$

Using

$$M_N = 940, \qquad M_\Sigma = 1190, \qquad M_\Xi = 1320 \quad \text{MeV}, \tag{11.29}$$

one obtains

$$M_\Lambda \simeq 1107 \text{ MeV}, \tag{11.30}$$

close to the observed 1115 MeV.

11.3 Hadron resonances

The decuplet of spin-3/2 baryon resonances is represented by

$$\begin{array}{cccc} \Delta^- & \Delta^0 & \Delta^+ & \Delta^{++} \\ & \Sigma^{*-} & \Sigma^{*0} & \Sigma^{*+} \\ & \Xi^{*-} & \Xi^{*0} & \\ & & \Omega^- & \end{array}$$

Figure 11.1: *
(11.31)

For the decuplet,

$$\langle B^* | H_{MS} | B^* \rangle, \tag{11.32}$$

where

$$B^{*ijk}\,|ijk\rangle \tag{11.33}$$

labels the general decuplet state. Since only one reduced matrix element occurs, the mass splitting is proportional to hypercharge:

$$M_{\Sigma^*} - M_\Delta = M_{\Xi^*} - M_{\Sigma^*} = M_{\Omega^-} - M_{\Xi^*}. \tag{11.34}$$

Experimentally,

$$M_{\Sigma^*} = 1385 \text{ MeV}, \qquad M_{\Xi^*} = 1530 \text{ MeV}. \tag{11.35}$$

This led to the prediction of the Ω^- near 1680 MeV.

11.4 Quarks

The light quarks transform as the defining 3 of $SU(3)$:

$$\begin{array}{c} u \\ d \\ s \end{array} \tag{11.36}$$

Baryons are built from three quarks because

$$3 \otimes 3 \otimes 3 = 10 \oplus 8 \oplus 8 \oplus 1. \tag{11.37}$$

Since $3 \otimes 3 = 6 \oplus \overline{3}$,

$$3 \otimes 3 \otimes 3 = (6 \otimes 3) \oplus (\overline{3} \otimes 3). \tag{11.38}$$

The decomposition

$$u^{ij}v^k = \frac{1}{3}(u^{ij}v^k + u^{ik}v^j + u^{kj}v^i) + \frac{1}{3}\left(\epsilon^{ik\ell}\epsilon_{\ell mn}u^{mj}v^n + \epsilon^{jk\ell}\epsilon_{\ell mn}u^{im}v^n\right) \tag{11.39}$$

shows $6 \otimes 3 = 10 \oplus 8$. Antiquarks transform as

$$\begin{matrix} \overline{u} \\ \overline{d} \\ \overline{s} \end{matrix} \tag{11.40}$$

The quark electric-charge matrix is

$$Q = T_3 + \frac{Y}{2} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 2/3 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & -1/3 \end{pmatrix}. \tag{11.41}$$

The magnetic moments of the baryon octet obey an octet-operator structure:

$$\mu(B) = \alpha \operatorname{Tr}\left(BB^\dagger Q\right) + \beta \operatorname{Tr}\left(B^\dagger BQ\right). \tag{11.42}$$

Problems

11.A. What does the Gell-Mann–Okubo argument imply for masses of particles transforming as a 6 of $SU(3)$?

11.B. Compare the probability for Σ^+ production in $\pi^0 p \to \Sigma^+$ with the probability for Ξ^{*0} production in $K^- p \to \Xi^{*0}$, assuming $SU(3)$ symmetry of the S-matrix.

11.C. Use the $SU(3)$ argument above to reproduce the Coleman–Glashow predictions for the spin-1/2 baryon magnetic moments in terms of $\mu(p)$ and $\mu(n)$.

Chapter 12

Young Tableaux

Young tableaux provide a compact way to encode the symmetry of tensors and therefore the irreducible representations of $SU(3)$. They also give a direct algorithm for Clebsch–Gordan decompositions and for branching from $SU(3)$ to $SU(2) \times U(1)$.

12.1 Raising the indices

An irreducible (n, m) tensor may be written as

$$A^{i_1 \cdots i_n}{}_{j_1 \cdots j_m}, \tag{12.1}$$

separately symmetric in the upper and lower indices and traceless under the contraction of any upper index with any lower index. Raise each lower index with an invariant epsilon tensor:

$$a^{i_1 \cdots i_n k_1 \ell_1 \cdots k_m \ell_m} = \epsilon^{j_1 k_1 \ell_1} \cdots \epsilon^{j_m k_m \ell_m} A^{i_1 \cdots i_n}{}_{j_1 \cdots j_m}. \tag{12.2}$$

The tensor on the left is antisymmetric in each pair (k_r, ℓ_r), symmetric under interchange of such pairs, and symmetric in the original i indices. Its Young tableau is therefore

$$\begin{array}{|c|c|c|c|c|c|} \hline k_1 & \cdots & k_m & i_1 & \cdots & i_n \\ \hline \ell_1 & \cdots & \ell_m \\ \cline{1-3} \end{array} \tag{12.3}$$

Thus one first symmetrizes the entries in each row and then antisymmetrizes the entries in each column. The trace condition becomes

$$\epsilon_{i_1 k_1 \ell_1} a^{i_1 \cdots i_n k_1 \ell_1 \cdots k_m \ell_m} = 0, \tag{12.4}$$

which follows because the row symmetrization is incompatible with the total antisymmetry of the epsilon tensor.

For a general tensor $a^{j_1 j_2 k_1}$, the tableau

$$\begin{array}{|c|c|} \hline j_1 & j_2 \\ \hline k_1 \\ \cline{1-1} \end{array} \tag{12.5}$$

produces

$$a^{j_1 j_2 k_1} + a^{j_2 j_1 k_1} - a^{k_1 j_2 j_1} - a^{j_2 k_1 j_1}, \tag{12.6}$$

which transforms as the $(1, 1)$, or adjoint, representation.

The rule extends to tableaux with more rows. In $SU(3)$, a column with more than three boxes vanishes. A complete column of three boxes is proportional to the invariant epsilon tensor:

$$\epsilon^{ijk} = \begin{array}{|c|} \hline i \\ \hline j \\ \hline k \\ \hline \end{array}. \tag{12.7}$$

> Note that $\epsilon^{i_1 i_2 \cdots i_n} = 0$ if $n >$ dimension of $SU(N)$ because otherwise at least two of the index will always be same.

Consequently, a complete three-box column may be removed without changing the $SU(3)$ representation. The general form

$$\begin{array}{|c|c|c|c|c|c|c|c|c|}\hline & \cdots & & k_1 & \cdots & k_m & i_1 & \cdots & i_n \\ \hline & \cdots & & \ell_1 & \cdots & \ell_m \\ \cline{1-6} & \cdots & \\ \cline{1-3} \end{array} \tag{12.8}$$

describes the same representation as (12.3): the full columns only supply invariant epsilon factors.

12.2 Clebsch–Gordan decomposition

Let A and B be tableaux corresponding to irreducible representations α and β. Label the boxes in the first row of B by a, those in the second row by b, and so on. Build the product by adding the boxes of B to A subject to four rules:

1. at every stage the result must remain a legal Young tableau;
2. no two a boxes may lie in the same column, no two b boxes may lie in the same column, and so on;
3. add the a boxes first, then the b boxes, and continue row by row;
4. reading from right to left along successive rows, the number of a's encountered must never be smaller than the number of b's, the number of b's must never be smaller than the number of c's, etc.

The last condition is the Littlewood–Richardson reading-word condition and prevents double counting.

The first examples are

$$\begin{gathered} \begin{array}{|c|}\hline \\ \hline \end{array} \otimes \begin{array}{|c|}\hline a \\ \hline \end{array} = \begin{array}{|c|c|}\hline & a \\ \hline \end{array} \oplus \begin{array}{|c|}\hline \\ \hline a \\ \hline \end{array}, \\ \mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \overline{\mathbf{3}}. \end{gathered} \tag{12.9}$$

Likewise,

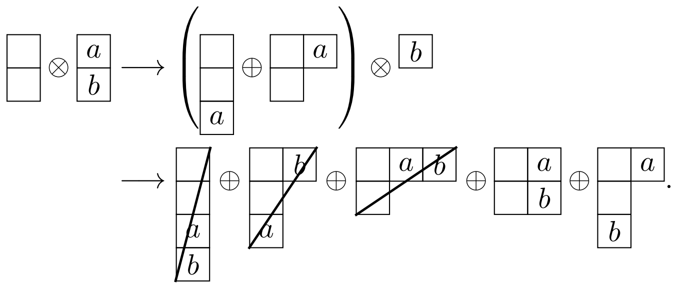
$$\begin{gathered} \begin{array}{|c|}\hline \\ \hline \\ \hline \end{array} \otimes \begin{array}{|c|}\hline a \\ \hline \end{array} = \begin{array}{|c|c|}\hline & a \\ \hline \\ \cline{1-1} \end{array} \oplus \begin{array}{|c|}\hline \\ \hline \\ \hline a \\ \hline \end{array}, \\ \overline{\mathbf{3}} \otimes \mathbf{3} = \mathbf{8} \oplus \mathbf{1}. \end{gathered} \tag{12.10}$$

Doing the same product in the less convenient order displays the rejected intermediate tableaux explicitly:

$$\begin{aligned} \begin{array}{|c|}\hline \\ \hline \end{array} \otimes \begin{array}{|c|}\hline a \\ \hline b \\ \hline \end{array} &\longrightarrow \left(\begin{array}{|c|c|}\hline & a \\ \hline \end{array} \oplus \begin{array}{|c|}\hline \\ \hline a \\ \hline \end{array} \right) \otimes \begin{array}{|c|}\hline b \\ \hline \end{array} \\ &\longrightarrow \cancel{\begin{array}{|c|c|c|}\hline & a & b \\ \hline \end{array}} \oplus \begin{array}{|c|c|}\hline & a \\ \hline b \\ \cline{1-1} \end{array} \oplus \cancel{\begin{array}{|c|c|}\hline & b \\ \hline a \\ \cline{1-1} \end{array}} \oplus \begin{array}{|c|}\hline \\ \hline a \\ \hline b \\ \hline \end{array}. \end{aligned} \tag{12.11}$$

The crossed tableaux fail the reading-word condition; the two survivors are again the octet and singlet.

Sometimes a tableau produced after adding the a boxes cannot be completed legally after the b boxes are added. For example,

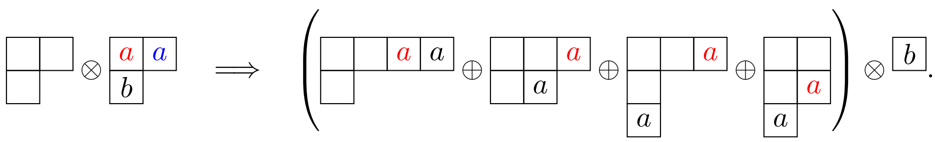


(12.12)

The first one vanishes because in $SU(3)$, no column can have more than 3 boxes. The second and third were rejected because we encountered *number of boxes containing '**b**' to be greater than number of boxes containing '**a**' when reading along rows and columns.* Removing full three-box columns from the two survivors yields

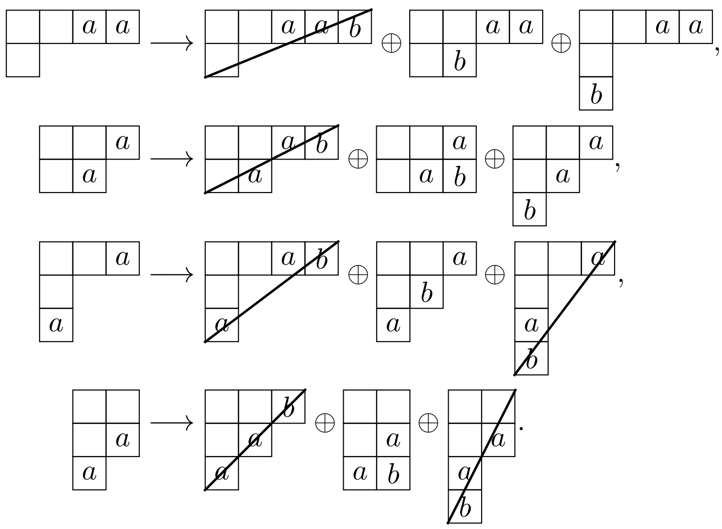
$$\mathbf{\overline{3}} \otimes \mathbf{\overline{3}} = \mathbf{\overline{6}} \oplus \mathbf{3}.$$

The product $\mathbf{8} \otimes \mathbf{8}$ illustrates the complete algorithm. The octet tableau is $[2, 1]$. The first stage, in which the two a boxes are added, produces four possible tableaux:

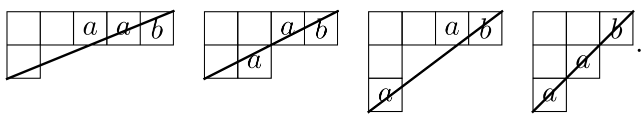


(12.13)

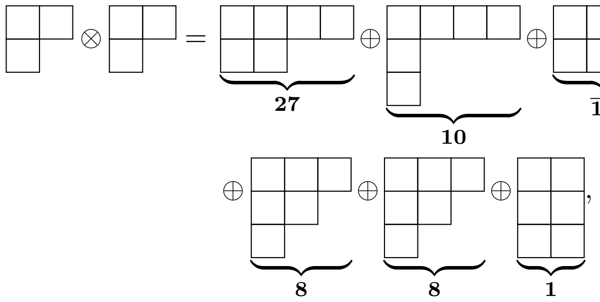
Adding the b box to each of these gives the following complete list of candidates:



The following entries were rejected because as we moved right to left on the first row, we found that number of boxes containing '**b**' were greater than number of boxes containing '**a**'.



and the remaining entries were rejected because number of boxes in **column 1** were greater than three. The legal tableaux are therefore



(12.14)

$$\mathbf{8}\otimes\mathbf{8}=\mathbf{27}\oplus\mathbf{10}\oplus\overline{\mathbf{10}}\oplus\mathbf{8}\oplus\mathbf{8}\oplus\mathbf{1}.$$

The dimensions satisfy $64 = 27 + 10 + 10 + 8 + 8 + 1$. The two octets correspond to the two inequivalent legal fillings of the same final tableau shape.

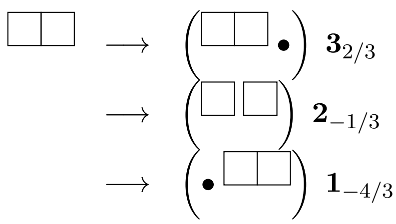
12.3 $SU(3) \to SU(2) \times U(1)$

A Young tableau may also be branched under the subgroup $SU(2) \times U(1)$. In the defining representation, the first two values of an index form an $SU(2)$ doublet with hypercharge 1/3, while the third value is an $SU(2)$ singlet with hypercharge $-2/3$. For a tableau with n boxes, if j boxes are assigned to the doublet and $n - j$ to the singlet, then

$$Y = \frac{j}{3} - \frac{2(n-j)}{3} = -\frac{2n}{3} + j. \tag{12.15}$$

The singlet boxes form one horizontal row. We write an $SU(2)$ multiplet of dimension $d = 2I+1$ and hypercharge Y as d_Y.

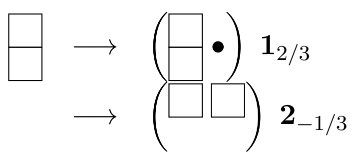
For the symmetric representation **6**:



(12.16)

Each subgroup representation appears once.

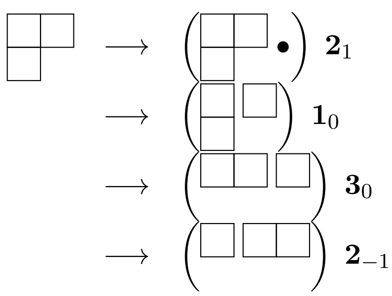
For the conjugate fundamental $\overline{\mathbf{3}}$:



(12.17)

Again, every term occurs once.

For the adjoint **8**:

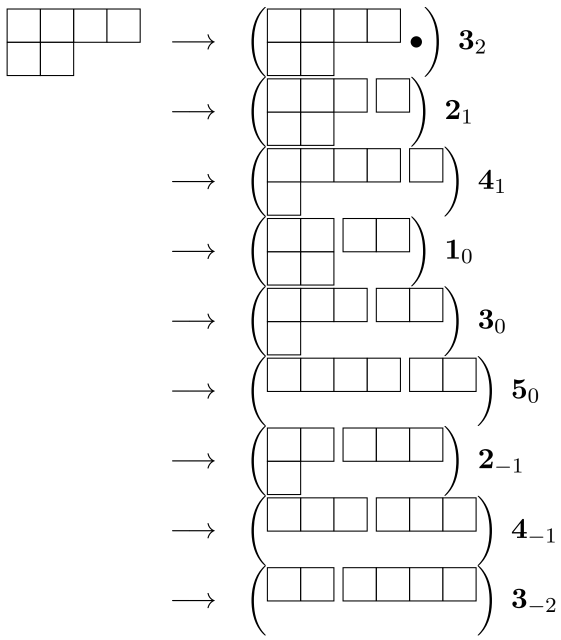


(12.18)

Thus

$$\mathbf{8} \longrightarrow \mathbf{2}_1 \oplus \mathbf{1}_0 \oplus \mathbf{3}_0 \oplus \mathbf{2}_{-1}.$$

Finally, for the **27**, whose tableau is $[4, 2]$, the complete branching diagram is



(12.19)

Therefore

$$\mathbf{27} \longrightarrow \mathbf{3}_2 \oplus \mathbf{2}_1 \oplus \mathbf{4}_1 \oplus \mathbf{1}_0 \oplus \mathbf{3}_0 \oplus \mathbf{5}_0 \oplus \mathbf{2}_{-1} \oplus \mathbf{4}_{-1} \oplus \mathbf{3}_{-2},$$

and the dimension check is

$$27 = 3 + 2 + 4 + 1 + 3 + 5 + 2 + 4 + 3.$$

Problems

12.A. Find $(2, 1) \otimes (2, 1)$. Determine, where possible, which irreducible representations occur symmetrically and which occur antisymmetrically under interchange of the two factors.

12.B. Find $\mathbf{10} \otimes \mathbf{8}$.

12.C. For any Lie group, the tensor product of the adjoint representation with any nontrivial irreducible representation D must contain D. Explain this from the action of the generators on the states of D, and verify it directly for $SU(3)$ with Young tableaux.

Chapter 13

$SU(N)$

13.1 Generalized Gell–Mann matrices

The group $SU(N)$ is the group of special unitary $N \times N$ matrices. Its Lie algebra consists of traceless Hermitian $N \times N$ matrices. We normalize the generators in the defining representation by

$$\mathrm{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}. \tag{13.1}$$

A convenient basis is obtained recursively, beginning with $SU(2)$ and enlarging to $SU(N)$. The off-diagonal generators can be chosen to have one nonzero off-diagonal entry, normalized by $1/\sqrt{2}$. Since the diagonal traceless real matrices form an $(N-1)$-dimensional space, the rank of $SU(N)$ is $N-1$.

A standard basis for the Cartan generators is

$$H_m = \frac{1}{\sqrt{2m(m+1)}}\,\mathrm{diag}(\underbrace{1, 1, \ldots, 1}_{m \text{ entries}}, -m, 0, \ldots, 0), \qquad m = 1, \ldots, N-1. \tag{13.2}$$

For the first few values,

$$H_1 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & -1 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad H_2 = \frac{1}{\sqrt{12}}\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & -2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{5.1}$$

$$H_3 = \frac{1}{\sqrt{24}}\begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & -3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad \ldots \tag{13.3}$$

Altogether there are $N^2 - 1$ independent traceless Hermitian generators.

The defining representation has dimension N. Let its weights be

$$\nu^j = (\nu_1^j, \nu_2^j, \ldots, \nu_{N-1}^j), \qquad j = 1, \ldots, N, \tag{13.4}$$

where ν_m^j is the jth diagonal entry of H_m. The squared length of each weight is

$$(\nu^j)^2 = \sum_{m=1}^{N-1}\left[\frac{1}{\sqrt{2m(m+1)}}\left(\sum_{k=1}^{m}\delta_{jk} - m\delta_{j,m+1}\right)\right]^2$$

$$= \frac{1}{2}\sum_{m=j}^{N-1}\frac{1}{m(m+1)} + \frac{(j-1)^2}{2j(j-1)}$$
$$= \frac{1}{2}\left(\frac{1}{j} - \frac{1}{N}\right) + \frac{j-1}{2j} = \frac{N-1}{2N}. \tag{13.5}$$

For $i < j$,

$$\nu^i \cdot \nu^j = \frac{1}{2}\sum_{m=j}^{N-1}\frac{1}{m(m+1)} - \frac{1}{2j} = -\frac{1}{2N}. \tag{13.6}$$

Thus all weights have the same length and the angle between any pair of distinct weights is the same:

$$(\nu^i)^2 = \frac{N-1}{2N}, \qquad \nu^i \cdot \nu^j = -\frac{1}{2N} \quad (i \neq j). \tag{13.7}$$

Equivalently,

$$\nu^i \cdot \nu^j = -\frac{1}{2N} + \frac{1}{2}\delta_{ij}. \tag{13.8}$$

The weights therefore form the vertices of a regular $(N-1)$-simplex.

Explicitly,

$$\begin{aligned}
\nu^1 &= \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}, \ldots, \frac{1}{\sqrt{2m(m+1)}}, \ldots, \frac{1}{\sqrt{2(N-1)N}}\right), \\
\nu^2 &= \left(-\frac{1}{2}, \frac{1}{2\sqrt{3}}, \ldots, \frac{1}{\sqrt{2m(m+1)}}, \ldots, \frac{1}{\sqrt{2(N-1)N}}\right), \\
\nu^{m+1} &= \left(0, 0, \ldots, -\frac{m}{\sqrt{2m(m+1)}}, \ldots, \frac{1}{\sqrt{2(N-1)N}}\right), \\
\nu^N &= \left(0, 0, \ldots, 0, -\frac{N-1}{\sqrt{2(N-1)N}}\right).
\end{aligned} \tag{13.9}$$

For convenience, positivity is chosen in the reverse lexicographic sense: a weight is positive when its last nonzero component is positive. With this convention,

$$\nu^1 > \nu^2 > \cdots > \nu^N. \tag{13.10}$$

The roots are differences of weights, $\nu^i - \nu^j$, and the positive roots are those with $i < j$. A convenient set of simple roots is

$$\alpha^i = \nu^i - \nu^{i+1}, \qquad i = 1, \ldots, N-1. \tag{13.11}$$

Using (13.8),

$$\alpha^i \cdot \alpha^j = \delta_{ij} - \frac{1}{2}\delta_{i,j+1} - \frac{1}{2}\delta_{i,j-1}. \tag{13.12}$$

Therefore all simple roots have unit length, neighboring simple roots meet at 120°, and all others are orthogonal. The Dynkin diagram is

$$\circ - \circ - \cdots - \circ - \circ. \tag{13.13}$$

The fundamental weights are

$$\mu^j = \sum_{k=1}^{j}\nu^k, \qquad j = 1, \ldots, N-1. \tag{13.14}$$

Indeed,

$$\frac{2\alpha^i \cdot \mu^j}{(\alpha^i)^2} = \delta_{ij}. \tag{13.15}$$

In particular, $\mu^1 = \nu^1$ is the highest weight of the defining representation.

13.2 $SU(N)$ tensors

As in $SU(3)$, one associates states with tensors carrying defining-representation indices:

$$T^{i_1 i_2 \cdots i_r}{}_{j_1 j_2 \cdots j_s}, \qquad i_a, j_b = 1, \ldots, N. \tag{13.16}$$

Arbitrary representations can be built from tensor products of defining representations.

Consider the completely antisymmetric product of m defining representations. The states are

$$A^{[i_1 i_2 \cdots i_m]} \left| i_1 i_2 \cdots i_m \right\rangle, \tag{13.17}$$

where $A^{[i_1 \cdots i_m]}$ is totally antisymmetric. Since no two indices may coincide, the highest-weight state has indices $1, 2, \ldots, m$, and therefore highest weight

$$\nu^1 + \nu^2 + \cdots + \nu^m = \mu^m. \tag{13.18}$$

Hence the antisymmetric rank-m tensor is the mth fundamental representation.

The highest weight of an arbitrary irreducible representation is

$$\mu = \sum_{k=1}^{N-1} q_k \mu^k, \qquad q_k \in \mathbb{Z}_{\geq 0}. \tag{13.19}$$

The corresponding Young tableau contains q_k columns of height k. Schematically,

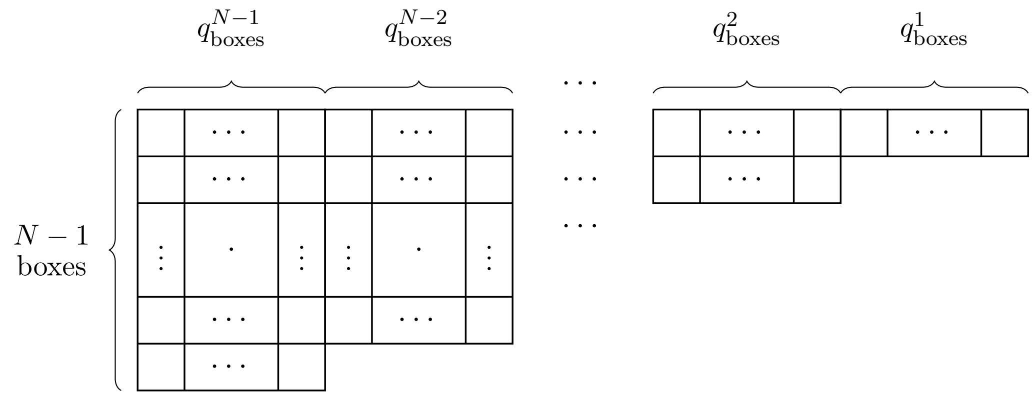


Figure 13.1: Young tableau associated with the Dynkin coefficients q_k.

The tensor is obtained by symmetrizing indices in each row and antisymmetrizing indices in each column. The highest-weight term has all 1s in the first row, all 2s in the second row, and so on. This is the same row-symmetrization/column-antisymmetrization construction used for irreducible representations of the permutation group.

A tableau containing a column longer than N gives zero because complete antisymmetrization over more than N values vanishes. A full column of exactly N boxes contributes an N-index Levi-Civita tensor and can be removed without changing the $SU(N)$ representation.

We denote a representation by the nonincreasing list of column lengths,

$$[\ell_1, \ell_2, \ldots], \tag{5.2}$$

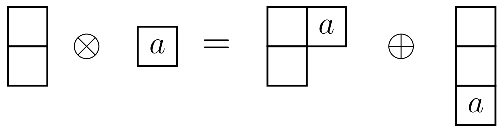
where each $\ell_i \leq N-1$. In this notation the jth fundamental representation is $[j]$. The adjoint representation has one upper and one lower index; raising the lower index with the Levi-Civita tensor gives the tableau

$$\mathrm{Adj}_{SU(N)} = [N-1, 1]. \tag{5.3}$$

The Young-tableau Clebsch–Gordan rules are the same as for $SU(3)$, except that one now uses letters $a, b, c, d, \dots$ for successive rows added from the second tableau, and requires that, when the labels are read from right to left along successive rows, the number of as never falls below the number of bs, the number of bs never falls below the number of cs, and so on.

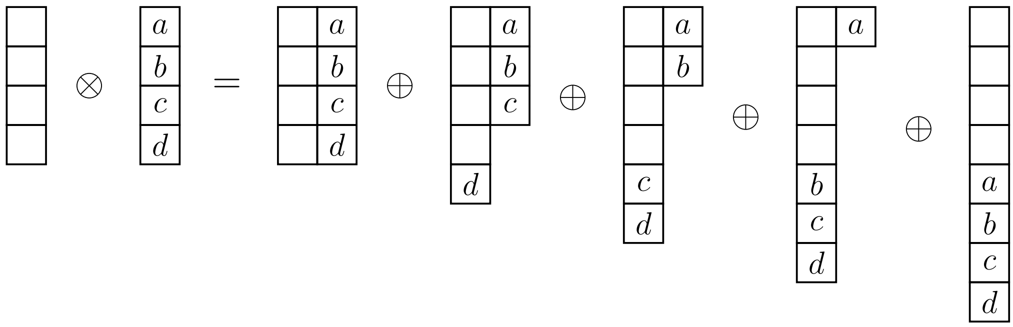
For example,

For example,



$$[2] \otimes [1] = [2,1] \oplus [3]$$

or



$$[4] \otimes [4] = [4,4] \oplus [5,3] \oplus [6,2] \oplus [7,1] \oplus [8]$$

If $N < 8$, some tableaux vanish and others reduce after removing columns of length N.

The only way boxes disappear in a tensor-product decomposition is through removal of full columns of N boxes. Therefore the total number of boxes is conserved modulo N. This is the $SU(N)$ analogue of triality in $SU(3)$ and is called the N-ality.

13.3 Dimensions

The dimension of the antisymmetric representation $[j]$ is simply the number of ways to choose j distinct indices out of N:

$$\dim[j] = \binom{N}{j} = \frac{N!}{j!(N-j)!}. \tag{13.23}$$

For example, using $[1] \otimes [2] = [2,1] \oplus [3]$,

$$\begin{aligned} \dim[2,1] &= \dim[1]\dim[2] - \dim[3] \\ &= N\binom{N}{2} - \binom{N}{3} = \frac{N(N+1)(N-1)}{3}. \end{aligned} \tag{13.24}$$

For a general tableau, the dimension is given by the factors-over-hooks rule. Place N in the upper-left box. Increase by 1 for each step to the right and decrease by 1 for each step downward. Let F be the product of these factors. For each box, define its hook as the box itself together with all boxes to its right in the same row and all boxes below it in the same column. Let H be the product of all hook lengths. Then

$$\dim R = \frac{F}{H}. \tag{13.25}$$

For $[2,1]$, the tableau has row lengths $(2,1)$. The factors are

$$\begin{array}{|c|c|}\hline N & N+1 \\ \hline N-1 & \multicolumn{1}{c}{} \\ \cline{1-1}\end{array}$$

so

$$F = N(N+1)(N-1). \tag{13.26}$$

The hook lengths are $3, 1, 1$, hence

$$H = 3. \tag{13.27}$$

Therefore

$$\dim[2,1] = \frac{F}{H} = \frac{N(N+1)(N-1)}{3}, \tag{13.28}$$

in agreement with (13.24).

13.4 Complex representations

Most irreducible representations of $SU(N)$ are complex. The defining representation has lowest weight ν^N. Since each Cartan generator is traceless,

$$\sum_{j=1}^{N} \nu^j = 0. \tag{13.29}$$

Hence

$$\nu^N = -\sum_{j=1}^{N-1} \nu^j = -\mu^{N-1}. \tag{13.30}$$

Thus the conjugate of the defining representation is the $(N-1)$st fundamental representation:

$$\overline{[1]} = [N-1]. \tag{13.31}$$

More generally, the lowest weight of $[m]$ is the sum of the m smallest defining weights,

$$\sum_{j=N-m+1}^{N} \nu^j = -\sum_{j=1}^{N-m} \nu^j = -\mu^{N-m}. \tag{13.32}$$

Therefore

$$\overline{[m]} = [N-m]. \tag{13.33}$$

For a general representation,

$$\overline{[\ell_1, \ell_2, \ldots, \ell_r]} = [N-\ell_r, \ldots, N-\ell_2, N-\ell_1], \tag{13.34}$$

after removing any full columns of length N.

Geometrically, the tableau of a representation and that of its complex conjugate fit together into an N-box-high rectangle:

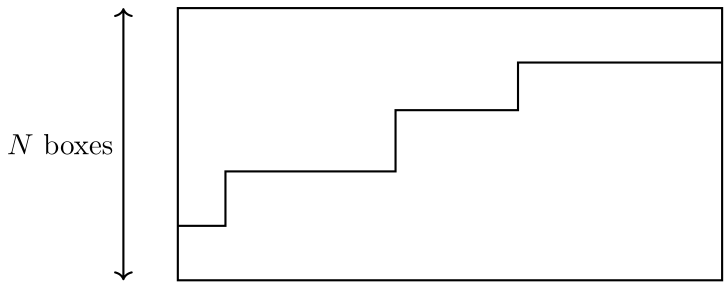


Figure 13.2:

13.5 $SU(N) \otimes SU(M) \otimes U(1) \subset SU(N+M)$

Consider the subgroup $SU(N) \otimes SU(M) \otimes U(1)$ of $SU(N+M)$, where $SU(N)$ acts on the first N components and $SU(M)$ on the last M. A commuting traceless $U(1)$ generator may be chosen to act with charge M on the first N components and charge $-N$ on the last M. Therefore the defining representation decomposes as

$$\mathbf{N}+\mathbf{M} = (\mathbf{N},\mathbf{1})_M \oplus (\mathbf{1},\mathbf{M})_{-N}. \tag{13.36}$$

Suppose an $SU(N+M)$ representation has $n+m$ boxes. To determine the multiplicity of an $SU(N) \otimes SU(M)$ pair (A, B), where A contains n boxes and B contains m boxes, temporarily let all indices run over all $N+M$ values and decompose the tensor product $A \otimes B$ as an $SU(N+M)$ representation. The multiplicity of the original $SU(N+M)$ representation in $A \otimes B$ is the multiplicity of (A, B) in the subgroup decomposition. The associated $U(1)$ charge is

$$Q = nM - mN. \tag{5.4}$$

When $M = 1$, the $SU(M)$ factor is trivial and the construction reduces to the familiar branching $SU(N+1) \to SU(N) \otimes U(1)$.

For the important example

$$SU(3) \otimes SU(2) \otimes U(1) \subset SU(5), \tag{5.5}$$

the defining representation decomposes as

$$\mathbf{5} = (\mathbf{3},\mathbf{1})_2 \oplus (\mathbf{1},\mathbf{2})_{-3}. \tag{13.37}$$

The adjoint of $SU(5)$ is $[4, 1]$, or equivalently the traceless part of $\mathbf{5} \otimes \overline{\mathbf{5}}$. Using the above decomposition,

$$\begin{aligned}\mathbf{24} = {} & (\mathbf{8},\mathbf{1})_0 \oplus (\mathbf{1},\mathbf{3})_0 \oplus (\mathbf{1},\mathbf{1})_0 \\ & \oplus (\mathbf{3},\mathbf{2})_{-5} \oplus (\overline{\mathbf{3}},\mathbf{2})_5.\end{aligned} \tag{13.38}$$

The dimensions check:

$$8 + 3 + 1 + 6 + 6 = 24.$$

Problems

13.A. Show explicitly that the $SU(n)$ algebra contains an $SU(n-1)$ subalgebra. Determine how each fundamental representation of $SU(n)$ decomposes under $SU(n-1)$.

13.B. Find the decomposition of $[3] \otimes [1]$ in $SU(5)$ and verify the dimensions.

13.C. Find the decomposition of $[3, 1] \otimes [2, 1]$.

13.D. Under

$$SU(N) \otimes SU(M) \otimes U(1) \subset SU(N+M),$$

with

$$\mathbf{N} + \mathbf{M} = ([1],[0])_M \oplus ([0],[1])_{-N},$$

determine the decomposition of all fundamental representations of $SU(N+M)$. Also determine the decomposition of the adjoint representation.

13.E. Find $[2] \otimes [1,1]$ in $SU(N)$ and use the factors-over-hooks rule to verify the dimensions for arbitrary N.

Chapter 14

3-D Harmonic Oscillator

This chapter uses the three-dimensional harmonic oscillator as a clean example of how an $SU(N)$ symmetry can arise in quantum mechanics.

14.1 Raising and lowering operators

With $\hbar = 1$, the Hamiltonian is

$$H = \frac{1}{2m}p^2 + \frac{m\omega^2}{2}r^2 = \omega\left(a_k^\dagger a_k + \frac{3}{2}\right), \tag{14.1}$$

where

$$a_k = \frac{1}{\sqrt{2m\omega}}(m\omega r_k + ip_k), \qquad a_k^\dagger = \frac{1}{\sqrt{2m\omega}}(m\omega r_k - ip_k), \tag{14.2}$$

for $k = 1, 2, 3$. The oscillator operators satisfy

$$[a_k, a_\ell^\dagger] = \delta_{k\ell}, \qquad [a_k, a_\ell] = [a_k^\dagger, a_\ell^\dagger] = 0, \qquad [a_k^\dagger a_k, a_\ell^\dagger] = a_\ell^\dagger, \qquad [a_k^\dagger a_k, a_\ell] = -a_\ell. \tag{14.3}$$

The ground state obeys

$$a_k |0\rangle = 0. \tag{14.4}$$

The energy eigenstates are

$$a_{k_1}^\dagger \cdots a_{k_n}^\dagger |0\rangle, \tag{14.5}$$

with energy

$$\omega\left(n + \frac{3}{2}\right). \tag{14.6}$$

The degeneracy is the number of symmetric combinations of n indices:

$$\frac{(n+1)(n+2)}{2}. \tag{14.7}$$

This is the dimension of the $(n, 0)$ representation of $SU(3)$. The corresponding $SU(3)$ generators are

$$Q_a = a_k^\dagger [T_a]_{k\ell} a_\ell, \tag{14.8}$$

where $T_a = \lambda_a/2$. They commute with the Hamiltonian:

$$[Q_a, H] = 0. \tag{14.9}$$

Also

$$Q_a |0\rangle = 0, \tag{14.10}$$

so the ground state is an $SU(3)$ singlet. The raising operators transform as a 3:

$$[Q_a, a_k^\dagger] = a_\ell^\dagger [T_a]_{\ell k}, \tag{14.11}$$

and the lowering operators transform as a $\overline{3}$:

$$[Q_a, a_k] = -[T_a]_{k\ell} a_\ell = -a_\ell [T_a^T]_{\ell k} = -a_\ell [T_a^*]_{\ell k}. \tag{14.12}$$

14.2 Angular momentum

Angular momentum is an $SU(2)$ subgroup of this $SU(3)$ symmetry, generated by

$$L_i = -i\epsilon_{ijk} a_j^\dagger a_k. \tag{14.13}$$

For $n > 1$, a single $SU(3)$ multiplet contains states of several angular momenta. For example, the $n = 2$ oscillator level transforms as a 6 of $SU(3)$, which decomposes under rotations as $5 \oplus 1$. Thus the harmonic oscillator has a degeneracy larger than what rotational invariance alone implies. Anharmonic terms such as $(x^2)^2$ preserve rotations but break this enlarged $SU(3)$ symmetry.

14.3 A more complicated example

Consider the two-particle Hamiltonian

$$H = \frac{\omega_1}{2}(p^2 + r^2) + \frac{\omega_2}{2}(P^2 + R^2) + \Delta(r \cdot R + p \cdot P)^2 + \Delta(r \cdot P - p \cdot R)^2. \tag{14.14}$$

Define

$$a = \frac{r + ip}{\sqrt{2}}, \qquad a^\dagger = \frac{r - ip}{\sqrt{2}}, \qquad b = \frac{R + iP}{\sqrt{2}}, \qquad b^\dagger = \frac{R - iP}{\sqrt{2}}. \tag{14.15}$$

Then

$$H = \omega_1 \left(a_k^\dagger a_k + \frac{3}{2} \right) + \omega_2 \left(b_k^\dagger b_k + \frac{3}{2} \right) + 4\Delta(a_k b_k)(a_\ell^\dagger b_\ell^\dagger). \tag{14.16}$$

This Hamiltonian commutes with

$$Q_a = a_k^\dagger [T_a]_{k\ell} a_\ell - b_k^\dagger [T_a^*]_{k\ell} b_\ell. \tag{14.17}$$

It also commutes with the number operators

$$N_a = a_k^\dagger a_k, \qquad N_b = b_k^\dagger b_k. \tag{14.18}$$

The states transform as arbitrary $SU(3)$ irreducible representations:

$$|(n,m),K\rangle = (a_{i_1}^\dagger \cdots a_{i_n}^\dagger b_{j_1}^\dagger \cdots b_{j_m}^\dagger - \text{traces})(a^\dagger \cdot b^\dagger)^K |0\rangle. \tag{14.19}$$

For example, octet states have the form

$$\left(a_i^\dagger b_j^\dagger - \frac{1}{3}\delta_{ij} a_k^\dagger b_k^\dagger \right) |0\rangle. \tag{14.20}$$

Highest-weight representatives are

$$(a_1^\dagger)^n (b_3^\dagger)^m (a^\dagger \cdot b^\dagger)^K |0\rangle. \tag{14.21}$$

For the interaction term,

$$(a^\dagger \cdot b^\dagger)(a \cdot b)\, |(n,m),K\rangle = (a) + (b) + (c), \tag{14.22}$$

where the first contribution gives

$$(a) = nK\,|(n,m),K\rangle\,, \tag{14.22a}$$

the second gives

$$(b) = mK\,|(n,m),K\rangle\,, \tag{14.23}$$

and the third gives

$$(c) = K(K+2)\,|(n,m),K\rangle\,. \tag{14.24}$$

The energy eigenvalue is therefore

$$\omega_1\left(n+K+\frac{3}{2}\right)+\omega_2\left(m+K+\frac{3}{2}\right)+4\Delta K(n+m+K+2). \tag{14.25}$$

The useful viewpoint is that the harmonic part has a larger $SU(3)\times SU(3)$ symmetry, while the interaction term preserves only the diagonal $SU(3)$ under which $a_k b_k$ and $a_k^\dagger b_k^\dagger$ are singlets.

Problems

14.A. Show that the operators built as the traceless part of products of creation and annihilation operators transform as tensor operators in the $(2,1)$ representation.

14.B. Calculate the nonzero matrix elements of the operator in 14.A between appropriate oscillator states.

14.C. Show that (14.13) generates the standard angular momentum $r\times p$.

14.D. Show that the generators (14.17) commute with the Hamiltonian (14.16).

Chapter 15

$SU(6)$ and the Quark Model

The low-lying baryon octet with spin 1/2 and decuplet with spin 3/2 lie close enough in mass to suggest a larger symmetry connecting flavor and spin. This motivates $SU(6)$, combining $SU(3)$ flavor with $SU(2)$ spin.

15.1 Including the spin

The quark state space is a tensor product of flavor and spin:

$$|q,i\rangle \in 3 \otimes 2. \tag{15.1}$$

The $SU(6)$ generators include the flavor generators

$$\frac{1}{2}\lambda_a, \tag{15.2}$$

with the identity in spin space, and the spin generators

$$\frac{1}{2}\sigma_j, \tag{15.3}$$

with the identity in flavor space. The remaining generators are the products

$$\frac{1}{2}\lambda_a\sigma_j. \tag{15.4}$$

There are $8+3+24=35$ generators, as expected for $SU(6)$. The defining 6 transforms under the subgroup $SU(3)\times SU(2)$ as $(3,2)$.

15.2 $SU(N)\times SU(M)\subset SU(NM)$

The general embedding is $SU(N)\times SU(M)\subset SU(NM)$. In the defining representation, the single NM index is replaced by a pair (i,x), where $i=1,\ldots,N$ and $x=1,\ldots,M$. The subgroup generators are

$$[T_a]_{ij}\delta_{xy}, \qquad \delta_{ij}[t_b]_{xy}. \tag{15.5}$$

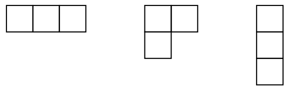
A representation of $SU(NM)$ with K boxes can contain an $SU(N)\times SU(M)$ representation (D_1,D_2) only if

$$K = K_1 \mod N = K_2 \mod M. \tag{15.6}$$

The 56 of $SU(6)$ corresponds to the completely symmetric tableau

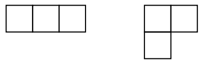
$$\square\square\square \tag{15.7}$$

The possible three-box $SU(3)$ tableaux are



(15.8)

The possible three-box $SU(2)$ tableaux are



(15.9)

The decuplet with spin 3/2 is

$$(10, 4), \tag{15.10}$$

while the octet with spin 1/2 is

$$(8, 2). \tag{15.11}$$

Thus the 56 contains precisely the low-lying baryon multiplets.

15.3 The baryon states

The decuplet states are symmetric in both flavor and spin. Examples are

$$\begin{aligned}
\left|\Delta^{++}, 3/2\right\rangle &= |uuu\rangle \,|+++\rangle\,, \\
\left|\Delta^{+}, 1/2\right\rangle &= \frac{1}{3}\left(|uud\rangle + |udu\rangle + |duu\rangle\right)\left(|++-\rangle + |+-+\rangle + |-++\rangle\right), \\
\left|\Sigma^{*0}, 1/2\right\rangle &= \frac{1}{6}\left(|uds\rangle + |dus\rangle + |sud\rangle + |sdu\rangle + |dsu\rangle + |usd\rangle\right) \\
&\quad \times \left(|++-\rangle + |+-+\rangle + |-++\rangle\right).
\end{aligned} \tag{15.12}$$

The octet states are mixed-symmetry states. For example,

$$\begin{aligned}
|\Lambda, 1/2\rangle =& \frac{\sqrt{3}}{6}\Big[(|uds\rangle - |dus\rangle)(|+-+\rangle - |-++\rangle) \\
&+ (|sud\rangle - |sdu\rangle)(|++-\rangle - |+-+\rangle) \\
&+ (|dsu\rangle - |usd\rangle)(|-++\rangle - |++-\rangle)\Big].
\end{aligned} \tag{15.13}$$

The proton spin-up state may be written as

$$\begin{aligned}
|P, 1/2\rangle =& \frac{\sqrt{2}}{6}\Big[\,|uud\rangle \left(2\,|++-\rangle - |+-+\rangle - |-++\rangle\right) \\
&+ |udu\rangle \left(2\,|+-+\rangle - |-++\rangle - |++-\rangle\right) \\
&+ |duu\rangle \left(2\,|-++\rangle - |++-\rangle - |+-+\rangle\right)\Big].
\end{aligned} \tag{15.14}$$

15.4 Magnetic moments

For a pointlike quark, the magnetic moment has the form

$$\frac{eQ\sigma}{2m}. \tag{15.15}$$

This transforms as an $SU(6)$ adjoint, so the relevant matrix elements are

$$\langle 56|35|56\rangle. \tag{15.16}$$

The tensor product contains a unique 56:

$$35 \otimes 56 \supset 56, \tag{15.17}$$

so the Wigner–Eckart theorem fixes the matrix elements up to one reduced matrix element:

$$\mu_{56} \propto \langle 56|Q\sigma|56\rangle. \tag{15.18}$$

For the proton,

$$\begin{aligned} Q\sigma_3\,|P,1/2\rangle = \frac{\sqrt{2}}{6}\Big\{ &\frac{2}{3}\,|uud\rangle\,(2\,|++-\rangle - |+-+\rangle + |-++\rangle) \\ &+ \frac{2}{3}\,|uud\rangle\,(2\,|++-\rangle + |+-+\rangle - |-++\rangle) \\ &- \frac{1}{3}\,|uud\rangle\,(-2\,|++-\rangle - |+-+\rangle - |-++\rangle) + \text{cyclic}\Big\}. \end{aligned} \tag{15.19}$$

Therefore

$$\langle P,1/2|Q\sigma_3|P,1/2\rangle = 3\frac{2}{36}\left[\frac{2}{3}(4+1-1) + \frac{2}{3}(4-1+1) - \frac{1}{3}(-4+1+1)\right] = 1. \tag{15.20}$$

For the neutron,

$$\langle N,1/2|Q\sigma_3|N,1/2\rangle = 3\frac{2}{36}\left[-\frac{1}{3}(4+1-1) - \frac{1}{3}(4-1+1) + \frac{2}{3}(-4+1+1)\right] = -\frac{2}{3}. \tag{15.21}$$

Thus

$$\frac{\mu_P}{\mu_N} = -\frac{3}{2}. \tag{15.22}$$

Experimentally,

$$\mu_P = 2.79, \qquad \mu_N = -1.91, \qquad \frac{\mu_P}{\mu_N} = -1.46. \tag{15.23}$$

The nonrelativistic quark-model magnetic moment is

$$\frac{eQ\sigma}{2m}. \tag{15.24}$$

For three quarks,

$$\sum_{\text{quarks}} \frac{eQ_i\sigma_i}{2m}. \tag{15.25}$$

The proton magnetic moment in nuclear magnetons obeys

$$\frac{e}{2m_p}\mu_P = \frac{e}{2m}\frac{\langle P|Q\sigma|P\rangle}{\langle P|\sigma|P\rangle}. \tag{15.26}$$

With $m \simeq m_p/3$,

$$\mu_P \simeq \frac{m_p}{m} = 3. \tag{15.27}$$

Coleman–Glashow gives

$$\mu_\Lambda = -\frac{1}{3}\mu_P = -0.93, \tag{15.28}$$

which corresponds to

$$\langle \Lambda,1/2|Q\sigma_3|\Lambda,1/2\rangle = -\frac{1}{3}. \tag{15.29}$$

Experimentally,

$$\mu_\Lambda \simeq -0.61. \tag{15.30}$$

The quark model gives

$$\frac{e}{2m_p}\mu_\Lambda = \sum_{\text{quarks}} \frac{eQ_a\sigma_a}{2m_{\text{quark}}}. \tag{15.31}$$

Only the strange quark contributes in the Λ state, so

$$\mu_\Lambda = -\frac{1}{3}\frac{m}{m_s}. \tag{15.32}$$

With

$$m_s \simeq \frac{3m_\Lambda - 2m_p}{3} \simeq 490\,\text{MeV}, \tag{15.33}$$

one obtains

$$\mu_\Lambda \simeq -0.64. \tag{15.34}$$

Problems

15.A. Find the $SU(6)$ wave functions for all spin-1/2 baryons except P, N, and Λ.

15.B. Use these wave functions to calculate magnetic moments: first in the $SU(6)$ limit, then including $m_s \neq m_{u,d}$.

15.C. Show that the $|\Lambda, 1/2\rangle$ state in (15.13) is an isospin singlet.

Chapter 16

Color

The simple quark model has two immediate problems. First, spin-1/2 quarks should obey Fermi statistics, while the low-lying baryon wave functions used in the $SU(6)$ model are symmetric in spin-flavor. Second, one needs an explanation for why only $q\overline{q}$ and qqq combinations appear as isolated hadrons. The resolution is color.

16.1 Colored quarks

Quarks transform as a 3 under color $SU(3)$:

$$q^i, \qquad i = 1, 2, 3. \tag{16.1}$$

A color-singlet baryon is made using the invariant epsilon tensor:

$$\epsilon_{ijk} q^i q^j q^k. \tag{16.2}$$

Because ϵ_{ijk} is antisymmetric, the remaining spin-flavor-space part of the baryon wave function can be symmetric.

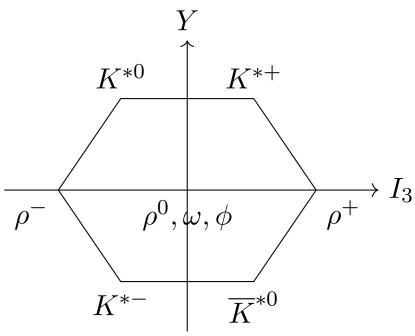
Antibaryons are similarly

$$\epsilon^{ijk} \overline{q}_i \overline{q}_j \overline{q}_k. \tag{16.3}$$

Mesons are color singlets of the form

$$q^i \overline{q}_i. \tag{16.4}$$

The vector-meson octet/singlet has the same weight pattern as the pseudoscalar mesons:



(16.5)

For two colored particles A and B,

$$T_a^A \left| r, A \right\rangle = \left| s, A \right\rangle [T_a^A]_{sr}, \qquad T_a^B \left| x, B \right\rangle = \left| y, B \right\rangle [T_a^B]_{yx}. \tag{16.6}$$

A two-particle state is

$$\left| v, A, B \right\rangle = v_{rx} \left| r, A \right\rangle \left| x, B \right\rangle. \tag{16.7}$$

The color interaction is proportional to

$$T_a^A T_a^B, \tag{16.8}$$

summed over a. The generator on the tensor-product space is

$$T_a^A + T_a^B. \tag{16.9}$$

It commutes with the interaction:

$$[T_a^A + T_a^B, T_b^A T_b^B] = if_{abc}(T_c^A T_b^B + T_b^A T_c^B) = 0. \tag{16.10}$$

The useful identity is

$$T_a^A T_a^B = \frac{1}{2}\left[(T_a^A + T_a^B)^2 - (T_a^A)^2 - (T_a^B)^2\right]. \tag{16.11}$$

The color force is most attractive in the least colorful channel, i.e. the representation with smallest quadratic Casimir. For $q\overline{q}$, the singlet is preferred. For qqq, the completely antisymmetric color singlet is preferred.

16.2 Quantum Chromodynamics

QCD is the quantum theory of quarks and gluons. It has a dimensional scale Λ_{QCD} of order a few hundred MeV. At short distances compared with $1/\Lambda_{\text{QCD}}$, the interaction is relatively weak; at long distances, it confines colored states into color singlets.

The light-quark mass term can be decomposed as

$$\begin{aligned} &m_u\overline{u}u + m_d\overline{d}d + m_s\overline{s}s \\ &= \frac{m_u + m_d + m_s}{3}(\overline{u}u + \overline{d}d + \overline{s}s) + \frac{m_u - m_d}{2}(\overline{u}u - \overline{d}d) \\ &+ \frac{2m_s - m_u - m_d}{6}(2\overline{s}s - \overline{u}u - \overline{d}d). \end{aligned} \tag{16.12}$$

The first term is flavor $SU(3)$ invariant. The second breaks isospin. The third transforms as the eighth component of an octet, explaining why Gell-Mann's medium-strong Hamiltonian ansatz works.

16.3 Heavy quarks

The heavier c, b, and t quarks are not part of Gell-Mann flavor $SU(3)$. Mesons containing one heavy antiquark and one light quark transform as a 3 of flavor $SU(3)$; mesons containing one heavy quark and one light antiquark transform as a $\overline{3}$; heavy-heavy mesons are flavor singlets. Baryons with one heavy quark and two light quarks transform as $6 \oplus \overline{3}$ under flavor $SU(3)$.

16.4 Flavor $SU(4)$ is useless!

Although it is tempting to enlarge flavor $SU(3)$ to include charm or bottom, this is not useful. The c and b masses are too large compared with Λ_{QCD}, so the breaking is not a small perturbation. Flavor $SU(4)$ and $SU(5)$ are therefore badly broken for this purpose.

Problems

16.A. Find a relation between the sum of products of color charges in a $q\bar{q}$ meson and a qq pair in a baryon.

16.B. Suppose a particle Q transforming as a 6 under color $SU(3)$ exists. What color-singlet bound states with one Q and additional quarks or antiquarks do you expect? How do they transform under flavor $SU(3)$?

16.C. For a representation D, define $C(D) = T_a T_a$. Use

$$\mathrm{Tr}(T_a T_a) = \dim(D) C(D) = 8k_D$$

and the rules

$$k_{D_1 \oplus D_2} = k_{D_1} + k_{D_2}, \qquad k_{D_1 \otimes D_2} = \dim(D_1) k_{D_2} + \dim(D_2) k_{D_1}$$

to compute $C(8)$, $C(10)$, and $C(6)$.

Chapter 17

Constituent Quarks

The color theory of hadrons is genuinely strongly interacting. Since there is no obvious small expansion parameter at hadronic scales, first-principles QCD calculations are difficult. Nevertheless, the nonrelativistic constituent-quark model gives a surprisingly good qualitative picture of light hadron masses.

17.1 The nonrelativistic limit

If all quark masses were large compared with $\Lambda_{\rm QCD}$, one could justify a nonrelativistic expansion. The leading contribution to a hadron mass would be the sum of constituent masses; the next term would be a spin-independent color potential; spin-dependent terms would be relativistic corrections suppressed by inverse powers of quark masses.

For ground-state hadrons, the most important spin-dependent interaction has the form

$$K \sum_{i<j} \frac{\sigma_i \cdot \sigma_j}{m_i m_j}, \tag{17.1}$$

where K includes the color and spatial wave-function dependence. In the flavor-symmetric limit, this depends only on the total spin:

$$\frac{K}{m^2} \sum_{i<j} \sigma_i \cdot \sigma_j = \frac{K}{2m^2} \left[\left(\sum_i \sigma_i \right)^2 - \sum_i \sigma_i^2 \right] = \frac{K}{2m^2} \left(4S(S+1) - 3n \right), \tag{17.2}$$

where $n = 3$ for baryons and $n = 2$ for mesons.

This interaction makes spin-one mesons heavier than spin-zero mesons and makes the baryon decuplet heavier than the octet. The smaller $K^* - K$ splitting compared with the $\rho - \pi$ splitting reflects the heavier strange quark. Similarly, the $\Sigma - \Lambda$ splitting reflects the different spin coupling of the u, d pair: in Σ, the u, d spins are aligned, while in Λ they are paired.

The success of this model is not fully understood from first principles, but it captures important features of QCD bound states and extends smoothly to hadrons containing heavy c and b quarks.

Problems

17.A. Suppose the quix Q from problem 16.B is a heavy spin-zero particle. Discuss the spectrum of ground-state qqQ states with light u, d, s quarks, giving their $SU(3)$ properties and spins.

17.B. Estimate $m_{u,d}/m_s$ from the $\rho - \pi$ and $K^* - K$ splittings. Make an independent estimate using suitable $\Sigma^* - \Sigma$ and $\Sigma - \Lambda$ mass differences.

17.C. Discuss semiquantitatively the mass spectrum of baryons with one charm quark and two light quarks.

Chapter 18

Unified Theories and $SU(5)$

The standard-model forces are associated with Lie algebras. This suggests the possibility of embedding the particle interactions into one simple algebra. The simplest successful-looking example is $SU(5)$.

18.1 Grand unification

The goal is to find a simple Lie algebra containing

$$SU(3)_{\text{color}} \times SU(2)_{\text{weak}} \times U(1)$$

as a subgroup, with the observed quarks and leptons fitting into representations of the larger group.

18.2 Parity violation, helicity and handedness

The weak interactions violate parity. For spin-1/2 particles, helicity is the component of spin along the direction of motion. Right-handed and left-handed components transform differently under the weak interaction.

For the right-handed particle creation operators, write

$$u^\dagger, \qquad d^\dagger, \qquad e^\dagger, \qquad \nu^\dagger, \tag{18.1}$$

including the corresponding antiquark and antilepton creation operators as needed. The positron and antineutrino form an $SU(2)$ doublet:

$$\ell_1^\dagger = e^\dagger, \qquad \ell_2^\dagger = \nu^\dagger. \tag{18.2}$$

Similarly,

$$\phi_1^\dagger = d^\dagger, \qquad \phi_2^\dagger = u^\dagger. \tag{18.3}$$

The electroweak commutators are

$$\begin{gathered}[S, u^\dagger] = \frac{2}{3}u^\dagger, \qquad [S, d^\dagger] = -\frac{1}{3}d^\dagger, \qquad [S, e^\dagger] = -e^\dagger, \\ [S, \phi_r^\dagger] = -\frac{1}{6}\phi_r^\dagger, \qquad [S, \ell_r^\dagger] = -\frac{1}{2}\ell_r^\dagger.\end{gathered} \tag{18.4}$$

The electric charge is

$$Q = R_3 + S. \tag{18.5}$$

Then

$$\begin{gathered}[Q, u^\dagger] = \frac{2}{3}u^\dagger, \qquad [Q, d^\dagger] = -\frac{1}{3}d^\dagger, \qquad [Q, e^\dagger] = -e^\dagger, \\ [Q, \overline{u}^\dagger] = -\frac{2}{3}\overline{u}^\dagger, \qquad [Q, \overline{d}^\dagger] = \frac{1}{3}\overline{d}^\dagger, \qquad [Q, e^{+\dagger}] = e^{+\dagger}, \qquad [Q, \nu^\dagger] = 0.\end{gathered} \tag{18.6}$$

18.3 Spontaneously broken symmetry

The electroweak $SU(2) \times U(1)$ cannot be an exact unbroken symmetry of the physical spectrum: the weak force is short-ranged, W and Z are massive, and fermion masses require left- and right-handed components. The symmetry is instead spontaneously broken to electromagnetism.

18.4 Physics of spontaneous symmetry breaking

A Higgs doublet ϕ with $S = 1/2$ transforms as

$$[S, \phi_1] = \frac{1}{2}\phi_1. \tag{18.7}$$

A simple invariant potential is

$$V(\phi) = \lambda(\phi^\dagger\phi - v^2)^2, \qquad \lambda > 0. \tag{18.8}$$

The vacuum may be chosen as

$$\phi_1 = 0, \qquad \phi_2 = v. \tag{18.9}$$

This leaves unbroken the generator

$$Q = R_3 + S. \tag{18.10}$$

The nonzero Higgs expectation value gives masses to the weak gauge bosons and permits fermion mass terms while preserving electromagnetism.

18.5 Is the Higgs real?

The Higgs mechanism is a model for spontaneous symmetry breaking. The group-theoretic point is that the vacuum transforms nontrivially under the full electroweak group, but is invariant under the electromagnetic $U(1)$ subgroup.

18.6 Unification and $SU(5)$

Including color, the right-handed creation operators transform under $SU(3) \times SU(2) \times U(1)$ as

$$(3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1} \oplus (\overline{3},2)_{-1/6} \oplus (1,2)_{1/2}. \tag{18.13}$$

The left-handed creation operators transform as the complex conjugate:

$$(\overline{3},1)_{-2/3} \oplus (\overline{3},1)_{1/3} \oplus (1,1)_{1} \oplus (3,2)_{1/6} \oplus (1,2)_{-1/2}. \tag{18.14}$$

The possible five-dimensional subset for an $SU(5)$ fundamental is

$$(\overline{3},1)_{1/3} \oplus (1,2)_{-1/2}. \tag{18.15}$$

The alternative

$$(3,1)_{2/3} \oplus (1,2)_{1/2} \tag{18.16}$$

does not work because the $U(1)$ generator would not be traceless. Embed $SU(3)$ in the first three indices of the 5:

$$\begin{pmatrix} T_a & 0 \\ 0 & 0 \end{pmatrix}, \tag{18.17}$$

and $SU(2)$ in the last two:

$$\begin{pmatrix} 0 & 0 \\ 0 & \tau_i/2 \end{pmatrix}. \tag{18.18}$$

The commuting $U(1)$ generator is

$$S = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \tag{18.19}$$

Thus the $\overline{5}$ fields may be written as

$$\lambda_x^\dagger = d_x^\dagger \quad (x = 1, 2, 3), \qquad \lambda_4^\dagger = e^\dagger, \qquad \lambda_5^\dagger = \nu^\dagger. \tag{18.20}$$

The remaining fields transform as

$$(3, 1)_{2/3} \oplus (1, 1)_{-1} \oplus (\overline{3}, 2)_{-1/6}. \tag{18.21}$$

The antisymmetric product of two 5s gives

$$\left[(3, 1)_{-1/3} \oplus (1, 2)_{1/2}\right]_{\mathrm{AS}}^{\otimes 2} = (\overline{3}, 1)_{-2/3} \oplus (1, 1)_1 \oplus (3, 2)_{1/6}. \tag{18.22}$$

Thus the remaining right-handed creation operators fit into a 10:

$$\begin{aligned} &\chi^{ab\dagger} = \epsilon^{abc} u_c^\dagger, \qquad a, b, c = 1, 2, 3, \\ &\chi^{a4\dagger} = u_a^\dagger, \qquad \chi^{a5\dagger} = d_a^\dagger, \qquad\qquad a = 1, 2, 3, \\ &\chi^{45\dagger} = e^\dagger. \end{aligned} \tag{18.23}$$

So one standard-model family fits into $\overline{5} \oplus 10$ of $SU(5)$.

18.7 Breaking $SU(5)$

Breaking $SU(5)$ to $SU(3) \times SU(2) \times U(1)$ can be done by a Higgs field in the adjoint 24 with a vacuum expectation value in the S direction. This is analogous to using the hypercharge direction in $SU(3)$ to preserve isospin and hypercharge.

Fermion masses require Higgs representations appearing in the tensor products of the fermion representations and containing an electroweak Higgs doublet. Since

$$[3] \otimes [3] = [1] \oplus [4, 2] \oplus [3, 3] = 5 \oplus 45 \oplus 50, \tag{18.24}$$

only the 5 and 45 contain components useful for ordinary electroweak mass generation; the 50 does not contain the required doublet.

18.8 Proton decay

Because quarks and leptons occur in common irreducible $SU(5)$ multiplets, some $SU(5)$ gauge interactions violate baryon number. If the $SU(5)$-breaking scale is very high, the corresponding gauge bosons are heavy and proton decay is rare. Minimal nonsupersymmetric $SU(5)$ is strongly constrained, but elaborations such as supersymmetric $SU(5)$ remain important models.

Problems

18.A. Check explicitly that electron and quark mass terms are allowed in the electroweak model with Higgs doublet (18.7).

18.B. Find the symmetric tensor product of (18.15) with itself.

18.C. Do the same for (18.21).

18.D. Consider a baryon-number violating operator made from two quark annihilation operators and a lepton creation operator. Show that it can mediate proton decay into $\pi^0 e^+$ with the right charge and color properties.

18.E. Decompose the 45 and 50 of $SU(5)$ under $SU(3) \times SU(2) \times U(1)$.

Chapter 19

The Classical Groups

There are four infinite series of simple Lie algebras, the classical algebras. The $SU(N)$ algebras are the A_n series, with Dynkin diagrams

$$\circ - \circ - \cdots - \circ - \circ. \tag{19.1}$$

The rank-n algebra $SU(n+1)$ is denoted A_n.

19.1 The $SO(2n)$ algebras

The group $SO(2n)$ is generated by imaginary antisymmetric $2n \times 2n$ matrices. A convenient Cartan basis is

$$H_m = \mathrm{diag}(0, \ldots, 0, \sigma_2, 0, \ldots, 0), \tag{19.2}$$

where σ_2 acts in the mth two-dimensional plane. The eigenvectors of σ_2 are

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}. \tag{19.3}$$

Thus the eigenvectors of H_m may be written

$$[\pm e_k]_j = \delta_{j,2k-1} \pm i\delta_{j,2k}, \tag{19.4}$$

with

$$H_m \left|\pm e_k\right\rangle = \pm\delta_{km} \left|\pm e_k\right\rangle . \tag{19.5}$$

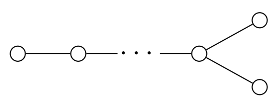
The weights are therefore

$$\pm e_k, \qquad [e_k]_j = \delta_{kj}. \tag{19.6}$$

The roots are

$$\begin{aligned} \text{roots:} &\quad \pm e_j \pm e_k, \qquad j \neq k, \\ \text{positive roots:} &\quad e_j \pm e_k, \qquad j < k, \\ \text{simple roots:} &\quad e_j - e_{j+1}, \quad j = 1, \ldots, n-1, \qquad e_{n-1} + e_n. \end{aligned} \tag{19.7}$$

The Dynkin diagram is



(19.8)

This is the D_n algebra.

19.2 The $SO(2n+1)$ algebras

For $SO(2n+1)$, the Cartan generators may be chosen as before, but there is one additional zero weight from the extra dimension. The roots are

$$\begin{aligned}
\text{roots:} &\quad \pm e_j \pm e_k \quad (j \neq k), \qquad \pm e_j, \\
\text{positive roots:} &\quad e_j \pm e_k \quad (j < k), \qquad e_j, \\
\text{simple roots:} &\quad e_j - e_{j+1}, \quad j = 1, \ldots, n-1, \qquad e_n.
\end{aligned} \tag{19.9}$$

The Dynkin diagram is

$$\circ - \circ - \cdots - \circ \Rightarrow \circ. \tag{19.10}$$

This is the B_n algebra.

19.3 The $Sp(2n)$ algebras

Consider Hermitian $2n \times 2n$ matrices of the form

$$1 \otimes A, \qquad \sigma_1 \otimes S_1, \qquad \sigma_2 \otimes S_2, \qquad \sigma_3 \otimes S_3, \tag{19.11}$$

where A is antisymmetric and the S_i are symmetric $n \times n$ matrices. The subset

$$1 \otimes A, \qquad \sigma_3 \otimes S_3 \tag{19.12}$$

contains an $SU(n)$ subalgebra of the form

$$\begin{pmatrix} T_a & 0 \\ 0 & -T_a^* \end{pmatrix}, \tag{19.13}$$

with representation $n \oplus \overline{n}$. The last Cartan generator may be taken as

$$H_n = \frac{1}{\sqrt{2n}} \sigma_3 \otimes 1. \tag{19.14}$$

The remaining roots come from matrices

$$(\sigma_1 \pm i\sigma_2) \otimes S_{k\ell}, \tag{19.15}$$

where $S_{k\ell}$ is symmetric. If v^{n+1} is a unit vector orthogonal to all v^j, the roots may be written

$$\begin{aligned}
\text{roots:} &\quad v^j - v^k \quad (j \neq k), \qquad \pm\left(v^j + v^k + \sqrt{\frac{2}{n}} v^{n+1}\right), \\
\text{positive roots:} &\quad v^j - v^k \quad (j < k), \qquad v^j + v^k + \sqrt{\frac{2}{n}} v^{n+1}, \\
\text{simple roots:} &\quad v^j - v^{j+1}, \quad j = 1, \ldots, n-1, \qquad 2v^n + \sqrt{\frac{2}{n}} v^{n+1}.
\end{aligned} \tag{19.16}$$

The Dynkin diagram is

$$\circ - \circ - \cdots - \circ \Leftarrow \circ, \tag{19.17}$$

with the $SU(n)$ roots shorter. This is the C_n algebra.

19.4 Quaternions

The classical groups may be regarded as rotation groups over different number systems. $SO(N)$ rotates real vectors, $SU(N)$ rotates complex vectors preserving the Hermitian norm, and $Sp(2N)$ may be regarded as rotations of quaternionic vectors.

A quaternion has the form

$$B + \mathbf{j} \cdot \mathbf{C}, \tag{19.18}$$

where the quaternionic units satisfy

$$j_a j_b = -\delta_{ab} + \epsilon_{abc} j_c. \tag{19.19}$$

A quaternionic vector is

$$q_i = B_i + \mathbf{j} \cdot \mathbf{C}_i. \tag{19.20}$$

The generator has the form

$$iX = iA + j_1 S_1 + j_2 S_2 + j_3 S_3, \tag{19.21}$$

or

$$iX = iA + \mathbf{j} \cdot \mathbf{S}. \tag{19.22}$$

The quaternionic units may be represented by Pauli matrices:

$$j_a \to -i\sigma_a. \tag{19.23}$$

A quaternionic N-vector can be represented as a $2N \times 2$ complex matrix:

$$\begin{pmatrix} B - iC_3 & -iC_1 - C_2 \\ -iC_1 + C_2 & B + iC_3 \end{pmatrix}. \tag{19.24}$$

The relevant absolute value satisfies

$$|z| = 0 \Rightarrow z = 0, \qquad |z_1 z_2| = |z_1||z_2|. \tag{19.25}$$

For quaternions,

$$|B + \mathbf{j} \cdot \mathbf{C}|^2 = B^2 + \mathbf{C}^2. \tag{19.26}$$

Equivalently,

$$|B + \mathbf{j} \cdot \mathbf{C}|^2 = (B - \mathbf{j} \cdot \mathbf{C})(B + \mathbf{j} \cdot \mathbf{C}). \tag{19.27}$$

The natural norm is

$$\|Q\|^2 = \sum_{j=1}^{N} |q_j|^2. \tag{19.28}$$

The only further division algebra is the octonions, whose multiplication is not associative. The exceptional Lie algebras are related to octonions.

Problems

19.A. Consider the 36 matrices built from independent Pauli matrices σ, τ, and η. Show they form a Lie algebra. Find the roots, simple roots, and Dynkin diagram.

19.B. Do the same for the corresponding set of 28 matrices. Identify the algebra.

Chapter 20

The Classification Theorem

We now classify the simple Lie algebras following Dynkin. The argument uses the geometry of simple roots and the master formula.

20.1 Π-systems

The simple roots of a simple Lie algebra obey:

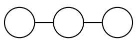
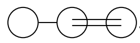
A. They are linearly independent vectors.

B. If α and β are distinct simple roots, then $2\alpha \cdot \beta/\alpha^2$ is a non-positive integer.

C. The simple-root system is indecomposable.

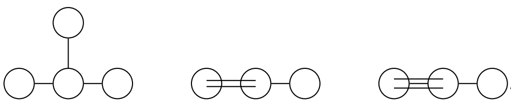
A root system is decomposable if it splits into mutually orthogonal subsystems. A decomposable system produces commuting invariant subalgebras, so the corresponding group is not simple but semisimple. Dynkin calls a vector system satisfying A, B, and C a Π-system.

Lemma 1. The only Π-systems of three vectors are

 and . (20.1)

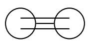
The reason is that the sum of the angles among three linearly independent vectors is less than $360°$, while allowed angles are $90°, 120°, 135°, 150°$.

There are diagrams satisfying B and C but not A:

. (20.2)

These fail because the vectors become coplanar.

A corollary is that no triple line occurs in a Π-system of three or more vectors. Hence the only Π-system with a triple line is

 (20.3)

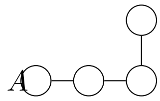
corresponding to G_2.

Lemma 2. If a Π-system contains two vectors connected by a single line, shrinking the line and replacing the two vectors by their sum gives another Π-system. If α and β are joined by a single line and γ is connected to α but not β, then

$$\gamma \cdot (\alpha + \beta) = \gamma \cdot \alpha, \qquad \gamma' \cdot (\alpha + \beta) = \gamma' \cdot \beta. \tag{20.4}$$

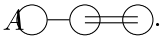
This implies that no Π-system has more than one double line and no Π-system contains a closed loop.

Lemma 3. If the configuration



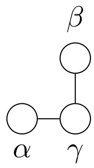
(20.5)

is a Π-system for some subdiagram A, then the diagram obtained by merging the two branches into a double line is also a Π-system:



(20.6)

Labeling the vectors as



(20.7)

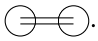
one has

$$\alpha \cdot \beta = 0, \qquad \frac{2\alpha \cdot \gamma}{\gamma^2} = \frac{2\alpha \cdot \gamma}{\alpha^2} = \frac{2\beta \cdot \gamma}{\gamma^2} = \frac{2\beta \cdot \gamma}{\beta^2} = -1, \tag{20.8}$$

and therefore

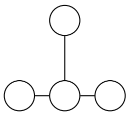
$$\frac{2\gamma \cdot (\alpha + \beta)}{\gamma^2} = -2, \qquad \frac{2\gamma \cdot (\alpha + \beta)}{(\alpha + \beta)^2} = -1. \tag{20.9}$$

The shrunken diagram is then



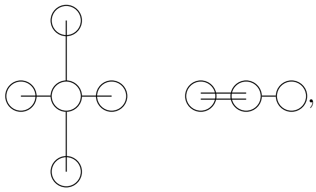
(20.10)

Thus branches only have the form



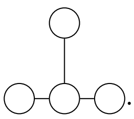
(20.11)

and not more complicated branchings such as



(20.12–20.13)

because they shrink to forbidden diagrams:



(20.14)

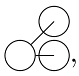
Lemma 4. No Π-system contains certain exceptional forbidden subdiagrams. The reason is that there exist nontrivial coefficients μ_j such that

$$\sum_j \mu_j \alpha^j = 0. \tag{20.15}$$

Thus linear independence fails.

The complete list of allowed indecomposable Dynkin diagrams is

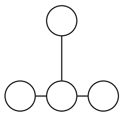
$$\begin{array}{rcl} A_n & : & \circ - \circ - \cdots - \circ - \circ, \\ B_n & : & \circ - \circ - \cdots - \circ \Rightarrow \bullet, \\ C_n & : & \bullet - \bullet - \cdots - \bullet \Leftarrow \circ, \\ D_n & : & \circ - \circ - \cdots - \end{array}$$



$$\begin{array}{rcl} G_2 & : & \circ \Rrightarrow \circ, \\ F_4 & : & \bullet - \bullet \Rightarrow \circ - \circ, \\ E_6, E_7, E_8 & : & \text{the three exceptional simply-laced branched diagrams.} \end{array} \tag{20.16}$$

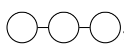
There are some coincidences: $A_1 = B_1 = C_1$, all giving $SU(2)$; $B_2 = C_2$; and

D_4 :



(20.17)

while D_3 degenerates into



(20.18)

so $D_3 = A_3$, while D_2 is not simple and corresponds to $SU(2) \times SU(2)$.

20.2 Regular subalgebras

A regular subalgebra R of a simple Lie algebra A is a subalgebra whose roots are a subset of the roots of A, with Cartan generators linear combinations of those of A. It is maximal if its rank is the same as that of A.

To find semisimple maximal regular subalgebras, add the lowest root α^0 to the simple roots. Since α^0 is the lowest root,

$$\frac{2\alpha^0 \cdot \alpha^i}{\alpha^{i2}}, \qquad \frac{2\alpha^i \cdot \alpha^0}{\alpha^{02}} \tag{20.19}$$

are non-positive integers. The resulting object is an extended Π-system, or extended Dynkin diagram. Removing one vector from an extended diagram gives the simple roots of a regular maximal subalgebra.

The extended diagrams are summarized schematically as

$$\begin{array}{rcl} A'_n & : & \text{closed loop of } n+1 \text{ simply-laced nodes,} \\ B'_n & : & \text{extended } B_n, \\ C'_n & : & \text{extended } C_n, \\ D'_n & : & \text{extended } D_n, \\ G'_2 & : & \circ \Rrightarrow \circ - \circ, \\ F'_4 & : & \bullet - \bullet \Rightarrow \circ - \circ - \circ, \\ E'_6, E'_7, E'_8 & : & \text{extended exceptional diagrams.} \end{array} \tag{20.20}$$

For $SU(2)$ the extension requires a 180° notation, schematically

$$\circ \Longleftrightarrow \circ. \tag{20.21}$$

Examples: A_n has no semisimple regular maximal subalgebras of this sort; removing a node from A'_n gives A_n again. For B_n, removing different nodes gives B_n, D_n, or products $B_k \times D_{n-k}$. For C_n, one obtains $C_k \times C_{n-k}$, including the degenerate case $A_1 \times C_{n-1}$. For D_n, one obtains $D_k \times D_{n-k}$. The exceptional groups have a finite list obtained by deleting nodes from their extended diagrams.

20.3 Other Subalgebras

The general problem of classifying subalgebras is more complicated. The useful principle is that each algebra has a simplest representation from which other representations can be built. Knowing how this simplest representation transforms under a proposed subalgebra determines the branching of all other representations.

Problems

20.A. Prove that decomposable Π-systems yield decomposable root systems.

20.B. Find the regular maximal subalgebras of E_6.

20.C. Find the regular maximal subalgebras of $SO(12)$. To find them all, apply the extended Dynkin diagram algorithm repeatedly.

Chapter 21

$SO(2n+1)$ and Spinors

The $SO(N)$ algebras have spinor representations. This chapter constructs them explicitly for $SO(2n+1)$ and discusses their reality properties.

21.1 Fundamental weights of $SO(2n+1)$

Label the generators as

$$M_{ab} = -M_{ba}, \qquad a, b = 1, \ldots, 2n+1. \tag{21.1}$$

In the defining representation,

$$[M_{ab}]_{xy} = -i(\delta_{ax}\delta_{by} - \delta_{bx}\delta_{ay}). \tag{21.2}$$

The commutation relations are

$$[M_{ab}, M_{cd}] = -i(\delta_{bc}M_{ad} - \delta_{ac}M_{bd} - \delta_{bd}M_{ac} + \delta_{ad}M_{bc}). \tag{21.3}$$

A Cartan subalgebra is generated by

$$H_j = M_{2j-1,2j}, \qquad j = 1, \ldots, n. \tag{21.4}$$

One may take the roots to be represented by

$$E_{\eta e_j} = \frac{1}{\sqrt{2}}\big(M_{2j-1,2n+1} + i\eta M_{2j,2n+1}\big),$$

$$E_{\eta e_j + \eta' e_k} = \frac{1}{2}\big(M_{2j-1,2k-1} + i\eta M_{2j,2k-1} + i\eta' M_{2j-1,2k} - \eta\eta' M_{2j,2k}\big), \tag{21.5}$$

where $\eta, \eta' = \pm 1$. They satisfy

$$[H_j, E_{\eta e_k}] = \eta [e_k]_j E_{\eta e_k} = \eta \delta_{jk} E_{\eta e_k}. \tag{21.6}$$

The simple roots are

$$\alpha^j = e_j - e_{j+1}, \qquad j = 1, \ldots, n-1, \qquad \alpha^n = e_n, \tag{21.7}$$

corresponding to the Dynkin diagram

$$\circ - \circ - \cdots - \circ \Rightarrow \bullet. \tag{21.8}$$

The fundamental weights are

$$\mu_j = \sum_{k=1}^{j} e_k, \qquad j = 1, \ldots, n-1, \qquad \mu_n = \frac{1}{2}\sum_{k=1}^{n} e_k. \tag{21.9}$$

The last fundamental representation is the spinor. Its weights are

$$\frac{1}{2}(\pm e_1 \pm e_2 \cdots \pm e_n). \tag{21.10}$$

There are 2^n such weights. Write the spinor space as a tensor product of n two-dimensional spaces, and define Pauli matrices σ_a^j acting in the jth factor:

$$|\pm e_1/2 \pm e_2/2 \cdots \pm e_n/2\rangle = |\pm e_1/2\rangle \otimes |\pm e_2/2\rangle \otimes \cdots \otimes |\pm e_n/2\rangle\,,$$

$$\sigma_a^j\, |x e_j\rangle = \left|x' e_j\right\rangle [\sigma_a]_{x'x}, \qquad x, x' = \pm\frac{1}{2}. \tag{21.11}$$

Then the Cartan generators are

$$H_j = \frac{1}{2}\sigma_3^j. \tag{21.12}$$

They satisfy

$$H_j^2 = \frac{1}{4}. \tag{21.13}$$

Since any M_{ab} could have been chosen as a Cartan generator,

$$M_{ab}^2 = \frac{1}{4}, \qquad a \neq b. \tag{21.14}$$

Now consider

$$E_{e_j} = \frac{1}{\sqrt{2}}(M_{2j-1,2n+1} + iM_{2j,2n+1}). \tag{21.15}$$

Since the corresponding state can be raised at most once,

$$(E_{e_j})^2 = 0. \tag{21.16}$$

This implies

$$\{M_{2j-1,2n+1}, M_{2j,2n+1}\} = 0. \tag{21.17}$$

More generally,

$$\{M_{ab}, M_{cd}\} = 0 \qquad \text{when all four indices are distinct.} \tag{21.18}$$

The raising operator obeys

$$E_{e_1} ket{-e_1/2 + x_2 e_2 + \cdots + x_n e_n} = f(x_2, \ldots, x_n)\, |e_1/2 + x_2 e_2 + \cdots + x_n e_n\rangle\,,$$

$$E_{-e_1}\, |-e_1/2 + x_2 e_2 + \cdots + x_n e_n\rangle = 0. \tag{21.19}$$

The norm is

$$\begin{aligned} |f(x_2, \ldots, x_n)|^2 &= \langle -e_1/2 + \cdots |\, E_{-e_1} E_{e_1}\, | -e_1/2 + \cdots \rangle \\ &= \langle -e_1/2 + \cdots |\, \{E_{-e_1}, E_{e_1}\}\, | -e_1/2 + \cdots \rangle\,. \end{aligned} \tag{21.20}$$

Using

$$\{E_{-e_1}, E_{e_1}\} = \frac{1}{2}, \tag{21.21}$$

one obtains

$$|f(x_2, \ldots, x_n)|^2 = \frac{1}{2}. \tag{21.22}$$

Choosing phases,

$$E_{e_1}\, |-e_1/2 + x_2 e_2 + \cdots + x_n e_n\rangle = \frac{1}{\sqrt{2}}\, |e_1/2 + x_2 e_2 + \cdots + x_n e_n\rangle\,. \tag{21.23}$$

Thus

$$E_{\pm e_1} = \frac{1}{\sqrt{2}}\sigma^1_\pm. \tag{21.24}$$

Similarly,

$$E_{e_2}\left|x_1e_1 - e_2/2 + \cdots + x_ne_n\right\rangle = h(x_1, x_3, \ldots, x_n)\left|x_1e_1 + e_2/2 + \cdots + x_ne_n\right\rangle,$$
$$|h(x_1, x_3, \ldots, x_n)|^2 = \frac{1}{2}. \tag{21.25}$$

For $x_1 = 1/2$, phases may be chosen so that

$$E_{e_2}\left|e_1/2 - e_2/2 + \cdots\right\rangle = \frac{1}{\sqrt{2}}\left|e_1/2 + e_2/2 + \cdots\right\rangle. \tag{21.26}$$

Using anticommutation with $E_{\pm e_1}$ gives

$$\begin{aligned} E_{e_2}\left|-e_1/2 - e_2/2 + \cdots\right\rangle &= \sqrt{2}E_{e_2}E_{-e_1}\left|e_1/2 - e_2/2 + \cdots\right\rangle \\ &= -\sqrt{2}E_{-e_1}E_{e_2}\left|e_1/2 - e_2/2 + \cdots\right\rangle \\ &= -\frac{1}{\sqrt{2}}\left|-e_1/2 + e_2/2 + \cdots\right\rangle. \end{aligned} \tag{21.27}$$

Therefore

$$E_{\pm e_2} = \frac{1}{\sqrt{2}}\sigma^1_3\sigma^2_\pm. \tag{21.28}$$

Continuing,

$$E_{\pm e_3} = \frac{1}{\sqrt{2}}\sigma^1_3\sigma^2_3\sigma^3_\pm,$$
$$E_{\pm e_j} = \frac{1}{\sqrt{2}}\sigma^1_3\cdots\sigma^{j-1}_3\sigma^j_\pm. \tag{21.29}$$

The Hermitian generators may be written

$$M_{2j-1,2n+1} = \frac{1}{2}\sigma^1_3\cdots\sigma^{j-1}_3\sigma^j_1, \qquad M_{2j,2n+1} = \frac{1}{2}\sigma^1_3\cdots\sigma^{j-1}_3\sigma^j_2. \tag{21.30}$$

All other generators follow from commutators,

$$M_{ab} = -i[M_{a,2n+1}, M_{b,2n+1}], \qquad a, b \neq 2n + 1. \tag{21.31}$$

21.2 Real and pseudo-real

Suppose T_a is an irreducible unitary representation equivalent to its complex conjugate:

$$T_a = -R^{-1}T^*_aR. \tag{21.32}$$

If another matrix Q also satisfies

$$T_a = -Q^{-1}T^*_aQ, \tag{21.33}$$

then

$$R^{-1}T^*_aR = Q^{-1}T^*_aQ, \tag{21.34}$$

so

$$T^*_aRQ^{-1} = RQ^{-1}T^*_a. \tag{21.35}$$

Equivalently,

$$[T^*_a, RQ^{-1}] = 0. \tag{21.36}$$

By Schur's lemma,

$$R = \lambda Q. \tag{21.37}$$

Thus R is unique up to scale. Hermiticity gives

$$T_a = -R^T T_a^* R^{-1T} \quad \Longrightarrow \quad [T_a, R^T R^{-1}] = 0. \tag{21.38}$$

Therefore

$$R^T = \lambda R. \tag{21.39}$$

Since transposing twice gives back the original matrix, $\lambda = \pm 1$. Thus

$$R = A = -A^T \qquad \text{or} \qquad R = S = S^T. \tag{21.40}$$

Representations with symmetric R are real-positive, or simply real; those with antisymmetric R are real-negative, or pseudo-real.

21.3 Real representations

Suppose the representation is equivalent to one generated by pure imaginary antisymmetric matrices:

$$T'_a = U^{-1} T_a U, \qquad T'_a = -T'^*_a. \tag{21.41}$$

Then the group representation is real:

$$\exp\bigl(i\theta_a T'_a\bigr) \tag{21.42}$$

has real matrix elements. In this case

$$T'_a = -T'^T_a = U^T T_a^T U^{-1T} = U^{-1} T_a U,$$

so

$$T_a = -(UU^T) T_a^* (UU^T)^{-1}. \tag{21.43}$$

Thus $R = UU^T$ is symmetric.

Conversely, for a unitary representation,

$$-T_a^* = R^{-1} T_a R = R^\dagger T_a R^{-1\dagger}. \tag{21.44}$$

Then

$$RR^\dagger T_a = T_a RR^\dagger, \tag{21.45}$$

so R may be taken unitary. If R is also symmetric, write

$$R = VV^T. \tag{21.46}$$

Then

$$T'_a = V^{-1} T_a V \tag{21.47}$$

is antisymmetric, since

$$-V^T T_a^* V^{-1T} = V^{-1} T_a V. \tag{21.48}$$

21.4 Pseudo-real representations

Pseudo-real representations are equivalent to their complex conjugates but cannot be transformed into purely imaginary antisymmetric form. The basic example is spin 1/2 of $SU(2)$:

$$-\sigma_a^* = \sigma_2^{-1} \sigma_a \sigma_2. \tag{21.49}$$

Here the intertwiner σ_2 is antisymmetric.

21.5 R is an invariant tensor

For unitary representations, (21.32) can be written as

$$R^{-1}T_aR = -T_a^*. \tag{21.50}$$

In components,

$$[T_a]_{ij}R_{jk} + R_{ij}[T_a]_{jk} = 0. \tag{21.51}$$

Thus R is an invariant tensor in $D \otimes D$. For a real irreducible representation, the trivial representation appears once in $D \otimes D$, with symmetric invariant tensor for real representations and antisymmetric invariant tensor for pseudo-real ones.

21.6 The explicit form for R

It is sufficient to satisfy (21.32) for

$$H_j, \qquad M_{2j-1,2n+1}, \qquad M_{2j,2n+1} \tag{21.52}$$

for all j. Build

$$R = r^1 \otimes r^2 \otimes \cdots \otimes r^n. \tag{21.53}$$

The conditions imply

$$-r^1\sigma_{1,2}^{1*}r^{1-1} = \sigma_{1,2}^1, \qquad -r^1\sigma_3^{1*}r^{1-1}\,r^2\sigma_{1,2}^{2*}r^{2-1} = \sigma_3^1\sigma_{1,2}^2, \tag{21.54}$$

and in general

$$-\prod_{k=1}^{j-1} r^k\sigma_3^{k*}r^{k-1}\,r^j\sigma_{1,2}^{j*}r^{j-1} = \prod_{k=1}^{j-1}\sigma_3^k\,\sigma_{1,2}^j. \tag{21.55}$$

Therefore choose alternating σ_2 and σ_1 factors:

$$R = R^{-1} = \prod_{j\,\mathrm{odd}}\sigma_2^j\prod_{k\,\mathrm{even}}\sigma_1^k. \tag{21.56}$$

Then

$$R^T = (-1)^{n(n+1)/2}R.$$

Hence

Algebra	Spinors
$SO(8k+3)$	pseudo-real
$SO(8k+5)$	pseudo-real
$SO(8k+7)$	real
$SO(8k+1)$	real

(21.57)

Problems

21.A. Show that the set of ten matrices σ_a, $\sigma_a\tau_1$, $\sigma_a\tau_3$, and τ_2 generate the spinor representation of $SO(5)$. Find the matrix $R = R^{-1}$ such that $T_a = -RT_a^*R$.

21.B. Identify a convenient $SO(2n-1)$ subalgebra of $SO(2n+1)$ and determine how the spinor representation of $SO(2n+1)$ transforms under it.

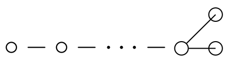
Chapter 22

$SO(2n + 2)$ Spinors

The spinor representations of $SO(2n + 2)$ can be obtained from the $SO(2n + 1)$ construction.

22.1 Fundamental weights of $SO(2n + 2)$

The Dynkin diagram for $SO(2n + 2)$ is



(22.1)

where

$$\alpha^j = e_j - e_{j+1}, \qquad j = 1, \ldots, n, \qquad \alpha^{n+1} = e_n + e_{n+1}. \tag{22.2}$$

All roots have the form

$$\pm e_j \pm e_k. \tag{22.3}$$

The two spinor representations correspond to the last two fundamental weights,

$$\mu_n = \frac{1}{2}(e_1 + \cdots + e_n - e_{n+1}), \qquad \mu_{n+1} = \frac{1}{2}(e_1 + \cdots + e_n + e_{n+1}). \tag{22.4}$$

Call the corresponding representations D_n and D_{n+1}. The weights of D_n are

$$\frac{1}{2}(\eta_1 e_1 + \cdots + \eta_{n+1} e_{n+1}), \tag{22.5}$$

where

$$\prod_{j=1}^{n+1} \eta_j = -1, \qquad \eta_j = \pm 1. \tag{22.6}$$

The weights of D_{n+1} have the same form but obey

$$\prod_{j=1}^{n+1} \eta_j = 1. \tag{22.7}$$

Since roots change the signs of two η's, these two sets are separately preserved.

For odd n, the spinors of $SO(2n+2) = SO(4m)$ are real or pseudo-real, because $-\mu_n$ remains in D_n and $-\mu_{n+1}$ remains in D_{n+1}. For even n, $-\mu_n$ lies in D_{n+1} and $-\mu_{n+1}$ lies in D_n, so the two spinor representations are complex conjugates.

Now consider the $SO(2n + 1)$ subgroup generated by

$$M_{jk}, \qquad j, k \leq 2n + 1. \tag{22.8}$$

This removes the Cartan generator

$$H_{n+1} = M_{2n+1,2n+2}. \tag{22.9}$$

The $4n$ generators with weights

$$\pm e_j \pm e_{n+1}, \qquad j \leq n, \tag{22.10}$$

which are linear combinations of

$$M_{j,2n+1} \qquad \text{and} \qquad M_{j,2n+2}, \qquad j \leq 2n, \tag{22.11}$$

collapse to the $2n$ generators $M_{j,2n+1}$ with weights

$$\pm e_j, \qquad j \leq n. \tag{22.12}$$

Under this $SO(2n+1)$ subgroup both D_n and D_{n+1} transform like the spinor representation of $SO(2n+1)$. Ignoring the $(n+1)$th component, their weights are

$$\frac{1}{2}\sum_{j=1}^{n} \eta_j e_j, \qquad \eta_j = \pm 1. \tag{22.13}$$

In tensor-product notation, the subgroup generators are

$$M_{2j-1,2n+1} = \frac{1}{2}\sigma_3^1 \cdots \sigma_3^{j-1}\sigma_1^j, \qquad M_{2j,2n+1} = \frac{1}{2}\sigma_3^1 \cdots \sigma_3^{j-1}\sigma_2^j. \tag{22.14}$$

For D_n,

$$\eta_{n+1} = -\prod_{j=1}^{n} \eta_j = -\sigma_3^1 \cdots \sigma_3^n, \tag{22.15}$$

so

$$H_{n+1} = M_{2n+1,2n+2} = -\frac{1}{2}\sigma_3^1 \cdots \sigma_3^n. \tag{22.16}$$

For D_{n+1},

$$\eta_{n+1} = \prod_{j=1}^{n} \eta_j = \sigma_3^1 \cdots \sigma_3^n, \tag{22.17}$$

and therefore

$$H_{n+1} = M_{2n+1,2n+2} = \frac{1}{2}\sigma_3^1 \cdots \sigma_3^n. \tag{22.18}$$

As before, define

$$R = R^{-1} = \prod_{j\,\text{odd}} \sigma_2^j \prod_{k\,\text{even}} \sigma_1^k. \tag{22.19}$$

Then examine the complex-conjugate representation

$$-RT_a^* R^{-1}. \tag{22.20}$$

For the $SO(2n+1)$ subgroup,

$$-RM_{jk}^* R^{-1} = M_{jk}, \qquad j,k \leq 2n+1. \tag{22.21}$$

If n is odd,

$$-RM_{2n+1,2n+2}^* R^{-1} = M_{2n+1,2n+2}, \tag{22.22}$$

so each spinor is equivalent to its complex conjugate. For $n = 4k+1$, R is antisymmetric, so the spinors are pseudo-real; for $n = 4k+3$, R is symmetric, so the spinors are real.

For even n,

$$-RM^*_{2n+1,2n+2}R^{-1} = -M_{2n+1,2n+2}, \tag{22.23}$$

so D_n and D_{n+1} are interchanged and are complex conjugates.

The complete pattern is

Algebra	Spinors
$SO(8k+3)$	pseudo-real
$SO(8k+4)$	pseudo-real
$SO(8k+5)$	pseudo-real
$SO(8k+6)$	complex
$SO(8k+7)$	real
$SO(8k)$	real
$SO(8k+1)$	real
$SO(8k+2)$	complex

(22.24)

Problems

22.A. Show that $SO(2n)$ has a regular maximal subalgebra $SO(2m) \times SO(2n-2m)$. How do the spinor representations of $SO(2n)$ transform under it?

22.B. Show that $SO(2n+1)$ has a regular maximal subalgebra $SO(2m) \times SO(2n-2m+1)$. How do the spinor representations of $SO(2n+1)$ transform under it?

22.C. Show that $SO(4)$ has the same algebra as $SU(2) \times SU(2)$. Explain how the arguments of this chapter apply despite non-simplicity.

22.D. Show that the $SO(6)$ and $SU(4)$ algebras are equivalent, with the 4 of $SU(4)$ corresponding to a spinor representation of $SO(6)$. In $SU(4)$, $4 \otimes 4 = 6 \oplus 10$. The 6 is the vector representation of $SO(6)$. What is the 10 in $SO(6)$ language?

Chapter 23

$SU(n) \subset SO(2n)$

To discuss the embedding of $SU(n)$ in $SO(2n)$ efficiently, it is useful to introduce another language for spinor representations: Clifford algebras. The same construction simultaneously exposes the tensor products of spinors and the $SU(n)$ subgroup of $SO(2n)$.

23.1 Clifford algebras

A Clifford algebra is a set of operators satisfying

$$\{\Gamma_j, \Gamma_k\} = 2\delta_{jk}, \qquad j, k = 1, \ldots, N. \tag{23.1}$$

Given such a set, define

$$M_{jk} = \frac{1}{4i}[\Gamma_j, \Gamma_k]. \tag{23.2}$$

The Jacobi identity and the Clifford relation imply that the M_{jk} satisfy the $SO(N)$ algebra. In addition, the Γ's transform as a vector tensor operator:

$$[M_{jk}, \Gamma_\ell] = i\big(\delta_{j\ell}\Gamma_k - \delta_{k\ell}\Gamma_j\big) = \Gamma_m\,[M_{jk}^{D^1}]_{m\ell}, \tag{23.3}$$

where

$$[M_{jk}^{D^1}]_{\ell m} = -i\big(\delta_{j\ell}\delta_{km} - \delta_{jm}\delta_{k\ell}\big) \tag{23.4}$$

generate the defining vector representation D^1, whose highest weight is μ^1.

For $SO(2n+1)$, a Clifford algebra acting on the 2^n-dimensional spinor space can be chosen as

$$\begin{aligned}
\Gamma_1 &= \sigma_2^1\sigma_3^2\cdots\sigma_3^n, & \Gamma_2 &= -\sigma_1^1\sigma_3^2\cdots\sigma_3^n,\\
\Gamma_3 &= \sigma_2^2\sigma_3^3\cdots\sigma_3^n, & \Gamma_4 &= -\sigma_1^2\sigma_3^3\cdots\sigma_3^n,\\
&\;\vdots & &\;\vdots\\
\Gamma_{2n-1} &= \sigma_2^n, & \Gamma_{2n} &= -\sigma_1^n,\\
\Gamma_{2n+1} &= \sigma_3^1\sigma_3^2\cdots\sigma_3^n.
\end{aligned} \tag{23.5}$$

Here σ_a^j denotes the Pauli matrix σ_a acting in the jth two-dimensional tensor factor. The product of all $2n+1$ Clifford generators is proportional to the identity:

$$\Gamma_1\Gamma_2\cdots\Gamma_{2n+1} = i^n\mathbf{1}. \tag{23.6}$$

The same matrices are closely related to the generators $M_{j,2n+2}$ in the spinor representation of $SO(2n+2)$: under the $SO(2n+1)$ subgroup, they transform as the components of a vector. Although the set (23.5) does not contain enough Clifford generators to build the full $SO(2n+2)$ algebra using (23.2), it does contain enough to construct $SO(2n)$ by omitting Γ_{2n+1}.

The resulting $SO(2n)$ representation is reducible, because

$$\Gamma_{2n+1} = (-i)^n \Gamma_1 \Gamma_2 \cdots \Gamma_{2n} = \prod_{j=1}^{n} \sigma_3^j \tag{23.7}$$

commutes with all $SO(2n)$ generators. Thus

$$\begin{aligned} &\frac{1}{2}(1 - \Gamma_{2n+1}) \quad \text{projects onto} \quad D^{n-1}, \\ &\frac{1}{2}(1 + \Gamma_{2n+1}) \quad \text{projects onto} \quad D^{n}. \end{aligned} \tag{23.8}$$

The product $\prod_{j=1}^{n} \sigma_3^j$ has eigenvalue -1 on the first subspace and $+1$ on the second.

23.2 Γ_m and R as invariant tensors

The usefulness of the Clifford construction is that the Γ matrices themselves are invariant tensors. Writing (23.3) in components gives

$$[M_{jk}]_{xx'}[\Gamma_\ell]_{x'y} - [\Gamma_\ell]_{xy'}[M_{jk}]_{y'y} + [M_{jk}^{D^1}]_{\ell m}[\Gamma_m]_{xy} = 0. \tag{23.9}$$

The first spinor index transforms in D^n, the second in the conjugate representation $\overline{D^n}$, and the vector index in D^1.

For $SO(2n+1)$, $\overline{D^n}$ is equivalent to D^n. Let R implement this equivalence:

$$M_{jk} = -R^{-1} M_{jk}^* R. \tag{23.10}$$

Write the matrix elements of M_{jk} as

$$[M_{jk}]^x{}_y. \tag{23.11}$$

Multiplying (23.10) by R and displaying the spinor indices gives

$$R^{xz}[M_{jk}]^y{}_z + [M_{jk}]^x{}_z R^{zy} = 0, \tag{23.12}$$

or, in a more symmetric form,

$$[M_{jk}]^x{}_z R^{zy} + [M_{jk}]^y{}_z R^{xz} = 0. \tag{23.13}$$

Thus R^{xy} is an invariant tensor with two upper spinor indices.

The Clifford matrices naturally have one upper and one lower spinor index,

$$[\Gamma_j]^x{}_y. \tag{23.14}$$

Multiplying by R raises the lower index:

$$[\Gamma_j R]^{xy} = [\Gamma_j]^x{}_z R^{zy}. \tag{23.15}$$

Unlike Γ_j itself, the tensor $\Gamma_j R$ has two identical kinds of spinor index, so it makes sense to ask whether it is symmetric or antisymmetric.

Equation (23.9) may be rewritten as

$$[M_{jk}]^x{}_z[\Gamma_\ell]^z{}_y + [R^{-1} M_{jk} R]^z{}_y [\Gamma_\ell]^x{}_z + [M_{jk}^{D^1}]_{\ell m}[\Gamma_m]^x{}_y = 0. \tag{23.16}$$

Multiplying by R^{yw} and using $R^T = \pm R$ yields

$$[M_{jk}]^x{}_z[\Gamma_\ell R]^{zw} + [M_{jk}]^w{}_z[\Gamma_\ell R]^{xz} + [M_{jk}^{D^1}]_{\ell m}[\Gamma_m R]^{xw} = 0. \tag{23.17}$$

Therefore $\Gamma_\ell R$ is an invariant tensor in $D^n \otimes D^n \otimes D^1$.

Define signs η_R and $\eta_{\Gamma R}$ by

$$R^T = \eta_R R, \qquad (\Gamma_j R)^T = \eta_{\Gamma R} \Gamma_j R, \qquad \eta_R, \eta_{\Gamma R} = \pm 1. \tag{23.18}$$

It is enough to inspect $\Gamma_{2n+1} R$. In every two-dimensional factor, multiplication by σ_3 interchanges the roles of σ_1 and σ_2 up to phases, and hence

$$\eta_R \eta_{\Gamma R} = (-1)^n. \tag{23.19}$$

From the explicit form of R found in Chapter 21,

$$\eta_R = (-1)^{n(n+1)/2}, \tag{23.20}$$

and therefore

$$\eta_{\Gamma R} = (-1)^{n(n-1)/2}. \tag{23.21}$$

23.3 Products of Γ's

Products of Clifford matrices also furnish invariant tensors. For example, the commutator of two Γ's transforms as an antisymmetric rank-two tensor. This follows directly from the Jacobi identity:

$$\begin{aligned} [M_{jk}, [\Gamma_\ell, \Gamma_m]] &= [[M_{jk}, \Gamma_\ell], \Gamma_m] + [\Gamma_\ell, [M_{jk}, \Gamma_m]] \\ &= [\Gamma_{m'}, \Gamma_m][M_{jk}^{D^1}]_{m'\ell} + [\Gamma_\ell, \Gamma_{m'}][M_{jk}^{D^1}]_{m'm}. \end{aligned} \tag{23.22}$$

More generally, for $r \leq n$, define the fully antisymmetrized product

$$\Gamma_{[j_1} \Gamma_{j_2} \cdots \Gamma_{j_r]} \equiv \frac{1}{r!} \sum_{p(j_1 \cdots j_r)} (\operatorname{sgn} p) \Gamma_{k_1} \Gamma_{k_2} \cdots \Gamma_{k_r}. \tag{23.23}$$

The Clifford relation eliminates all symmetric contractions, so these antisymmetric products are the genuinely new tensors. Since distinct Γ's anticommute, the antisymmetrization simply ensures that no two vector indices coincide.

For $SO(2n+1)$, products with more than n indices are not independent, because the product of all $2n+1$ Clifford generators is proportional to the identity. Consequently, the product $D^n \otimes \overline{D^n}$ contains antisymmetric tensors of rank $0, 1, \ldots, n$. The dimensions match because

$$\begin{aligned} 2^{2n+1} &= (1+1)^{2n+1} = 2 \sum_{k=0}^{n} \binom{2n+1}{k}, \\ 2^n \times 2^n &= \sum_{k=0}^{n} \binom{2n+1}{k}. \end{aligned} \tag{23.24}$$

The first equality uses

$$\binom{2n+1}{k} = \binom{2n+1}{2n+1-k}, \tag{23.25}$$

so the lower half of the binomial sum equals the upper half.

To determine whether the rank-k tensor lies in the symmetric or antisymmetric product of the two spinors, multiply by R:

$$\Gamma_{j_1} \Gamma_{j_2} \cdots \Gamma_{j_k} R = (\Gamma_{j_1} R) R^{-1} (\Gamma_{j_2} R) R^{-1} \cdots (\Gamma_{j_k} R). \tag{23.26}$$

Taking the transpose gives

$$\begin{aligned}
&(\Gamma_{j_k} R)^T R^{-T} \cdots (\Gamma_{j_2} R)^T R^{-T} (\Gamma_{j_1} R)^T \\
&\quad = \eta_R^{k-1} \eta_{\Gamma R}^k R(\Gamma_{j_k} R) R^{-1} \cdots (\Gamma_{j_2} R) R^{-1} (\Gamma_{j_1} R) \\
&\quad = \eta_R^{k-1} \eta_{\Gamma R}^k \Gamma_{j_k} \cdots \Gamma_{j_2} \Gamma_{j_1} R.
\end{aligned} \tag{23.27}$$

To restore the original order requires

$$(k-1) + (k-2) + \cdots + 1 = \frac{k(k-1)}{2} \tag{23.28}$$

transpositions. Each interchange contributes a minus sign, so

$$\left(\Gamma_{[j_1} \Gamma_{j_2} \cdots \Gamma_{j_k]} R\right)^T = (-1)^{k(k-1)/2} \eta_R^{k-1} \eta_{\Gamma R}^k \Gamma_{[j_1} \Gamma_{j_2} \cdots \Gamma_{j_k]} R. \tag{23.29}$$

The sign can be rewritten as

$$\begin{aligned}
(-1)^{k(k-1)/2} \eta_R^{k-1} \eta_{\Gamma R}^k &= (-1)^{k(k-1)/2} \eta_R (\eta_R \eta_{\Gamma R})^k \\
&= (-1)^{k(k-1)/2} (-1)^{n(n+1)/2} (-1)^{nk} \\
&= (-1)^{k(k-1)/2} (-1)^{n(n+1)/2} (-1)^{-nk} \\
&= (-1)^{(n-k)(n-k+1)/2}.
\end{aligned} \tag{23.30}$$

Thus the symmetry pattern repeats with period four:

k \ n	1	2	3	4
0	$-$	$-$	$+$	$+$
1	$+$	$-$	$-$	$+$
2	$+$	$+$	$-$	$-$
3	$-$	$+$	$+$	$-$

(23.31)

Here $+$ means symmetric and $-$ antisymmetric in the two spinor indices. Thus, for example, in $SO(3)$ the scalar is antisymmetric while the vector is symmetric; in $SO(5)$ the scalar and vector are antisymmetric while the rank-two tensor is symmetric; and the pattern continues periodically.

For $SO(2n)$, the 2^n-dimensional Clifford space is reducible. Introduce

$$P_\pm = \frac{1}{2}(1 \pm \Gamma_{2n+1}). \tag{23.32}$$

Then P_+ projects onto D^n and P_- onto D^{n-1}. A general $2^n \times 2^n$ matrix transforms under commutation like

$$\left(D^n \oplus D^{n-1}\right) \otimes \left(\overline{D^n} \oplus \overline{D^{n-1}}\right). \tag{23.33}$$

For an arbitrary matrix K,

$$\begin{array}{lll}
P_+ K P_+ & \text{transforms like} & D^n \otimes \overline{D^n}, \\
P_- K P_- & \text{transforms like} & D^{n-1} \otimes \overline{D^{n-1}}, \\
P_+ K P_- & \text{transforms like} & D^n \otimes \overline{D^{n-1}}, \\
P_- K P_+ & \text{transforms like} & D^{n-1} \otimes \overline{D^n}.
\end{array} \tag{23.34}$$

Since Γ_{2n+1} anticommutes with Γ_j for $j = 1, \ldots, 2n$,

$$\Gamma_j P_+ = P_- \Gamma_j, \qquad j = 1, \ldots, 2n. \tag{23.35}$$

Therefore odd products of Γ's occur between opposite projectors, whereas even products occur between equal projectors. As before, ranks larger than n are related by duality to lower ranks.

23.4 Self-duality

There is an additional subtlety for rank-n antisymmetric tensors. Not all components in $D^n \otimes \overline{D^n}$ or $D^{n-1} \otimes \overline{D^{n-1}}$ are independent. For example,

$$\begin{aligned} P_+\Gamma_1\Gamma_2\cdots\Gamma_n P_+ &= P_+\Gamma_1\Gamma_2\cdots\Gamma_n\Gamma_{2n+1}P_+ \\ &= (-i)^n P_+\Gamma_1\cdots\Gamma_n\Gamma_1\cdots\Gamma_{2n}P_+ \\ &= (-i)^n(-1)^{n(n-1)/2}P_+\Gamma_{n+1}\Gamma_{n+2}\cdots\Gamma_{2n}P_+ \\ &= (-i)^{n^2}\frac{1}{n!}\epsilon_{12\cdots n\, j_1\cdots j_n}P_+\Gamma_{j_1}\cdots\Gamma_{j_n}P_+. \end{aligned} \tag{23.36}$$

Thus rank-n tensors satisfy a duality constraint. If n is even, the constraint is real. The tensor in $D^n \otimes \overline{D^n}$ is self-dual,

$$A_{j_1\cdots j_n} = \frac{1}{n!}\epsilon_{j_1\cdots j_n k_1\cdots k_n}A_{k_1\cdots k_n}, \tag{23.37}$$

whereas the corresponding tensor in $D^{n-1} \otimes \overline{D^{n-1}}$ is anti-self-dual,

$$A_{j_1\cdots j_n} = -\frac{1}{n!}\epsilon_{j_1\cdots j_n k_1\cdots k_n}A_{k_1\cdots k_n}. \tag{23.38}$$

If n is odd, the duality condition is complex:

$$A_{j_1\cdots j_n} = \pm\frac{i}{n!}\epsilon_{j_1\cdots j_n k_1\cdots k_n}A_{k_1\cdots k_n}. \tag{23.39}$$

This complex self-duality is how the complex nature of the $SO(4m+2)$ spinor representations appears in the ordinary tensor representations.

Let (r) denote the rank-r antisymmetric tensor representation. The spinor products are summarized by

$$\begin{aligned} SO(2n+1): &\qquad D^n \otimes D^n = \sum_{k=0}^{n}(k), \\ SO(4m): &\qquad D^{2m} \otimes D^{2m-1} = \sum_{k=0}^{m-1}(2k+1), \\ &\qquad D^{2m} \otimes D^{2m} = \sum_{k=0}^{m-1}(2k) + (2m)_+, \\ &\qquad D^{2m-1} \otimes D^{2m-1} = \sum_{k=0}^{m-1}(2k) + (2m)_-, \\ SO(4m+2): &\qquad D^{2m+1} \otimes D^{2m} = \sum_{k=0}^{m}(2k), \\ &\qquad D^{2m+1} \otimes D^{2m+1} = \sum_{k=0}^{m-1}(2k+1) + (2m+1)_1, \\ &\qquad D^{2m} \otimes D^{2m} = \sum_{k=0}^{m-1}(2k+1) + (2m+1)_2. \end{aligned} \tag{23.40}$$

Here $(2m)_\pm$ are self-dual and anti-self-dual respectively, while $(2m+1)_1$ and $(2m+1)_2$ satisfy the two complex self-duality conditions.

23.5 Example: $SO(10)$

For the 16-dimensional spinor D^5 of $SO(10)$,

$$16 \otimes 16 = (1) \oplus (3) \oplus (5)_1. \tag{23.41}$$

The rank-one representation is the 10-dimensional vector. The rank-three tensor has dimension

$$\binom{10}{3} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2} = 120.$$

The rank-five tensor is subject to complex self-duality, so its dimension is half of $\binom{10}{5}$:

$$\frac{1}{2}\frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2} = 126. \tag{23.42}$$

The symmetry factor in (23.30), evaluated with $n = 5$, shows that the (1) and $(5)_1$ pieces are symmetric under interchange of the two identical 16's, while the (3) is antisymmetric. In conventional dimension notation,

$$16 \otimes 16 = 10_s \oplus 120_a \oplus 126_s.$$

Likewise,

$$16 \otimes \overline{16} = (0) \oplus (2) \oplus (4) = 1 \oplus 45 \oplus 210. \tag{23.43}$$

23.6 The $SU(n)$ subalgebra

The $SU(n)$ subgroup of $SO(2n)$ becomes transparent after combining the Clifford generators into fermionic creation and annihilation operators:

$$A_j = \frac{1}{2}\big(\Gamma_{2j-1} - i\Gamma_{2j}\big), \qquad A_j^\dagger = \frac{1}{2}\big(\Gamma_{2j-1} + i\Gamma_{2j}\big). \tag{23.44}$$

The Clifford relation gives

$$\{A_j, A_k\} = \{A_j^\dagger, A_k^\dagger\} = 0, \qquad \{A_j, A_k^\dagger\} = \delta_{jk}. \tag{23.45}$$

Thus the A_j and $A_j^\dagger$ form a set of fermionic annihilation and creation operators.

Let $[T_a]_{jk}$ be the matrices of the defining representation of $SU(n)$. Define

$$T_a = \sum_{j,k} A_j^\dagger [T_a]_{jk} A_k. \tag{23.46}$$

These operators generate the $SU(n)$ algebra on the Clifford space. To verify that they lie inside $SO(2n)$, expand

$$\begin{aligned} A_j^\dagger A_k &= \frac{1}{2}\{A_j^\dagger, A_k\} + \frac{1}{2}[A_j^\dagger, A_k] \\ &= \frac{1}{2}\delta_{jk} + \frac{i}{2}M_{2j-1,2k-1} + \frac{1}{2}M_{2j-1,2k} - \frac{1}{2}M_{2j,2k-1} + \frac{i}{2}M_{2j,2k}. \end{aligned} \tag{23.47}$$

The δ_{jk} term drops out of (23.46), because each T_a is traceless. The remaining terms are linear combinations of $SO(2n)$ generators, so the T_a generate an $SU(n)$ subalgebra of $SO(2n)$.

Let $|0\rangle$ be annihilated by every A_j. In the explicit Clifford representation, the A_j are proportional to lowering operators, so $|0\rangle$ is the state for which all $\sigma_3^j = -1$. It lies in D^n when n is even and in D^{n-1} when n is odd.

The creation operators transform in the defining representation:

$$[T_a, A_j^\dagger] = \sum_k A_k^\dagger [T_a]_{kj}. \tag{23.48}$$

Because the $A_j^\dagger$ anticommute, a state obtained by acting with r creation operators transforms as the completely antisymmetric r-index representation $[r]$ of $SU(n)$.

The parity of r determines which of the two $SO(2n)$ spinors contains the state: Γ_{2n+1} anticommutes with each $A_j^\dagger$, so an even number of creation operators preserves chirality and an odd number reverses it. Therefore, for $SO(4n+2)$, whose subgroup is $SU(2n+1)$,

$$D^{2n+1} = \sum_{j=0}^{n} [2j+1], \qquad D^{2n} = \sum_{j=0}^{n} [2j]. \tag{23.49}$$

This is consistent with $D^{2n} = \overline{D^{2n+1}}$, because $[r]$ and $[2n+1-r]$ are conjugate $SU(2n+1)$ representations.

For $SO(4n)$, whose subgroup is $SU(2n)$,

$$D^{2n} = \sum_{j=0}^{n} [2j], \qquad D^{2n-1} = \sum_{j=0}^{n-1} [2j+1]. \tag{23.50}$$

There is one more $SO(2n)$ generator commuting with the entire $SU(n)$ subalgebra:

$$S = \sum_{j=1}^{n} M_{2j-1,2j} = \sum_{j=1}^{n} A_j^\dagger A_j - \frac{n}{2}. \tag{23.51}$$

It generates the commuting $U(1)$. Since

$$S\,|0\rangle = -\frac{n}{2}\,|0\rangle\,, \tag{23.52}$$

and $[S, A_j^\dagger] = A_j^\dagger$, the antisymmetric representation $[r]$ carries

$$S = r - \frac{n}{2}.$$

Thus the construction actually gives the subgroup $SU(n) \times U(1) \subset SO(2n)$, with the $U(1)$ charge determined by fermion number relative to half filling.

Problems

23.A. Verify the dimensions in (23.49) and (23.50) using the binomial theorem.

23.B. Use (23.9) and (23.44) to determine how the vector representation D^1 of $SO(2n)$ transforms under $SU(n)$. Explain why the result is geometrically obvious.

23.C. Let $u_{jk\ell}$ be a completely antisymmetric tensor of $SO(6)$. A self-duality condition has the form

$$u_{jk\ell} = \lambda \epsilon_{jk\ell mnp} u_{mnp}.$$

What values of λ are possible?

23.D. Determine how the spinor representations of $SO(14)$ transform under

(a) $SU(7)$;

(b) $SO(4) \times SU(5)$.